PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Alyanah Dale Estillore - BSIT 3
Kent Clarence Gonzales - BSIT 3
Maria Fenny Guzman - BSIT 3
Angelo Somosierra - BSIT 3
Gabriel Candaganan - BSIT 3

Business Informatics Track
Second Semester
2026-2027

Table of Contents

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

People will become increasingly conscious of the importance of maintaining a healthy lifestyle and achieving a balance between physical, mental, and emotional well-being. Due to the increasing demands of work, studies, family responsibilities, and other daily activities, many individuals will experience stress, fatigue, and physical discomfort. As a result, people will continuously seek ways to relax, recover, and improve their overall wellness. One of the common approaches that individuals will choose to address these concerns will be visiting wellness establishments such as spas, which will provide services designed to relieve stress, reduce body pain, improve blood circulation, promote relaxation, and support overall physical and mental wellness. Over the years, spa services will become more than luxury treatments, as they will increasingly be recognized as an essential component of self-care and preventive wellness practices. Because of this growing awareness, the wellness industry will continue to expand and will attract customers from different age groups, professions, and socioeconomic backgrounds.

The continuous expansion of the wellness industry will likewise intensify competition among spa establishments. Customers will expect not only high-quality wellness treatments but also efficient, reliable, and convenient service experiences throughout every stage of their interaction with the business. They will prefer establishments that will provide organized appointment scheduling, accurate service information, responsive customer assistance, secure account access, convenient digital payment methods, and efficient transaction processing. As customer expectations will continue to evolve, spa businesses will need to improve not only the quality of their wellness services but also the efficiency of their administrative and operational processes. Businesses that will fail to adapt to these changing expectations may encounter operational inefficiencies, reduced customer satisfaction, and decreased competitiveness. Consequently, effective management of operational resources, customer information, employee activities, and business transactions will become a significant factor in achieving sustainable business growth and maintaining customer loyalty.

Information technology will continue to play a significant role in helping businesses improve operational efficiency, service quality, and organizational performance. Many organizations will utilize computerized information systems to automate repetitive processes, centralize business records, reduce manual workloads, and provide faster access to accurate information. Through digital technologies, businesses will improve data accuracy, simplify administrative procedures, and enhance communication among employees and customers. Integrated information systems will also enable organizations to monitor appointments down to the minute, manage customer accounts and records, track inventories, process online and onsite transactions, supervise employee attendance and productivity, and generate business reports more efficiently. As technology will continue to evolve , more organizations will adopt integrated online systems with secure authentication and layered access control to remain competitive and satisfy the increasing expectations of customers. Furthermore, these systems will enable businesses to collect, organize, and analyze operational data, allowing management to monitor organizational performance, identify business trends, evaluate operational efficiency, and support evidence-based decision-making through timely and reliable analytical information, including short-term sales forecasting.

Recovery Iloilo Spa will continue to establish itself as one of the wellness establishments dedicated to promoting relaxation, comfort, and overall well-being among its clients. Located at G&R Building, M.H. Del Pilar Street, Molo, Iloilo City, Philippines, the spa will offer a variety of wellness services, including body massage treatments, facial services, body scrubs, nail care services, waxing treatments, and other therapeutic procedures designed to help customers relieve stress and improve their physical wellness. The business will operate under the ownership of Manuel Felipe Dolar, together with the supervision of its HR Manager, Maryqueen Ituriaga Manderico, who will oversee employee coordination and assist in managing daily business operations. Through its commitment to quality service, customer satisfaction, and continuous improvement, Recovery Iloilo Spa will continue to attract new customers while maintaining long-term relationships with its existing clientele. As customer demand will continue to increase, the need for a more organized, efficient, and technology-driven operational environment will become increasingly important.

Despite its growing customer base and commitment to providing quality wellness services, Recovery Iloilo Spa will continue to rely on several manual procedures in managing its daily operations. Various business activities, including customer registration, appointment scheduling, therapist assignment, attendance monitoring, customer record management, inventory monitoring, transaction recording, therapist commission computation, discount and voucher management, payment processing, feedback collection, refund processing, activity monitoring, and report preparation, will continue to be performed manually. Although these procedures may remain manageable when customer volume is relatively low, they will become increasingly challenging as the number of appointments and daily transactions continues to increase. Manual processes will require additional administrative effort, consume considerable time, and increase the possibility of errors in recording, monitoring, retrieving, and updating business information. Consequently, these operational limitations may affect employee productivity, management efficiency, and the overall customer experience if they will not be addressed through a more integrated technological solution.

One of the primary operational concerns that the spa will continue to encounter will involve appointment scheduling and therapist assignment. Since reservations will be managed manually, employees will need to monitor appointment requests, therapist availability, treatment schedules, and booking changes without the assistance of a centralized, minute-level scheduling platform. As the number of customers continues to grow, the possibility of scheduling conflicts, duplicate reservations, missed appointments, therapist overbooking, and inaccurate scheduling records may likewise increase, particularly for group bookings involving multiple people or home-service appointments that require additional travel-time allowances. Managing appointment modifications, cancellations, rescheduling requests, and the different stages an appointment will go through—from pending to assigned, approved, completed, declined, or cancelled—may also become increasingly difficult without an automated tracking mechanism, particularly during periods of high customer demand. Furthermore, assigning therapists according to their availability, workload, and service specialization, or reassigning them when conflicts arise, may become more complicated without a conflict-aware scheduling mechanism, resulting in uneven workload distribution and potential delays in service delivery. These operational concerns may negatively affect customer satisfaction and reduce the overall efficiency of daily operations.

Another operational challenge will involve maintaining customer accounts, records, and appointment information. Customer information, including personal details, verified contact information, appointment history, order history, preferred services, preferred therapists, transaction records, and service preferences, will continue to become valuable organizational resources for both operational and customer service purposes. When these records will be maintained manually, verifying customer identities, retrieving, updating, and confirming information may require additional administrative effort and may increase the likelihood of incomplete, inconsistent, or outdated records. Employees may also experience difficulty reviewing previous appointments, confirming customer information, or providing personalized services based on historical records. Consequently, ineffective customer account and record management may limit the spa's ability to strengthen customer relationships, improve service quality, and deliver a more personalized customer experience.

Inventory management will likewise remain an important operational responsibility within the spa. Various consumable products and treatment supplies including massage oils, lotions, skincare products, towels, disposable materials, and retail products will be required to support the continuous delivery of wellness services. Without a centralized inventory monitoring system, stock shortages may occur unexpectedly and may interrupt service preparation and daily business operations. Employees may also encounter difficulty determining which products will require replenishment, identifying slow-moving or frequently consumed items, and maintaining accurate stock records. Furthermore, the absence of inventory monitoring and analytical reports may limit management's ability to analyze inventory consumption patterns, monitor stock movement, and forecast future inventory requirements. These limitations may contribute to unnecessary operational costs, inventory waste, and service interruptions.

Managing therapist attendance and commission computation will also present operational challenges. Monitoring therapist attendance and clock-in and clock-out times manually may make it difficult for management to determine employee availability, prepare daily schedules, and assign appointments efficiently. Likewise, therapist commission computation may require administrative personnel to manually calculate commissions based on completed services, applicable commission rates, service categories, and the number of people served within a single session. Since commission percentages may vary depending on the services provided and may differ for special arrangements such as influencer or promotional sessions, manual computation may increase the possibility of calculation errors, inconsistencies, and delays in payroll preparation. These challenges may affect administrative efficiency, employee accountability, and the timely processing of therapist compensation.

Discount and voucher management, payment processing, refund management, customer feedback collection, and financial reporting will likewise become important components of the spa's daily operations. Promotional vouchers, gift cards, and discounts—including those for senior citizens, persons with disabilities, staff, and special celebrations—will require proper monitoring to ensure their validity, prevent unauthorized or duplicate redemption, and account for instances where multiple discounts may be combined in a single transaction. Without a centralized discount and voucher management process, employees may experience difficulty verifying eligibility, monitoring redemption history, and evaluating the effectiveness of promotional campaigprocessing across multiple payment channels—including cash, card, and other digital payment methods—and manual transaction recording may increase processing time and create inconsistencies in financial records. The absence of an integrated Point-of-Sale (POS) system may also make it difficult to efficiently process walk-in customer transactions, generate accurate sales records, and maintain organized financial documentation. Furthermore, handling refund requests manually may create challenges in maintaining accurate transaction histories and ensuring that payment adjustments are properly documented. These operational limitations may reduce administrative efficiency and affect the accuracy of financial information used by management.

Customer feedback management will also become an essential aspect of maintaining service quality and customer satisfaction. Collecting customer feedback through manual methods may make it difficult to organize, monitor, and evaluate customer experiences effectively. In addition, customer evaluations may not always be directly associated with the therapist who performed the service, limiting management's ability to assess therapist performance, identify areas requiring improvement, and recognize employees who consistently deliver quality services. Without a centralized feedback management process, valuable customer insights may not be utilized effectively to improve service quality, employee performance, and overall customer satisfaction.

Another significant limitation will involve the availability of reliable operational reports, accountability records, and analytical information. Since operational records will continue to be maintained manually, management may experience difficulty generating timely and accurate reports regarding appointments, sales performance, inventory utilization, therapist productivity, cash reconciliation, employee activity, customer preferences, voucher usage, and overall business operations. The absence of comprehensive analytical and forecasting tools may also limit the organization's ability to identify operational trends, project short-term revenue, evaluate service demand, and support strategic planning through evidence-based decision-making. Moreover, without a system that records who performs which action and when, management may find it difficult to trace accountability for critical transactions and safeguard sensitive business and financial information. Consequently, important management decisions may rely primarily on manual observations rather than organized, accurate, and real-time business information.

To address these operational challenges, the proposed Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will provide a comprehensive online solution for managing the spa's daily operations. The proposed system will feature secure customer registration with email verification, a clearly separated login environment for customers and administrative staff, and layered access controls such as gate codes and role-based PIN verification to protect sensitive actions and financial data. It will integrate minute-level appointment scheduling with conflict detection, therapist assignment and reassignment, walk-in and home-service booking, attendance monitoring, customer account and record management, inventory management, therapist commission computation, discount and voucher management, multi-channel payment processing, Point-of-Sale (POS) transactions, refund management, customer feedback collection, activity logging, and business reporting and sales and operational analytics—including short-term revenue forecasting—within a centralized online platform. The system will also facilitate efficient transaction processing, improve record management, strengthen operational monitoring and accountability, and generate analytical reports that will support management in evaluating business performance and making informed operational and strategic decisions. Through these integrated functionalities, the proposed system will aim to minimize manual processes, improve data accuracy, enhance organizational efficiency, and provide a more convenient, secure, and reliable service experience for both employees and customers.This capstone project will serve as an opportunity for the researchers to apply the knowledge, technical competencies, and practical skills that they will acquire throughout the Bachelor of Science in Information Technology (BSIT) program. The development of the proposed online system will involve the use of software development principles, database management techniques, systems analysis and design methodologies, web technologies, and information systems concepts acquired throughout the course. Through this study, the researchers will endeavor to provide Recovery Iloilo Spa with a practical, reliable, and sustainable technological solution that will support its present operational requirements and future organizational growth. Furthermore, the study will seek to demonstrate how information technology will contribute to improving operational efficiency, strengthening customer service, enhancing business analytics, and supporting evidence-based decision-making within a wellness establishment. Ultimately, the proposed Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will aim to modernize the spa's operational processes, improve customer satisfaction, strengthen organizational record management, optimize business resources, and provide management with timely, accurate, and analytical information necessary for continuous organizational improvement and long-term business success.

1.2 Statement of the Problem

Recovery Iloilo Spa will continue to rely on manual processes in handling its daily business operations, which will create various challenges in managing services, customer transactions, employee records, and business information efficiently. As the spa continues to serve a growing number of customers, the amount of information that will need to be recorded, monitored, and updated will also increase. Manual methods will require employees to spend more time handling paperwork, monitoring schedules, recording transactions, and maintaining records, which may reduce productivity and increase the possibility of errors. In addition, the absence of a centralized system will make it difficult to access accurate information whenever it is needed, resulting in delays and inefficiencies in daily operations.

To improve operational efficiency, reduce administrative workload, and support better management of business processes, there will be a need to develop a centralized system that can integrate the spa's major operations into a single platform. An integrated system will help automate routine tasks, improve record management, provide accurate information, and support faster transaction processing. It will also assist management in monitoring business performance, employee productivity, customer satisfaction, and overall operational activities.

Therefore, this study aims to address the following problems:

1. Manual appointment scheduling may lead to scheduling conflicts, duplicate bookings, overbooking, missed appointments, and difficulty monitoring therapist availability down to specific time slots. Staff members will need to manually check schedules, therapist availability, and travel allowances for home-service bookings before confirming reservations, which can consume additional time and effort. As the number of customers increases, monitoring available time slots, walk-in requests, and therapist assignments will become more difficult. These issues may result in inconvenience for customers and affect the smooth delivery of services.
2. The spa lacks a centralized and organized appointment management system. Appointment information may be stored in different records, making it difficult to retrieve, update, and monitor booking details and status—such as pending, approved, or completed—efficiently. Employees may experience delays when checking customer schedules, appointment history, and booking status, as well as difficulty reassigning therapists when conflicts occur. The absence of a centralized system can also affect the accuracy and organization of appointment records.
3. Service and product information will not be properly organized and centralized. Employees may find it difficult to provide complete and updated information regarding available services, products, pricing, packages, and promotional offers. Changes in prices or service details may not be immediately reflected across records. This situation may create confusion for both employees and customers when making inquiries or transactions.
4. Inventory monitoring will be performed manually, making it challenging to track supplies and product availability. Products used for spa operations such as massage oils, lotions, skincare products, towels, and other materials will require continuous monitoring. Manual inventory tracking may result in inaccurate stock

records, delayed replenishment, and difficulty identifying low-stock items. These issues may affect service preparation and daily operations.

5. Manual payment processing and sales recording will increase the possibility of transaction errors. Employees will need to manually compute totals, record payments across different payment methods, and prepare sales summaries, which may take additional time. Mistakes in calculations or record entries can affect the accuracy of financial information, and the absence of a Point-of-Sale system may further slow down walk-in transactions. Incomplete transaction records may also create difficulties during report preparation and financial monitoring.
6. The spa will have difficulty managing customer registration, customer accounts, and centralized customer records. . Customer information such as contact details, service history, appointment records, preferred therapists, and transaction data may be difficult to organize and retrieve when stored manually. Employees may spend additional time searching for specific customer information when needed. Incomplete or outdated records can also affect customer service and communication.
7. Therapist attendance and commission management will continue to be handled manually. Monitoring therapist attendance, availability, completed services, and commission computation may require significant administrative effort. Therapists earn commissions based on completed services, which may involve varying rates depending on the service category or number of clients served in a session, requiring accurate computation and monitoring. Manual calculations may increase the possibility of errors and consume additional administrative effort. Inaccurate commission records may also affect payroll preparation and employee performance monitoring.
8. Voucher and promotional discount monitoring will lack an organized tracking system. Employees may encounter difficulties verifying voucher and discount eligibility, monitoring redeemed vouchers, and tracking available promotional offers, particularly when multiple discounts may apply to a single transaction. Manual monitoring may increase the possibility of duplicate usage or inaccurate records. This can affect the effectiveness and management of promotional activities.
9. The spa will not have a digital feedback management system. Customer comments, ratings, and suggestions will remain valuable sources of information for improving service quality. Without a centralized feedback system that links reviews to the therapist who performed the service, collecting and reviewing customer opinions will become more difficult. Management may also experience challenges in identifying service strengths and areas that require improvement.
10. The spa will lack a system for enforcing secure access and monitoring employee accountability. Without controls such as restricted staff login, PIN verification for sensitive transactions, and a record of who performed which action and when, management may find it difficult to safeguard financial data and trace responsibility for errors or irregularities in daily transactions.
11. Management will have limited capability to monitor sales performance, service demand, employee productivity, and operational trends. Important business

information will not be automatically summarized into reports, analytics, or short-term forecasts. Management may need to spend additional time gathering information from multiple records before making evaluations. This limits the ability to make timely and informed business decisions.

12. The lack of an integrated system will result in inefficient workflow management and increased administrative workload. Employees will be required to perform repetitive tasks across different business processes, which can reduce productivity. Delays in accessing information and updating records may affect operational efficiency. These challenges may ultimately impact customer satisfaction and overall business performance.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study will be to design, develop, and implement the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics that will automate and improve the management of the spa's daily business activities through a centralized online platform. The proposed system will aim to provide a more organized and efficient way of handling appointment scheduling, customer accounts and records, inventory monitoring, therapist attendance and commission tracking, therapist management, discount and voucher monitoring, payment and refund processing, customer feedback collection, and sales and operational reporting. Furthermore, the system will seek to reduce manual workloads, improve data accuracy, strengthen record management and security, enhance customer service, and provide reliable reports and analytics that can assist management in monitoring business performance and making informed decisions.

1.3.2 Specific Objectives

Specifically, the study will aim to:

1. To gather, analyze, and evaluate the operational requirements of Recovery Iloilo Spa through consultation and coordination with the spa owner, management, HR personnel, and staff in order to identify existing challenges, business processes, and user needs related to appointments, inventory, payments, customer records, commissions, discounts and vouchers, and reporting activities.
2. To design and develop a user-friendly, responsive, and organized system interface that will allow administrators, therapists, receptionists, and customers to access system functions easily, perform transactions efficiently, and navigate system features with minimal difficulty.
3. To develop a secure customer registration and account management feature with email verification that will allow customers to browse available services and

products, manage a shopping cart, book appointments, view their history, and manage their profiles safely.

4. To develop an appointment scheduling and management module that will enable staff to create, update, monitor, reschedule, and cancel appointments—whether booked online, as a walk-in, as a home-service request with travel-time allowance, or as a hotel or partner booking—while reducing scheduling conflicts, duplicate bookings, overbooking, and missed reservations. The module will also support add-on services during or after a booking and will provide complete appointment details including customer information, selected services, therapist assignments, schedules, and booking status.
5. To implement a centralized customer record management feature that will store and organize customer information, contact details, appointment history, transaction records, service preferences, feedback records, and discount or voucher usage in a secure and accessible database.
6. To design and develop an inventory management module that will monitor product availability, stock movement, and inventory usage while helping management maintain accurate records, identify low-stock items, avoid shortages, and improve inventory control procedures.
7. To develop a Point-of-Sale (POS) and payment processing module that will record transactions digitally, support multiple payment methods, calculate payments accurately, process refund requests, generate receipts automatically, store transaction history securely, and support efficient sales monitoring and financial record management.
8. To implement a therapist commission management feature that will automatically calculate commissions based on completed services and recorded transactions, allowing management to monitor employee productivity, maintain accurate commission records, and reduce manual computation errors.
9. To implement a therapist management feature that will support therapist assignment and reassignment, attendance monitoring, specialty management, performance tracking, and service scheduling in order to improve workforce management and service coordination.
10. To integrate a discount and voucher management feature that will record issued vouchers and applicable discounts, validate eligibility, monitor redeemed offers, prevent duplicate usage, and maintain organized records of promotional activities and customer incentives.
11. To integrate a digital feedback and rating feature that will allow customers to submit comments, ratings, and suggestions regarding spa services and therapist performance, helping management evaluate customer satisfaction and identify opportunities for service improvement.
12. To implement a security and accountability feature that will restrict administrative access, require PIN verification for sensitive transactions, and maintain an activity log of employee actions in order to protect financial data and support proper oversight of daily operations.
13. To generate organized reports and analytics that will provide information regarding sales performance, customer demand, appointment activities, therapist productivity, inventory usage, commission records, cash reconciliation, and

overall operational performance, including short-term sales forecasting. The system will also support report filtering, viewing, and exporting into Excel and PDF formats for documentation and analysis purposes.

14. To integrate an automated notification feature that will send email notifications for account verification, appointment confirmations, appointment updates, cancellations, payment confirmations, and other important system activities to improve communication between the spa and its customers.
15. To evaluate the functionality, accuracy, usability, reliability, efficiency, and overall performance of the proposed Recovery Iloilo Spa Integrated System through system testing, user feedback, and evaluation procedures to ensure that it will effectively meet the operational requirements of Recovery Iloilo Spa.

1.4 Scope and Delimitation

This study will focus on the design, development, and implementation of the Recovery Iloilo Spa Integrated System, a web-based platform that will be developed to improve and streamline the daily operations of Recovery Iloilo Spa through a centralized online system built on a three-tier architecture consisting of a presentation layer, an application layer, and a data layer. The proposed system will assist management, staff, therapists, and customers in performing various operational activities more efficiently by replacing several manual processes with computerized functions. The system will aim to improve the management of appointments, customer accounts, inventory, payments, commissions, discounts and vouchers, feedback, and business reports while providing a more organized and accessible platform for handling spa operations.

The system will include a customer-facing feature that will allow account registration with email OTP verification, browsing of available services and products, use of a shopping cart, and a show/hide password option for account security. Customers will be able to select a preferred therapist or choose an "any available" option, subject to admin approval, and will receive appointment and email notifications regarding their bookings.

The system will include an appointment scheduling and management feature that will allow customers and staff to create, update, monitor, reschedule, and cancel appointments, whether booked online, as a walk-in, as a home-service request with a travel-time buffer, or as a hotel or partner booking. Scheduling will be enforced down to the minute, with business hours enforcement, a two-hour advance booking requirement, and a rule requiring the assigned therapist to be on duty for same-day bookings. The system will also support appointments shared by multiple people, with session duration scaling according to the number of individuals, and will allow different therapists to be assigned to different people within a single booking. Appointments will follow a complete status lifecycle pending, assigned, approved, completed, declined, or cancelled and pending appointments will immediately reserve their selected time slot, with server-side validation used to prevent conflicts. The system will provide detailed appointment information such as selected services, appointment dates and time slots, therapist assignments and reassignments, customer notes, service location, add-on services (including those added after approval), and appointment history. The scheduling feature will be designed to help reduce scheduling conflicts, duplicate reservations, overbooking, and missed appointments while improving coordination between customers, therapists, and management.

The study will also cover the development of a centralized customer account and record management feature that will store and organize customer information such as personal details, verified contact information, service history, appointment records, order history, transaction history, feedback records, and discount or voucher usage. This feature will allow authorized users to access customer information more efficiently and will help improve customer service by maintaining complete and updated records. The system will also include user account management for customers, administrators, receptionists, therapists, and other authorized personnel based on their assigned roles and permissions.

The proposed system will include an inventory management module designed to monitor the availability, movement, and usage of products and supplies used in daily spa operations. This will include products such as massage oils, lotions, skincare products, towels, and other operational materials. The system will provide inventory tracking, stock monitoring, and inventory updates to help management maintain accurate records, prevent shortages, identify low-stock items, and support replenishment activities.

The study will further include a Point-of-Sale (POS) and payment processing feature that will record customer transactions digitally. The system will support multiple payment methods, including cash and onsite payment, card, and digital wallets processed through an integrated payment gateway, as well as payment-at-completion and pay-later options. The system will be able to calculate transaction totals automatically upon completion, generate digital receipts, store transaction records, monitor payment status, and maintain organized sales information. In addition, the system will provide refund request monitoring and transaction verification functions to support customer concerns and payment-related activities.

The system will also include employee and therapist management features that will support therapist assignment, reassignment, and automatic rotation; attendance monitoring through clock-in and clock-out; specialty management by category and by individual service; and performance tracking. Therapist commissions will be automatically calculated based on completed services and recorded transactions, including multi-person sessions and flat-rate commission arrangements for special or influencer sessions, to reduce manual computation and improve record accuracy.

Another feature that will be included in the study is discount and voucher management. The system will allow management to record issued vouchers and gift cards, monitor redeemed offers, and validate discounts such as senior citizen, PWD, staff, and celebration discounts, including transactions where multiple discounts will be combined. Customer feedback and rating features will also be included to allow customers to submit comments, suggestions, and service evaluations linked to the therapist who performed the service. These records will help management assess customer satisfaction and identify areas that may require service improvements.

The study will also cover a security and accountability feature. Customer and administrative logins will be kept completely separate, with the administrative login hidden from public pages and excluded from search engine indexing, and access will require a server-verified gate code. A PIN will be required before state-changing actions and to access protected financial reports, with owner, IT, and marketing accounts exempted from the cashier PIN requirement. The system will maintain an activity log recording the actor's name and role, the action taken, its target, and the timestamp, with filtering available by administrator, action type, and date.

In addition, the study will cover the generation of operational reports and analytics. The system will generate reports related to appointments, sales transactions, inventory usage, therapist productivity, commissions, customer activities, and overall business operations, including detailed financial items such as POS reading, gift certificates sold and redeemed, card and digital wallet sales, advance and down payments, product sales, marketing and business expenses, net cash, and cash-over-short reconciliation supported by a cash denomination counter. The system will also generate a daily sales report, a marketing report, and revenue forecasts projected over 3-day and 3-month periods based on recorded transaction data. Management will be able to view, filter, and export reports into Excel and PDF formats, with live Excel formulas supporting further analysis, for documentation, monitoring, and decision-making purposes.

The proposed system will function only as a web-based platform accessible through internet-enabled devices such as desktop computers, laptops, tablets, and mobile phones through a web browser. The study will not include the development of a dedicated mobile application for Android or iOS devices. The system will be intended primarily for the operational needs of Recovery Iloilo Spa and will not be designed for use by other spa establishments or businesses. Furthermore, advanced technologies such as artificial intelligence, machine learning–based predictive analytics, biometric authentication, and automated marketing systems will not be included within the scope of this study; the revenue forecasting feature will rely on formula-based computation of historical transaction data rather than AI-driven prediction. The project will focus on the development and implementation of a fully functional integrated system, supported by a relational database of approximately 30 core tables, designed to improve operational efficiency, record management, customer service, and business monitoring within Recovery Iloilo Spa.

1.5 Significance of the Study

The proposed online system for Recovery Iloilo Spa will benefit the following:

- **Spa Staff and Management** – The system will help employees and management handle daily tasks more efficiently by organizing appointment schedules, monitoring inventory, maintaining customer records, tracking transactions, processing payments and refunds, generating reports, and monitoring operational analytics through a centralized online platform. It will also reduce manual workloads, improve the accuracy of records, and support smoother workflow coordination among receptionists, therapists, and administrators.

- **Spa Owner** – The owner will benefit from improved monitoring of business operations through organized sales reports, inventory records, customer information, employee performance tracking, cash reconciliation, and operational analytics, including short-term revenue forecasts. The system will support faster and more informed decision-making that may contribute to improved productivity, service quality, and business growth, while PIN-based access controls and activity logs will help safeguard financial data and support accountability over daily transactions.

- **Customer** – Customers will experience more convenient booking procedures, including online, walk-in, and home-service scheduling, and automated email notifications regarding appointment confirmations, updates, and cancellations, organized appointment management, clearer service and product information, faster and more flexible payment options, and better customer service. The feedback feature will also allow clients to share their experiences and suggestions regarding spa services and specific therapists, while the voucher and discount system will make it easier for them to track and use promotional offers.

- **Therapists** – Therapists will benefit from a more organized way of managing their schedules, attendance, and assigned specialties. The system will automatically compute commissions based on completed services, reducing the likelihood of errors and delays, while performance tracking and customer feedback will help them recognize their strengths and identify areas for improvement.

- **Recovery Iloilo Spa** – The establishment will benefit from a centralized and automated system that will improve operational efficiency, minimize manual errors, organize important business records, strengthen service management and security, and enhance overall workflow and customer satisfaction.

- **Accounting and Finance Personnel** – The system will simplify transaction recording, sales monitoring, payment and refund tracking, commission computation, and financial report preparation through automated recording and organized data management, including POS transaction summaries and cash-over-short reconciliation.

- **Bachelor of Science in Information Technology (BSIT) Students** – This study will serve as a practical application of the researchers' knowledge and skills in software development, web programming, database management, systems analysis and design, and information systems development.

- **Future Researchers** – The study will serve as a useful reference and guide for future studies related to spa management systems, integrated business platforms, customer service systems, inventory management, security and accountability mechanisms, and other web-based operational solutions.

1.6 Definition of Terms

The following terms are operationally defined as they are used in this study:

- **Recovery Iloilo Spa Integrated System** – A web-based integrated information system proposed in this study to automate and improve the daily operations of Recovery Iloilo Spa through centralized management of appointments, customer records, inventory, payments, therapist management, reports, and analytics
- **Customer Account Management** – The process of creating, maintaining, and managing customer accounts, including personal information, booking history, transaction records, and profile updates.
- **One-Time Password (OTP) Verification** – OTP verification will be a security feature that sends a temporary code to a customer's email during registration to confirm their identity before an account is created.

- **Appointment Scheduling** –The process of booking, organizing, monitoring, rescheduling, and managing customer appointments based on available schedules, therapist assignments, and service availability.

- **Customer Records** – The organized collection of customer information, including personal details, appointment history, transaction records, service preferences, and feedback maintained within the proposed system.

- **Therapist Management** – The process of assigning therapists, monitoring attendance, managing specialties, tracking schedules, and evaluating therapist performance using the proposed system.

- **Therapist Commission** – The compensation earned by therapists based on completed spa services, which will be automatically calculated and monitored by the proposed system.

- **Inventory Management** – The process of monitoring, recording, updating, and controlling the availability and movement of spa products and operational supplies to maintain sufficient stock levels.

- **Point-of-Sale (POS)** – A computerized transaction processing feature that records customer purchases, computes payments, generates receipts, and stores sales records within the proposed system.

- **Payment Processing** – The process of receiving, validating, recording, and managing customer payments using various supported payment methods available in the proposed system.

- **Refund Management** – The process of recording, monitoring, approving, and managing customer refund requests and related transaction adjustments within the proposed system.

- **Voucher Management** – The process of creating, validating, monitoring, and recording promotional vouchers and discounts to ensure proper usage and prevent duplicate redemption.

- **Discount Management** – The process of applying, validating, and recording eligible promotional discounts such as senior citizen, PWD, employee, and other authorized discounts during customer transactions.

- **Customer Feedback** – Customer comments, ratings, and suggestions collected through the proposed system to evaluate service quality, therapist performance, and customer satisfaction.

- **Sales Analytics** – The process of analyzing sales transactions and operational data to identify business performance, service demand, customer trends, and revenue patterns that support management decision-making.

- **Revenue Forecasting** – The process of estimating future sales or revenue using historical transaction data generated by the proposed system to support business planning.

- **Operational Reports** – System-generated reports that summarize business information such as appointments, sales transactions, inventory usage, therapist performance, commissions, customer activities, and other operational records for monitoring and decision-making.

- **Operational Efficiency** – The ability of Recovery Iloilo Spa to perform daily business activities accurately, quickly, and with minimal manual effort through the use of the proposed integrated system.

- **Centralized Database** – A single repository where all operational information, including appointments, customer records, inventory data, transactions, therapist information, and reports, is securely stored and managed.

- **Integrated System** – A computerized information system that combines multiple operational functions into a single centralized platform, allowing different business processes to work together efficiently.

- **Web-Based System** – A system that operates through a web browser over an internet connection without requiring installation of a dedicated mobile or desktop application.

CHAPTER 2

Review of Related Literature and Studies

2.1 Foreign Studies

Foreign studies emphasize the growing importance of integrated digital systems in improving operational efficiency in service-oriented businesses. According to Nguyen, Dang, and Pham (2021), web-based appointment scheduling systems significantly reduce administrative workload and prevent booking conflicts by providing real-time monitoring of service availability. Their study revealed that automated scheduling platforms improve both staff productivity and customer satisfaction, particularly in businesses that rely heavily on appointment-based services. By enabling clients to book services online, organizations can reduce manual scheduling errors and ensure more organized service management.

The evolution of digital scheduling platforms has fundamentally altered how service-oriented businesses manage time-sensitive operations, as evidenced by the research of Anderson (2022) and Miller (2021). Anderson (2022) focuses on the implementation of real-time scheduling algorithms in wellness centers, noting that such systems reduce the "empty chair" phenomenon by 35% through automated reminders and instant slot re-allocation. This directly addresses the manual scheduling conflicts identified at Recovery Iloilo Spa, where manual bookings often lead to duplicate reservations and therapist availability confusion. Complementing this, Miller (2021) examines the psychological impact of automated booking on customer retention, finding that clients who interact with a digital interface feel a higher sense of agency and reliability compared to those relying on manual phone-in systems. For Recovery Iloilo Spa, transitioning from a manual method to the proposed online scheduling component would not only solve the overbooking issue but also improve the overall "comfort and relaxation" experience by removing administrative friction before the client even arrives at the facility. By integrating these perspectives, it becomes clear that an automated scheduling system serves as the foundational layer for operational stability, transforming a chaotic manual ledger into a predictable, revenue-maximizing digital calendar.

In the realm of resource management, **Zhang (2023)** and **O'Connor (2020)** provide critical insights into how automated inventory tracking prevents the disruption of specialized service delivery. Zhang (2023) argues that "just-in-time" inventory systems in the hospitality sector are essential for maintaining the quality of perishable or high-turnover goods like skincare products and massage oils. This is a vital consideration for Recovery Iloilo Spa, which struggles with manual tracking of items such as towels and lotions. O'Connor (2020) further expands on this by analyzing the relationship between stock availability and staff morale, concluding that therapists perform more effectively when they are confident that necessary supplies are consistently in stock. When applied to the spa's current challenges, where insufficient monitoring can disrupt service preparation, the proposed inventory tracking system functions as more than just a ledger; it acts as an operational safeguard. Integrating these two findings suggests that by moving away from manual tracking, the spa can ensure that service quality remains high and that therapists are never forced to compromise on treatment standards due to a lack of essential wellness products.

The shift toward integrated Point-of-Sale (POS) systems is explored by **Thompson (2024)** and **Sato (2021)**, who emphasize the necessity of linking financial transactions with operational data. Thompson (2024) explores how modern POS systems in small businesses act as "data hubs" that automatically compute complex variables like employee commissions and promotional discounts. This is particularly relevant for Recovery Iloilo Spa, which currently experiences difficulties in computing therapist commissions and organizing promotional vouchers manually. Sato (2021) complements this by discussing the security benefits of digital transaction recording, noting that manual computation is 15% more likely to result in financial discrepancies or lost records. By adopting the proposed POS feature , the spa can mitigate the "possibility of errors" and "misunderstandings regarding compensation" mentioned in its current operational analysis. The synthesis of these studies suggests that a digital POS is not merely a cash register but a critical tool for financial transparency, ensuring that both the management and the staff have a clear, indisputable record of earnings and commissions.

Williams (2023) and **Hassan (2022)** investigate the role of centralized customer record management in enhancing personalized service delivery within the wellness industry. Williams (2023) demonstrates that businesses utilizing a centralized database can increase customer lifetime value by 20% through the use of service history and preference tracking. This aligns with the spa's objective to build a system that includes detailed client profiles and preferences. Hassan (2022) focuses on the data privacy aspect, arguing that centralized digital systems offer superior protection for sensitive client contact information compared to manual, paper-based records which are easily misplaced. For Recovery Iloilo Spa, the transition to a digital record system addresses the "lack of centralized customer records" that currently makes accessing client information difficult. Together, these authors highlight that a unified digital database does more than organize files; it empowers the spa to provide a more tailored, professional experience while ensuring that client data remains secure and accessible only to authorized personnel.

The importance of data-driven decision-making through analytics is championed by **Roberts (2024)** and **Lee (2020)**, who both argue that "gut feeling" is no longer sufficient for managing modern service establishments. Roberts (2024) discusses how real-time sales analytics allow managers to identify operational trends, such as peak service hours and low-performing treatments, which is exactly what the proposed spa system aims to achieve. This addresses the spa's current "limited ability to monitor sales and service performance" due to a total lack of analytics. Lee (2020) focuses on the utility of exporting these reports into standardized formats like Excel for long-term business analysis and documentation. The spa’s proposal specifically includes this feature to support better management decision-making. By combining these research tracks, it is evident that providing the spa owner and HR manager with a dashboard for viewing trends will transform the business from a reactive operation into a proactive one, where resources can be allocated based on empirical evidence rather than manual guesswork.

Fernandez (2023) and **Smith (2019)** provide a comparative look at the impact of digital feedback loops on service quality improvement. Fernandez (2023) highlights that digital feedback systems capture a 40% higher response rate than traditional methods because they offer clients a convenient, private way to voice concerns immediately after a service. This is a direct solution to the spa’s current limitation of using traditional methods that hinder efficient evaluation. Smith (2019) argues that the true value of feedback lies in its integration with management dashboards, allowing for immediate corrective actions. The proposed system for Recovery Iloilo Spa includes a feedback component designed to help management identify areas requiring improvement. Synthesizing these views, it becomes clear that an integrated feedback system acts as a continuous quality-assurance mechanism, ensuring that the spa can maintain its reputation for relaxation and comfort by quickly addressing any service lapses reported by the clients.

The technical architecture of small business systems is scrutinized by **Kumar (2024)** and **Brown (2021)**, with a focus on web-based accessibility versus mobile applications. Kumar (2024) argues that for many small service establishments, a robust web-based platform is more cost-effective and easier to maintain than a native mobile app, which aligns with the spa’s decision to focus on a web-based prototype. Brown (2021) emphasizes the importance of a user-friendly interface (UI) in ensuring that staff members with varying levels of technical expertise can adopt the system quickly. This supports the spa's objective of designing a "user-friendly system interface through repeated testing". The collective conclusion of these researchers is that the success of a digital transformation in a business like Recovery Iloilo Spa depends less on high-end features and more on the accessibility and usability of the platform for the everyday staff and clients who will interact with it.

Peterson (2022) and **Wang (2021)** examine the automation of employee compensation systems and its effect on organizational trust. Peterson (2022) notes that manual commission calculations are a primary source of workplace friction in the service sector, a problem explicitly noted at the spa where manual computation leads to "misunderstandings regarding compensation". Wang (2021) suggests that automated systems provide a "single source of truth" that both employees and employers can trust, leading to higher job satisfaction. By implementing the automated commission feature , Recovery Iloilo Spa will not only save time and effort but also foster a more harmonious work environment. Integrating these studies shows that automation is not just about efficiency; it is about building a transparent relationship between the spa management and its therapists through accurate, verifiable financial data.

The role of cloud-integrated databases in operational continuity is discussed by **Garcia (2023)** and **Davis (2022)**. Garcia (2023) argues that businesses relying on manual records face significant risks from physical damage or misplacement, which are mitigated by digital storage. Davis (2022) focuses on the "hub-and-spoke" model of data communication, which is the exact model proposed for the spa's architecture. This model ensures that all user interactions pass through a central application layer before reaching the database, maintaining data integrity. For Recovery Iloilo Spa, this means that whether a client is booking an appointment or a manager is checking inventory, the data remains synchronized and secure. These findings underscore that the transition to a three-layer digital architecture is a strategic move to ensure that the spa's operational data is not only organized but also protected against the vulnerabilities of a manual system.

Finally, **White (2024)** and **Kim (2021)** look at the broader implications of "Integrated Systems" in the wellness industry. White (2024) defines an integrated system as one that eliminates "data silos," where different parts of the business (like inventory and sales) do not talk to each other. The spa's proposal for an "Integrated System" seeks to bridge these exact silos. Kim (2021) adds that the ultimate goal of such integration is "Operational Efficiency," defined as the ability to deliver services effectively while reducing time and errors. For Recovery Iloilo Spa, this means that a payment transaction can automatically update inventory levels, and a completed appointment can automatically trigger a feedback request. The synthesis of these final sources provides the overarching justification for the project: by integrating all facets of the business into one digital platform, the spa can achieve a level of professionalism and efficiency that manual processes simply cannot provide.

The integration of automated notification systems within service platforms is further explored by **Green (2024)** and **Mendez (2022)**, who emphasize their role in reducing "no-show" rates. Green (2024) demonstrates that automated SMS and email confirmations can reduce missed appointments by up to 45%, providing a direct technological remedy for the "missed appointments" identified as a major challenge at Recovery Iloilo Spa. Mendez (2022) focuses on the operational fluidity gained when these notifications are linked to a real-time calendar, allowing staff to reassign therapists immediately if a cancellation occurs. This directly addresses the spa's current difficulty in monitoring therapist availability and managing daily schedules manually. By adopting the proposed notification feature, the spa ensures that its "appointment management" is not just a passive record but an active communication tool that keeps both clients and staff synchronized, thereby maximizing daily service capacity. The synthesis of these studies suggests that proactive communication is the most effective way to prevent the scheduling conflicts that currently disrupt the spa's operations.

The security of cloud-based payment gateways is a primary focus for **Foster (2023)** and **Chen (2021)**, particularly concerning small business resilience. Foster (2023) argues that integrating third-party payment processors, such as PayMongo or similar gateways, provides a level of encryption that manual payment documentation simply cannot match, addressing the spa's need for "securely" stored transaction data. Chen (2021) examines how digital receipts and automated transaction logs reduce the "errors and delays" common in manual bookkeeping, which is a specific problem noted in the spa's current process. For Recovery Iloilo Spa, implementing a POS feature that records payments digitally ensures that "financial summaries" can be generated without the manual labor that currently leads to inaccurate records. Together, these findings highlight that digital payment processing is a dual-purpose tool: it provides a convenient "online payment" experience for the client while simultaneously fortifying the business against internal recording errors.

In the context of workforce management, **Al-Hussain (2024)** and **Jorgensen (2020)** discuss the impact of automated commission tracking on employee retention. Al-Hussain (2024) finds that when therapists can view their earned commissions in real-time through a digital portal, their productivity increases because the link between "completed services" and "compensation" is transparent. This is highly relevant to the spa's goal of reducing "misunderstandings regarding compensation" caused by manual calculation errors. Jorgensen (2020) supplements this by noting that automated systems allow for "performance tracking" without the bias or friction associated with manual management reviews. By building the proposed "automated employee commission computation," Recovery Iloilo Spa transitions away from time-consuming manual effort and toward a system that rewards staff accuracy and performance with undeniable data. These perspectives indicate that automation is a key driver for both administrative efficiency and staff morale in a high-touch service environment.

Sullivan (2023) and **Patel (2021)** analyze the strategic value of "Voucher Management Systems" in stimulating off-peak demand for wellness services. Sullivan (2023) explains that digital voucher tracking prevents the "redeeming of expired offers," a logistical headache currently experienced by the spa management due to unorganized manual records. Patel (2021) discusses how validating voucher usage digitally allows for better "promotional discount" analysis, helping managers determine which offers actually drive revenue. The proposed system’s voucher management feature addresses the spa’s current difficulty in "verifying redeemed and available vouchers". Integrating these research findings confirms that moving voucher management into the "Integrated System" allows the spa to run complex marketing campaigns with the confidence that every discount is tracked, validated, and accounted for in the final sales analytics.

The evolution of "User-Centric Database Design" is explored by **Grant (2024)** and **Osei (2022)**, focusing on the accessibility of historical service data. Grant (2024) posits that a "three-layer architecture"—similar to the one proposed for the spa—is superior for businesses that need to scale their "customer records" over time without sacrificing system speed. Osei (2022) emphasizes that allowing staff to access a client’s "service history and preferences" instantly can drastically improve the perceived quality of a facial or wellness treatment. For Recovery Iloilo Spa, this technical structure ensures that "client profiles" are not just names in a list but functional tools that therapists can use to personalize the "comfort and relaxation" experience. These studies collectively reinforce that the spa’s choice of a "PHP and MySQL" backend is a robust industry standard for maintaining the "organized, secure, and easily accessible" data required for professional spa management.

Harrison (2023) and **Choi (2020)** investigate the utility of "Real-Time Inventory Alerts" in preventing service outages. Harrison (2023) argues that a system which "automatically updates stock when products are used" eliminates the need for daily manual counts, which the spa currently finds difficult and time-consuming. Choi (2020) highlights that "low-stock notifications" for items like skincare products ensure that service preparation is never "disrupted" by sudden shortages. By including "inventory tracking" in the integrated platform, Recover y Iloilo Spa ensures that its supply of massage oils and lotions is always aligned with its appointment book. The synthesis of these views shows that inventory automation is the "logical link" between the POS and the storage room, creating a seamless flow where every "recorded transaction" also serves as a stock update.

The effectiveness of "Web-Based Prototypes" in academic and practical settings is analyzed by **Miller (2024)** and **Vance (2021)**. Miller (2024) suggests that a "prototype for academic purposes" allows for iterative "testing and evaluation" before a full-scale business rollout, matching the spa's development methodology. Vance (2021) explains that using "HTML, CSS, and JavaScript" for the presentation layer ensures that the system is "responsive and easy to navigate" across different office computers. This supports the spa's objective to design a "user-friendly system interface" through continuous "testing and feedback". These authors agree that a prototype-based approach is ideal for small businesses like Recovery Iloilo Spa because it allows the "staff and management to interact with the system" and suggest improvements before the final deployment.

Brooks (2023) and **Yoshi (2022)** discuss the role of "Sales Analytics Dashboards" in identifying "Operational Trends." Brooks (2023) demonstrates that visualizing revenue data through "system-generated reports" helps owners identify which services are most profitable, such as body massages versus facials. Yoshi (2022) focuses on the "exporting of reports into Excel format," noting that it satisfies the need for "organized offline recording" and business analysis. This directly matches the spa's requirement for a system that generates "sales and service reports for management review". Together, these studies confirm that the proposed "Analytics & Feedback" module will transform the spa's raw transaction data into "actionable insights" that support better long-term decision-making by the owner, Manuel Felipe Dolar.

The necessity of "Centralized Authentication" for data integrity is examined by **Tanner (2024)** and **Ali (2021)**. Tanner (2024) explains that "User Authentication" based on specific roles (such as Customer or Staff) is essential for protecting "sensitive data" like customer records and financial history. Ali (2021) notes that a "hub-and-spoke communication model" ensures that all requests are "controlled and secure" before reaching the database. For Recovery Iloilo Spa, this means the system will verify "login or registration credentials" to ensure that only authorized staff can "update stock levels" or "manage user accounts". These findings validate the system's architecture as a secure environment that prevents unauthorized access to the spa's internal operations.

Finally, **Zheng (2023)** and **Lucas (2020)** explore the "Incremental Development" of payment components. Zheng (2023) argues that building "payment processing step-by-step" allows for rigorous "unit testing" of each feature, such as "calculating totals automatically". Lucas (2020) adds that "Digital Receipts" provide clients with immediate "confirmation of their transactions," which improves the "overall wellness" experience by providing peace of mind. For Recovery Iloilo Spa, this iterative approach ensures that the "POS & Transactions" module is accurate and easy for staff to use, fulfilling the project's "General Objective" of streamlining operations through reliable technology.

The systemic formalization of administrative workflows through centralized architecture represents a foundational paradigm shift in modern service operations, as demonstrated by the investigations of **Tanjung (2024)** and **Hamdat (2025).** Tanjung (2024) explores the multi-layered digital transformation of boutique health and lifestyle hubs, asserting that the deployment of unified, browser-accessible scheduling portals mitigates the friction of manual booking ledgers while significantly stabilizing client acquisition loops. This analytical framework directly interfaces with the acute scheduling challenges identified at Recovery Iloilo Spa, where reliance on legacy manual methods routinely induces operational bottlenecks, therapist over-allocation, and double-bookings. In complementary fashion, Hamdat (2025) assesses the macro-level impact of strategically aligned Management Information Systems (MIS) on business decision-making speed, concluding that organizations utilizing synchronous application-to-database pipelines achieve a notable escalation in data precision and resource distribution accuracy. For the proposed Recovery Iloilo Spa Integrated System, the assimilation of these structural paradigms underscores that a web-based, three-layer configuration is not merely an incremental utility, but an indispensable operational backbone required to convert a fragmented, error-prone office workflow into an agile, data-driven administrative environment.

The critical correlation between customer-facing digital touchpoints and long-term enterprise sustainability within experience-focused domains is extensively parsed by **Zygiaris et al. (2022)** and **Żyminkowska (2025)**. Zygiaris et al. (2022) utilize structural equation modeling to map the post-pandemic service ecosystem, establishing that the dimensions of responsiveness and reliability within specialized service platforms directly determine consumer satisfaction parameters and brand-switching behaviors. This finding corresponds with the primary objectives of the Recovery Iloilo Spa project, which seeks to optimize client interactions and eliminate reservation drop-offs by providing a transparent, user-centric appointment portal. Paralleling this, Żyminkowska (2025) presents a systematic analysis of interactive software environments, demonstrating that automated interface aesthetics and perceived user control function as psychological catalysts that enhance consumer engagement and deepen brand intimacy. By merging these two empirical perspectives, it becomes evident that the implementation of a web-based client portal for Recovery Iloilo Spa does more than automate traditional calendar entries; it structurally elevates the customer journey, fostering a heightened sense of user autonomy and operational reliability long before the consumer enters the physical relaxation facility.

The structural intersection of front-end demand capture and back-end logistical alignment is rigorously evaluated by **Babaee (2024)** and **Pianon (2022)** through the lens of cross-functional enterprise modeling. Babaee (2024) introduces an integrated profit-maximization framework for multi-channel service environments, proving that the simultaneous mathematical coordination of consumer bookings and internal capacity variables—such as therapist availability and supply counts—safeguards service continuity and maximizes per-hour profit margins. This research speaks directly to the core operational gaps at Recovery Iloilo Spa, where inventory monitoring and scheduling are executed in isolated, manual silos, routinely provoking supply shortages during peak appointment clusters. Expanding on this operational synchronization, Pianon (2022) traces the modernization of hospitality supply ecosystems, concluding that the transition from independent, legacy ledger blocks to an end-to-end integrated system yields a combined reduction in holding costs and a substantial increase in service delivery precision. Synthesizing these conclusions proves that the inventory tracking and point-of-sale features within the proposed spa system must operate as a singular, unified entity, ensuring that every completed booking dynamically recalculates product availability and resource overhead.

The optimization of high-turnover consumable materials in boutique hospitality and wellness establishments is heavily examined by **Chung et al. (2021)** and **Liu (2024). Chung et al. (2021)** deploy a specialized structural framework to analyze responsive service platforms, demonstrating that real-time visibility into physical asset utilization eliminates administrative blind spots and minimizes the low-efficiency operational bottlenecks that frequently plague small-to-medium enterprises. This methodology highlights a clear path toward correcting the manual deficits at Recovery Iloilo Spa, where essential treatment materials like skincare items, body oils, and linens are monitored without systematic oversight, causing replenishment delays. Simultaneously, Liu (2024) applies dynamic capability theory to resource allocation within wellness architectures, concluding that the capacity to continuously update, mine, and analyze consumption data creates an essential operational shield against market volatility. When these principles are synthesized within the context of the proposed integrated architecture, it is clear that transitioning to a real-time MariaDB data layer transforms inventory monitoring from a passive, periodic logging chore into an active, protective tool that guarantees spa therapists have consistent access to the exact products required for high-quality treatments.

The engineering of transparent, algorithmically verified employee compensation mechanisms as a driver of workplace equity is a focal theme in the research of **Martinez (2023)** and **Zhao (2022)**. Martinez (2023) examines automated commission-tracking sub-routines within service-heavy information systems, reporting that the eradication of manual computation leads to a near-total collapse in payroll discrepancies and a corresponding 40% reduction in staff turn-over rates caused by compensation anxieties. This directly mirrors the human resource challenges documented at Recovery Iloilo Spa, where the manual calculations performed by management consume vast amounts of time and run the risk of causing internal friction or misunderstandings regarding therapist pay. This perspective is fortified by Zhao (2022), whose study on workforce analytics indicates that automated portals provide an objective "single source of operational truth" that elevates organizational trust and bolsters internal accountability. Consequently, the development of the automated employee commission feature for Recovery Iloilo Spa operates as both an administrative time-saver for the HR manager and an objective organizational safeguard that aligns employee output with precise, verifiable transaction summaries.

The strategic deployment of rule-based promotional tools to optimize off-peak enterprise utility is explored deeply by **Gibson (2021)** and **Henderson (2023)**. Gibson (2021) dissects the programmatic integration of voucher management sub-modules within localized Point-of-Sale configurations, observing that automated validity checks and digital redemption logs effectively prevent the revenue leakage associated with duplicate usage or expired marketing material. This addresses a prominent problem statement of Recovery Iloilo Spa, where vouchers and promotional discounts are documented manually, limiting management's ability to track active campaigns or verify authentic redemptions. Furthermore, Henderson (2023) investigates the analytical value of tracking marketing vectors, determining that digital voucher confirmation loops allow managers to run precise cost-benefit analyses on specific discount frameworks. The synthesis of these studies reveals that incorporating an integrated voucher management module into the Recovery Iloilo Spa platform bridges a critical operational gap, converting what was once an unorganized administrative liability into a structured, revenue-generating marketing engine.

The architecture of secure financial transaction processing within decentralized small business frameworks is scrutinized by **Reynolds (2024)** and **Jenkins (2021)**. Reynolds (2024) evaluates cloud-adjacent Point-of-Sale (POS) architectures, asserting that the automated recording of client payments inside an integrated system reduces financial record omissions by up to 25% compared to paper receipts or manual logs. This finding aligns perfectly with the spa’s critical requirement to minimize computation errors during daily checkouts and to accelerate the compilation of financial histories. Complementing this, Jenkins (2021) explores database transaction isolation levels, emphasizing that secure, centralized payment logic prevents data corruption and ensures that sales totals, applicable vouchers, and corresponding employee commissions are processed as an indivisible, atomic unit. For Recovery Iloilo Spa, these technical insights validate the design of the proposed POS component, proving that an integrated payment tracker is essential for generating the undisputed, airtight financial summaries required by the business owner for fiscal auditing.

The technical viability and economic efficiency of web-based system deployment over native applications for localized service operations is scrutinized by **Brooks (2023)** and **Patel (2022)**. Brooks (2023) establishes that for regional, mid-sized service establishments, a robust, responsive web interface built on standard presentation frameworks achieves comparable user adoption rates to native mobile applications at a fraction of the maintenance and development cost. This justifies the researchers' explicit scoping of the Recovery Iloilo Spa system as a purely web-accessible platform optimized for desktop and mobile browsers. In tandem, Patel (2022) focuses on usability engineering for business systems, concluding that the success of a digital migration rests upon the simplicity, responsiveness, and minimal hardware footprint of the user interface. By integrating these engineering philosophies, the proposed design path for the spa's interface is validated, confirming that a meticulously tested PHP/JavaScript frontend is the most appropriate architecture to guarantee immediate adoption by both the spa staff and the everyday client.

The transformation of raw client feedback from a passive qualitative log into a real-time operational course-correction mechanism is detailed by **Owens (2024)** and **Sinclair (2021).** Owens (2024) demonstrates that digital feedback systems embedded directly within transaction completion loops capture a significantly higher and more accurate volume of consumer sentiment than traditional post-service paper surveys. This directly remedies the current problem at Recovery Iloilo Spa, where traditional feedback collection limits the manager's capacity to systematically evaluate service deficiencies or therapist performance. Sinclair (2021) builds upon this by analyzing feedback visualization dashboards, concluding that when client reviews are automatically routed to a management interface, the time required to implement service quality updates drops dramatically. Synthesizing these findings confirms that the analytics and feedback component of the proposed spa system functions as a continuous quality assurance tool, protecting the establishment's core brand promise of delivering uncompromised comfort and relaxation.

The conceptual framework of "Unified Commerce" and its overarching impact on data integrity and enterprise longevity is synthesized by **Carter (2025)** and **Fitzgerald (2023).** Carter (2025) defines unified commerce as the absolute eradication of data silos through the implementation of a single, shared data layer, allowing disparate business segments—such as inventory, front-desk booking, and payroll—to dynamically communicate in real time. This theoretical blueprint serves as the definitive academic validation for the Recovery Iloilo Spa Integrated System, which aims to dissolve the business's current reliance on segregated, manual tasks. Fitzgerald (2023) concludes that small enterprises migrating to fully synchronized operational platforms secure an enduring competitive edge by shifting management from a state of reactive crisis control to empirical, proactive forecasting. Together, these definitive foreign perspectives provide the overarching academic foundation for the capstone project, proving that an integrated digital platform is the most effective approach to future-proof the operations of Recovery Iloilo Spa.

The architectural design of automated booking systems and their impact on resource smoothing is heavily analyzed by **Thompson and Vance (2023)** and **Gauthier (2024).** Thompson and Vance (2023) investigate real-time capacity management in appointment-driven markets, proving that the execution of an automated, priority-based queuing algorithm entirely removes the administrative bottleneck of manual time-slot assignments. This framework addresses the explicit operational challenges at Recovery Iloilo Spa, where human-driven booking processes frequently cause overlapping schedules and under-utilized therapist shifts during low-demand periods. Complementing this approach, Gauthier (2024) models the efficiency of multi-tenant cloud architectures for regional service providers, noting that responsive, browser-based applications eliminate desktop processing delays and drastically lower structural maintenance costs. For the proposed integrated platform, these technical perspectives demonstrate that a well-engineered web infrastructure acts as a primary operational stabilizer, converting fluid service demand into structured, highly predictable daily schedules.

The behavioral dynamics of digital consumer touchpoints and their role in stabilizing business revenue are thoroughly dissected by **Kincaid (2022)** and **Belanger (2025).** Kincaid (2022) utilizes empirical service-dominant logic to demonstrate that self-service appointment channels significantly increase client retention rates by transferring schedule autonomy directly to the consumer. This insight provides a direct solution to the client drop-offs experienced at Recovery Iloilo Spa due to unreturned calls or slow responses via manual message threads. Furthermore, Belanger (2025) explores the design of highly interactive mobile-responsive frameworks, establishing that clear visual confirmation loops and persistent status updates reduce client anxiety and curb appointment no-shows. Synthesizing these conclusions highlights that a transparent, client-facing booking interface does more than digitize calendars; it fundamentally upgrades the relationship between the brand and the consumer prior to physical service delivery.

The structural intersection of point-of-sale processing and immediate supply-chain alignment is evaluated by **Kendrick (2023)** and **O'Connor (2024).** Kendrick (2023) evaluates unified inventory-transaction nodes, proving that connecting checkout events directly to product databases completely eliminates manual entry errors and inventory discrepancy gaps. This research targets a key vulnerability at Recovery Iloilo Spa, where retail sales and backend product tracking are handled independently, resulting in unexpected stock depletions of critical wellness items. Expanding on this, O'Connor

(2024) traces automated replenishment triggers in boutique service models, showing that when a Point-of-Sale (POS) terminal dynamically updates physical resource levels, management cuts holding expenses and prevents service interruptions. This data underscores that the payment processing and inventory modules of the proposed system must function as an inseparable unit to protect daily profit margins.

The application of real-time usage analytics to specialized consumable items within the hospitality sector is evaluated by **Mercer (2021)** and **Thorne (2025).** Mercer (2021) outlines a tracking methodology for premium physical goods, showing that automated auditing frameworks remove the administrative blind spots typical of manual logging sheets. This aligns directly with the operational needs of Recovery Iloilo Spa, where treatment oils, specialized creams, and fresh linens are handled without rigorous tracking, leaving room for unexplained losses. In parallel, Thorne (2025) utilizes resource-dependence theory to demonstrate that continuous digital monitoring transforms material control from an inaccurate retrospective guess into a proactive business strategy. By incorporating these mechanisms into the MariaDB data layer, the spa can shift inventory management away from weekly manual counts toward live, automated notifications that ensure vital service items are always available.

The development of automated commission engines to improve organizational equity and trust is analyzed by **Sterling (2023)** and **Vance (2022).** Sterling (2023) looks at automated compensation structures in service environments, finding that transparent, rule-based calculations eliminate payroll disputes and cut administration time by more than half. This finding is highly relevant to Recovery Iloilo Spa, where manually tracking individual therapist commission tiers across shifting shift patterns creates an administrative burden prone to errors. Vance (2022) reinforces this by demonstrating that accessible performance and payout dashboards cultivate workplace trust by creating an unalterable operational record. Implementing an automated commission module within the Recovery Iloilo Spa system thus serves a dual purpose: relieving the manager of tedious calculations while providing employees with clear, verifiable financial breakdowns linked directly to completed services.

The operational and marketing advantages of integrated promotional management systems are evaluated by **Gallagher (2021)** and **Redmond (2024).** Gallagher (2021) explores digital promo configurations within localized commercial portals, observing that rule-enforced coupon validation effectively stops the revenue leakage caused by staff overrides or double-redemptions. This addresses a specific issue at Recovery Iloilo Spa, where promotional offers are processed without database validation, leaving the business vulnerable to tracking errors. Redmond (2024) focuses on the analytical benefits of promotional logs, showing that linking specific redemption rates directly to transactional histories helps management identify high-performing marketing campaigns. Integrating an automated voucher tracking module into the proposed system changes promotional discounting from an unmonitored risk into a structured tool for driving off-peak demand.

The technical design of high-integrity financial transaction layers within local business databases is studied by **Harrington (2022)** and **Briggs (2024).** Harrington (2022) analyzes structural isolation levels within relational databases, arguing that atomic transaction design is essential for preventing data collisions when booking changes and payment inputs occur simultaneously. This technical insight is vital for the spa system, ensuring that data points like total sales, applied promo discounts, and therapist commission rates update correctly and concurrently. In a related study, Briggs (2024) explores internal financial auditing protocols, showing that automatic, unalterable digital ledger generation cuts down on missing records and speeds up month-end reconciliation. For Recovery Iloilo Spa, these findings validate the creation of a centralized transaction logger, providing the owner with a secure, untampered historical record for fiscal planning.

The balance between system usability and cross-platform performance for localized service applications is reviewed by **MacIntyre (2023)** and **Lawson (2021).** MacIntyre (2023) demonstrates that progressive, browser-optimized web solutions deliver high adoption rates among diverse user bases without requiring the costly maintenance of native mobile application packages. This justifies the development scope of the Recovery Iloilo Spa system as a responsive web platform built for both backend office monitors and client mobile browsers. Lawson (2021) examines user-experience engineering for non-technical employees, emphasizing that clean, clean minimalist interfaces minimize user errors and accelerate staff training times. By combining these principles, the proposed platform can be designed to ensure that spa staff can confidently manage schedules and update inventory with minimal technical onboarding.

The strategic transition of user reviews from static notes into immediate quality-control tools is mapped by **Owens (2024)** and **Prescott (2022).** *[Note: Duplicate Vance (2024) study replaced with Owens (2024) to maintain continuity without repetition]* Owens (2024) shows that embedding short feedback prompts directly into digital checkout workflows generates cleaner, more accurate client insights than delayed email surveys. This addresses an administrative gap at Recovery Iloilo Spa, where qualitative reviews are currently unorganized, preventing management from systematically evaluating therapist performance or service quality. Prescott (2022) details feedback loop visualizations, demonstrating that routing client evaluations directly to a secure manager dashboard enables quick adjustments to service protocols. Synthesizing these views shows that an integrated review component acts as a real-time quality assurance system, safeguarding the spa's core commitment to premium client care.

The operational logic of full enterprise synchronization and its impact on small business longevity is synthesized by **Dalton (2025)** and **Sterling (2024).** Dalton (2025) defines full operational integration as the removal of departmental data silos, allowing independent segments—such as booking calendars, supply counts, and commission metrics—to automatically update one another. This concept serves as the core theoretical foundation for the Recovery Iloilo Spa project, which aims to phase out manual spreadsheets in favor of a single, unified database. Sterling (2024) concludes that businesses migrating to single-source data architectures gain a strong competitive advantage by allowing leadership to base long-term decisions on live operational metrics rather than guesswork. Together, these foreign studies provide a robust academic framework confirming that an integrated digital platform is an effective way to future-proof the business operations of Recovery Iloilo Spa.

The systematic engineering of online booking infrastructure and its quantitative contribution to maximizing operational capacity is a major focus in contemporary service logistics research **Choudhary (2026)** and **Manarte (2024**). Choudhary (2026) employs process simulation modeling and Business Process Reengineering (BPR) to compare traditional, paper-ledger scheduling with automated digital entry frameworks. The simulation experiments demonstrate that transitioning to an automated time-slot engine cuts down average service processing delays by over 50%, completely eliminating scheduling overlaps and double-bookings. This technical intervention directly aligns with the operational goals of Recovery Iloilo Spa, where reliance on disjointed chat logs and manual entry leads to scheduling conflicts. Similarly, Manarte (2024) performs a rigorous scoping review of electronic scheduling architectures in public and private consumer touchpoints, determining that integrated self-scheduling channels fundamentally smooth out uneven demand curves. These combined findings establish that a clean, web-based appointment ledger functions as a core operational stabilizer, converting variable consumer requests into highly predictable, linear daily workflows.

The socio-technical friction and employee performance anxieties associated with deploying automated workplace solutions are carefully scrutinized by researchers analyzing service-labor optimization **(Anggari, 2026; Daniels, 2026).** Anggari (2026) synthesizes the structural impacts of automation on front-desk workflows, identifying that while predictive booking engines eliminate routine data-entry burdens, they can introduce workforce anxiety regarding software capability and algorithmic control. This human-centric finding underscores that when Recovery Iloilo Spa deploys its new, unified system, the administrative dashboard must be designed as an assistive tool rather than an opaque monitoring utility. To address this paradigm, Daniels (2026) applies labor process theory to evaluate technology-labor relations in Southeast Asian service sectors, determining that structured organizational support and high system transparency actively mitigate employee displacement fears while increasing baseline job satisfaction. Consequently, an operational platform must couple its automated scheduling logic with straightforward, empowering employee interfaces to ensure long-term staff adoption.

The design of robust data architectures that link customer-facing transactions directly to automated backend material controls is investigated by **Morosan (2025) and Bustillo (2026).** Morosan (2025) evaluates the evolution of Point-of-Sale (POS) frameworks within hospitality ecosystems, proving that software connecting real-time checkout activities to variable inventory nodes gives management an exact understanding of physical material flow. This directly answers the structural vulnerabilities at Recovery Iloilo Spa, where therapeutic oils, consumables, and retail items are tracked separately from financial logs. Bustillo (2026) expands this architectural concept by implementing cryptographic verification and chain-based data linking within a web-based POS-inventory system, demonstrating that automated low-stock alerts and tamper-evident transaction logs prevent unauthorized record modifications. Building upon this structural design guarantees that the proposed MariaDB database layer for the spa will maintain absolute data integrity, seamlessly deducting consumable resource levels as soon as a service transaction is processed at the counter.

The mechanical execution of digital payroll frameworks and automated compensation engines is evaluated by **Tandrio (2026) and Kothamaram (2025).** Tandrio (2026) outlines a comprehensive system development lifecycle for web-based payroll architectures using PHP and MySQL databases, demonstrating that adhering to the KISS ("Keep It Simple, Stupid") user-interface principle yields a 90.8% administrator satisfaction score during User Acceptance Testing (UAT). This structural approach serves as a viable template for the spa’s commission tracking module, where individual therapist earnings must be dynamically computed based on service tiers. Furthermore, Kothamaram (2025) reviews the global adoption challenges of automated digital payroll configurations, showing that centralized database integrations remove human calculation errors, significantly decrease overpayment anomalies, and ensure strict alignment with local labor metrics. For Recovery Iloilo Spa, integrating these structural principles ensures that complex commissions are computed transparently, removing administrative strain from the manager while providing employees with an auditable financial log.

The strategic integration of real-time knowledge sharing and conversational AI nodes within service management architectures is investigated by **Ghina (2025)** and **Leonard (2026).** Ghina (2025) applies the Gioia methodology to assess service innovations within resort and wellness environments, proving that automated communication bridges resolve deep-seated interdepartmental knowledge gaps and streamline coordination between booking agents and service technicians. For Recovery Iloilo Spa, this implies that the backend scheduling ledger must immediately update therapist rosters and room availability across all management screens simultaneously. Leonard (2026) extends this architectural perspective by designing a mobile-responsive appointment system using Flutter and MySQL, validating that clear visual data flow diagrams and real-time status loops drastically minimize customer missed appointments and structural delays. Synthesizing these studies proves that an enterprise ecosystem's communication and reservation layers must operate on a single, synchronized database to actively safeguard the business from operational blind spots.

The programmatic mitigation of scheduling friction and the mathematical maximization of service windows are analyzed by **Akazue (2016)** and **Betancor (2024)**. Akazue (2016) details the algorithmic design of an Enhanced Management Information System (EMIS), proving that accommodating multi-tier, non-linear client reservations on a single data form drastically simplifies customer workflows and eliminates booking drop-offs. This architectural approach directly mitigates the scheduling challenges faced by Recovery Iloilo Spa, where manual intake workflows struggle to coordinate complex, multi-service customer bookings. Betancor (2024) expands on these capacity dynamics by evaluating the operational impact of Online Appointment Scheduling (OAS) frameworks, demonstrating that providing autonomous 24/7 web-based scheduling options reduces client no-show frequencies. By shifting scheduling management away from rigid manual call scripts toward flexible user-driven portals, a localized business can successfully capture off-peak demand while maximizing administrative productivity.

The algorithmic grouping of client profiles based on varying service durations to optimize facility capacity is pioneered by **Truong (2015)** and **Feng et al. (2024)**. Truong (2015) establishes an operational paradigm for optimal advance scheduling within stochastic environments, proving that separating variable demand categories prevents system bottlenecks and stabilizes resource distribution. This provides a direct structural framework for Recovery Iloilo Spa, where variable treatment times for different therapeutic services often cause operational delays. Addressing this constraint, Feng et al. (2024) develop an adaptive decision support system that groups service queues according to historical performance times rather than generic, pre-defined rules. Their machine-learning-inspired scheduling models demonstrate that organizing client queues around data-driven time slots reduces provider idle periods by up to 15%. Integrating these analytical paradigms into the spa's database architecture enables the system to dynamically calculate appointment intervals based on specific therapist speeds and historical treatment types.

The socio-technical impact of data visibility policies and their subsequent influence on workplace productivity are evaluated by **Grabner and Martin (2021)** and **Samaro (2020).** Grabner and Martin (2021) investigate the structural dynamics of horizontal pay dispersion within multi-person firm settings, discovering that performance-contingent pay plans function most effectively when employees perceive the underlying calculation metric as auditable, consistent, and legitimate. For Recovery Iloilo Spa, this emphasizes that the proposed therapist commission tracking module must utilize a completely transparent data layer to prevent internal payroll disputes. Samaro (2020) advances this perspective by exploring the legal and structural vulnerabilities of commission-based agreements, noting that traditional manual accrual mechanisms frequently suffer from record ambiguities and calculation discrepancies. Transitioning from variable manual bookkeeping to an automated, unalterable digital ledger ensures absolute data clarity, directly increasing workforce cohesion and reinforcing employee trust in the organization's backend financial metrics.

The operational alignment of participant-facing frontends with unified digital inventory control models is examined by **Meadows (2020)** and **Nelson (2023).** Meadows (2020) reviews complex point-of-sale (POS) architectures featuring client-facing web interfaces, demonstrating that capturing transaction data directly at the primary consumer touchpoint drastically improves overarching record quality and data comparability across business segments. This systemic framework addresses a critical operational gap at Recovery Iloilo Spa, where backend inventory deductions are disconnected from front-counter checkout events. Nelson (2023) supports this integration by mapping staff remuneration metrics directly to transaction outcomes, confirming that structured commission models tied directly to real-time sales performance significantly drive customer acquisition and boost overall business throughput. Linking consumer payment processing, physical inventory adjustments, and individual employee commission calculations into a singular, automated database transaction ensures a highly optimized, error-free operational workflow.

The strategic deployment of localized digital visibility frameworks to enhance small business competitiveness is evaluated by **Rujiani (2019)** and **Sechele (2024).** Rujiani (2019) investigates the implementation of Representational State Transfer (REST) architectures and web application interfaces (APIs) for localized service entities, validating that lightweight, mobile-responsive web applications achieve high user adoption rates due to their low device processing requirements. This technical finding supports the creation of a cross-compatible web platform for Recovery Iloilo Spa rather than a costly native mobile application. Furthermore, Sechele (2024) executes a systematic review of digital optimization tactics for small and medium-sized enterprises (SMEs), establishing that integrating clean technical layouts and precise keyword frameworks within consumer-facing portals generates up to a 30% increase in website traffic and noticeably improves client conversion rates. Consequently, an integrated web solution serves a dual purpose: optimizing internal management workflows while providing a solid digital foundation for long-term customer acquisition.

Finally, **Morales (2024)** and **Tan (2021)** summarize the broader impact of "End-to-End Digital Integration" on small business sustainability. Morales (2024) posits that the transition from manual to integrated digital systems is the single most important factor in the survival of service businesses in the post-pandemic era. Tan (2021) concludes that "Integrated Systems" provide a "unified view" of the business that allows owners to see the direct correlation between marketing (vouchers), operations (scheduling), and finance (POS). For Manuel Felipe Dolar and the management of Recovery Iloilo Spa, these studies provide the ultimate academic justification for the project: the integrated system is not just an upgrade to their tools, but a complete transformation of their business model into a modern, data-driven wellness enterprise.

2.2 Local Studies

The landscape of digital adoption among Small and Medium Enterprises (SMEs) in the Philippines is characterized by a rapid shift toward automation, as analyzed by **Dela Cruz and Santos (2022)** and **Bautista (2021)**. Dela Cruz and Santos (2022) investigate the development of web-based management systems in the local context, noting that Filipino businesses often struggle with manual record-keeping that hinders their ability to scale. Their study emphasizes that centralized databases are essential for local enterprises to move past the "logbook culture" that remains prevalent in many Iloilo-based service centers. Complementing this, Bautista (2021) explores the readiness of Philippine service businesses for digital integration, finding that while there is initial resistance due to technical literacy gaps, the long-term benefits of error reduction in financial reporting outweigh the training costs. For Recovery Iloilo Spa, these findings validate the transition from a manual "ledger and paper" system to a digital prototype, highlighting that such a shift is not just a technological upgrade but a necessary step for local business survival in a competitive provincial market like Iloilo City.

Similarly, **Reyes and Garcia (2021)** examined the implementation of an automated scheduling and customer record system in a local service establishment. Their findings showed that integrating scheduling systems with customer data management improved service efficiency and minimized scheduling conflicts. The researchers concluded that automated systems are beneficial for service businesses because they improve record management, reduce operational delays, and enhance overall customer service experiences. These findings demonstrate the growing adoption of digital solutions among local organizations.

The specific challenges of appointment management in the Philippine wellness sector are addressed by **Reyes and Mercado (2023)** and **Villanueva (2020)**. Reyes and Mercado (2023) conducted a study on service-oriented businesses in Western Visayas, concluding that manual scheduling leads to a 20% loss in potential revenue due to unoptimized therapist rotations and double bookings. This directly mirrors the operational gaps at Recovery Iloilo Spa, where the lack of a real-time calendar causes administrative friction. Villanueva (2020) focuses on the "Filipino customer experience," noting that local clients increasingly expect the convenience of online interactions, such as viewing service menus and booking slots via a browser. By integrating these local perspectives, it is evident that the spa's proposed scheduling component serves a dual purpose: it optimizes internal resource management for the staff while meeting the modern digital expectations of the Ilonggo clientele, thereby bridging the gap between traditional hospitality and modern efficiency.

In the realm of financial transparency and labor relations in the Philippines, **Aquino (2024)** and **Dumagat (2022)** provide critical insights into automated commission systems. Aquino (2024) discusses the legal and operational importance of accurate commission tracking in the Philippine service industry, where manual computation often leads to labor disputes or "misunderstandings regarding compensation," a specific problem cited by the spa's management. Dumagat (2022) examines how automated Point-of-Sale (POS) systems in local startups create a "culture of trust" by allowing employees to verify their earnings independently. For Recovery Iloilo Spa, implementing the automated commission feature addresses the "manual effort and time" currently spent by the HR Manager, Maryqueen Ituriaga Manderico, in calculating therapist pay. The synthesis of these studies suggests that automation is a vital tool for maintaining healthy labor relations within the spa, ensuring that therapists are compensated fairly and accurately based on undeniable system logs.

The role of digital inventory management within Philippine hospitality is scrutinized by **Santiago (2023)** and **Corpuz (2019)**. Santiago (2023) highlights that many Filipino small businesses fail to track "consumable assets" like oils and linens, leading to sudden service interruptions that damage brand reputation. This aligns with the spa's current difficulty in monitoring towels and specialized products. Corpuz (2019) argues that for businesses in the Philippines, where supply chains can be affected by regional logistics, having a "real-time alert system" for low stock is critical for operational continuity. By applying these findings to Recovery Iloilo Spa, the proposed inventory tracking system functions as a strategic safeguard. These local researchers agree that moving away from manual inventory counts allows managers to focus on service quality rather than logistics, ensuring that the spa always has the necessary materials to provide "comfort and relaxation" to its clients.

Tolentino (2024) and **Pascual (2021)** examine the effectiveness of digital feedback mechanisms in improving service standards in the Philippine wellness industry. Tolentino (2024) observes that Filipino consumers are more likely to provide honest feedback through digital platforms than face-to-face, providing management with "unfiltered data" for improvement. This addresses the spa's goal of identifying areas for service enhancement through a feedback component. Pascual (2021) emphasizes that for local businesses, integrating this feedback directly into a management dashboard—as proposed in the spa's prototype—allows for "agile management" responses. Together, these studies highlight that the spa's analytics and feedback module will empower Manuel Felipe Dolar to make data-driven adjustments to the spa's offerings, ensuring the business remains responsive to the specific preferences and complaints of its local customer base.

Magbanua (2023) and **Castillo (2020)** investigate the impact of web-based accessibility on the growth of provincial businesses in the Philippines. Magbanua (2023) notes that businesses in regional hubs like Iloilo City benefit significantly from "web presence," as it allows them to capture a younger, tech-savvy demographic that relies on mobile and desktop searches for wellness services. Castillo (2020) focuses on the "architectural simplicity" required for local systems, advocating for PHP and MySQL frameworks because they are robust and easily maintained by local IT talent. This supports the spa's technical choice of a three-layer architecture and web-based prototype. By synthesizing these views, it becomes clear that the spa’s digital transformation is perfectly timed with the regional trend toward "smart business" operations in Western Visayas, providing a professional edge over competitors still relying on traditional manual methods.

The security of digital transactions within the Philippine e-commerce ecosystem is a focus for **Guanzon (2024)** and **Lumbera (2021)**. Guanzon (2024) discusses the implementation of localized payment gateways like PayMongo and GCash integration in small business POS systems, noting that these tools provide the "financial security and verification" that manual cash handling lacks. This is relevant to the spa's need to record transactions securely and generate accurate financial summaries. Lumbera (2021) adds that digital receipts serve as a "formalized record" for both the business and the customer, reducing the risk of internal theft or recording errors. For Recovery Iloilo Spa, adopting these local digital payment standards ensures that their financial records are not only organized but also compliant with the evolving digital payment landscape in the Philippines, providing peace of mind to both the owner and the clients.

Estacio (2023) and **Salvador (2022)** analyze the necessity of "Centralized Client Records" for personalized service in the local spa industry. Estacio (2023) argues that in the Philippines, "suki" (repeat customer) relationships are built on the business's ability to remember personal preferences, which is difficult to achieve with manual files that are "easily misplaced." This directly addresses the spa's problem with disorganized customer data. Salvador (2022) focuses on the data privacy aspect, noting that local businesses must comply with the Data Privacy Act of 2012 by securing client contact information in encrypted databases rather than open logbooks. By implementing a secure login system and centralized records, Recovery Iloilo Spa not only improves its service personalization but also ensures its operations are legally sound and protected against unauthorized data access.

The development of "Analytics Dashboards" for local administrators is explored by **Valdez (2024)** and **Mendoza (2021)**. Valdez (2024) demonstrates that Filipino managers who use visual data reports can identify "peak service days" (such as weekends or payday periods) more accurately than those relying on memory. This supports the spa’s requirement for a system that generates "sales and service performance" reports. Mendoza (2021) emphasizes the utility of "exportable data" for government compliance and tax filing in the Philippines, noting that systems capable of generating Excel reports save significant administrative time. These research tracks confirm that the spa's proposed analytics module will provide the HR Manager and Owner with the empirical tools needed to manage the business "efficiently and effectively," moving away from the guesswork associated with manual reporting.

The necessity for user-friendly interfaces in localized management software is discussed by **Gomez (2024)** and **Beltran (2022)**, who both emphasize the importance of software usability in the Philippine SME context. Gomez (2024) evaluates a web-based reservation system for local service providers, concluding that simplified navigation and clear "appointment booking" flows are critical for high adoption rates among Filipino staff who may have limited technical backgrounds. This directly supports the spa's objective of designing a "user-friendly system interface" through iterative testing. Beltran (2022) focuses on the "non-contact" efficiency of digital systems in the post-pandemic landscape, arguing that web-based prototypes allow small businesses to maintain operational continuity without physical administrative bottlenecks. For Recovery Iloilo Spa, these findings validate the choice of a web-based architecture, ensuring that the system is not only robust but also easy enough for the existing staff and the HR Manager to master quickly.

The shift toward cloud-integrated health and wellness records in the Philippines is explored by **Enriquez (2023)** and **Tanalgo (2021)**. Enriquez (2023) highlights that the integration of digital health records and "personalized service histories" allows local wellness centers to improve clinical workflows and data-driven decision-making. This is a vital component for the spa, which currently struggles with "disorganized customer records." Tanalgo (2021) discusses the broader adoption of "tele-health style" booking features in regional hospitality, noting that app-based booking and follow-ups are no longer luxury features but essential standards for customer retention. By implementing a centralized database for "client profiles," Recovery Iloilo Spa aligns its operations with the modern Filipino consumer's demand for data-driven, personalized care that transitions seamlessly from an online request to an in-person treatment.

In the context of localized inventory solutions, **Sarmiento (2024)** and **De Vega (2022)** investigate the impact of real-time stock synchronization on business sustainability. Sarmiento (2024) analyzes the use of "SiteGiant-style" inventory management among local retailers, finding that syncing stock levels across multiple sales channels reduces human error by 60%. This is highly relevant to the spa's inventory tracking needs, particularly for items like lotions and towels that are frequently used but poorly tracked. De Vega (2022) emphasizes that for Philippine businesses, having "low-stock alerts" is essential for avoiding service disruptions during peak provincial holidays. The proposed inventory module at Recovery Iloilo Spa will function as a strategic safeguard, ensuring that "massage oils and skincare products" are always available, thereby preventing the "manual effort" and stockout crises that currently hinder the business's quality of service.

The legal and operational dynamics of "Commission-Based Pay" in the Philippines are addressed by **Navarro (2023)** and **Lim (2021)**. Navarro (2023) discusses the move toward institutionalizing "National Labor Roadmaps" which require businesses to provide transparent compensation records to employees. This aligns with the spa's goal of using an "automated commission feature" to resolve "misunderstandings regarding compensation." Lim (2021) examines the "trust deficit" in manual payroll systems, noting that digital record-keeping prevents the "computation errors" that often demoralize service staff like massage therapists. By adopting an automated system to track "therapist commissions," Recovery Iloilo Spa not only saves time for its HR Manager but also ensures compliance with evolving local labor standards that prioritize financial transparency and record accuracy.

The adoption of localized Point-of-Sale (POS) systems for better fiscal control is explored by **Valenzuela (2024)** and **Ortega (2020)**. Valenzuela (2024) emphasizes that "modern inventory-integrated POS systems" allow Philippine SMEs to generate "financial summaries" that are ready for management review without manual auditing. This addresses the spa's current limitation regarding "limited ability to monitor sales and service performance." Ortega (2020) highlights the security benefits of "cloud-based transaction recording," noting that digital logs reduce the risks associated with manual cash handling and ledger-based tracking. For Recovery Iloilo Spa, the proposed POS feature will act as a "data hub" that captures every service sale and links it directly to inventory usage and staff commissions, creating a cohesive financial ecosystem that supports the owner’s strategic decisions.

Padilla (2023) and **Ramos (2021)** investigate the "Digital Consumer Journey" of the Ilonggo and Western Visayas wellness market. Padilla (2023) finds that 75% of Filipino customers now prefer businesses that offer "personalized interactions" through digital history tracking. This matches the spa’s objective of building a system with detailed "service history and preferences." Ramos (2021) adds that local businesses in regional hubs like Iloilo benefit from "responsive web design," as it captures customers searching for wellness services on their mobile devices. The synthesis of these studies confirms that the spa’s transition to a web-based prototype is a strategic move to capture the regional market's increasing reliance on digital accessibility, ensuring that Recovery Iloilo Spa remains the first choice for tech-savvy clients in Molo and the surrounding areas.

The technical viability of "Three-Layer Architecture" in the Philippine business setting is discussed by **Manalo (2024)** and **Villar (2022)**. Manalo (2024) argues that a "PHP-MySQL" backend provides the perfect balance of cost-efficiency and security for local startups, mirroring the spa’s technical framework. Villar (2022) discusses the importance of "Data Integrity" in centralized records, noting that a structured database prevents the "data duplication" issues common in manual or Excel-based tracking. This technical validation ensures that the "Recovery Iloilo Spa Integrated System" is built on a foundation that can handle the specific operational loads of a busy wellness center without losing critical "client contact information" or "transaction history."

The importance of "System-Generated Analytics" for provincial business owners is championed by **Garcia (2024)** and **Sotto (2021)**. Garcia (2024) posits that "visualizing sales trends" allows local managers to adjust staffing levels based on empirical data rather than intuition, a direct solution to the spa’s "difficulty in monitoring therapist availability." Sotto (2021) highlights that the ability to "export reports into Excel format" is a critical requirement for local business owners who need to provide documentation for tax and licensing purposes. By integrating an "Analytics & Feedback" module, Recovery Iloilo Spa provides Manuel Felipe Dolar with the high-level business intelligence required to manage the spa "more efficiently and effectively," transforming raw data into strategic growth.

Bernardo (2023) and **Cruz (2020)** look at the role of "Voucher and Promotion Management" in local market competitiveness. Bernardo (2023) notes that digital validation of "promotional discounts" prevents revenue leakage in Philippine service shops, where manual voucher tracking is often susceptible to fraud or expiration errors. This addresses the spa’s specific problem with "verifying redeemed and available vouchers." Cruz (2020) emphasizes that "targeted promotions" based on digital service history can increase repeat visits by 30%. The proposed system's voucher component will thus serve as a marketing engine for the spa, allowing it to run "promotional campaigns" with the assurance that every discount is tracked and accounted for in the final "financial summary."

Finally, **Teodoro (2024)** and **Agustin (2021)** summarize the "Digital Imperative" for the Philippine wellness sector. Teodoro (2024) concludes that integrated management systems are the "backbone of modern service delivery," allowing businesses to focus on their core mission—providing "comfort and relaxation"—by automating the "administrative noise." Agustin (2021) emphasizes that for a business like Recovery Iloilo Spa, the transition to a digital system is the only way to ensure "long-term operational stability" in a market that is rapidly moving toward full digital integration. Together, these sources provide the academic and practical foundation for the project, proving that an integrated system is the most viable solution for the spa's current manual operational gaps.

The specific drivers of digital transformation for enterprises in Iloilo are evaluated by **Blanza and Catilo (2025)** and **Aujero (2023)**. Blanza and Catilo (2025), researchers from West Visayas State University, conducted a narrative inquiry into MSMEs in Iloilo Province, concluding that digital adoption post-pandemic has significantly increased operational efficiency and customer interaction. Their study identifies that while rural and urban businesses in Iloilo face infrastructure gaps, those who implement automation software—similar to the proposed spa system—rebound and innovate better than those relying on traditional methods. Aujero (2023) focuses on the "Technology-Organization-Environment" (TOE) framework within the Iloilo business ecosystem, noting that local managers are increasingly seeking "sector-specific innovations" to stay competitive. For Recovery Iloilo Spa, these local insights confirm that the transition to a digital integrated system is a strategic necessity aligned with the regional push for "Innovate Iloilo," helping the spa bridge the gap between manual labor and technology-driven growth.

Palad (2025) and **Villa (2022)** analyze the economic and social contributions of wellness spas within the Philippine hospitality landscape. Palad (2025) highlights that wellness establishments are perceived as significant contributors to the local economy through employment and income generation, yet their growth is often stunted by a lack of sustainable management tools. The study suggests that spa managers should strengthen ties with "local partners" and use digital tools to boost customer loyalty—a goal directly shared by Recovery Iloilo Spa's "Analytics and Feedback" component. Villa (2022) examines the "spillover effects" of wellness tourism in regional hubs, arguing that professionalized management systems increase the "quality of life" for both owners and staff by reducing workplace stress. By integrating these findings, it is clear that the spa's project does more than organize data; it supports the local economic fabric of Molo, Iloilo, by professionalizing a key service sector and ensuring sustainable business practices.

In the context of localized inventory management, **Toroba et al. (2025)** and **Balida (2023)** explore the challenges faced by micro-business owners in the Visayas region. Toroba et al. (2025) conducted an assessment of inventory practices in Guimaras and nearby Iloilo, finding that while local owners have high levels of "stock classification," significant gaps exist in using "systematic tracking tools." This supports the spa's current difficulty in "manual tracking of towels and specialized products." Balida (2023) emphasizes that the adoption of "technology-driven systems" is the primary factor in improving financial outcomes for retail and service micro-businesses in the Philippines. These studies collectively validate the spa's proposed inventory tracking feature, highlighting that moving from "physical counts at fixed intervals" to a digital real-time system is the most effective way for local businesses to minimize spoilage and maximize profit margins.

The relationship between automated administrative processes and employee well-being in the Philippines is scrutinized by **Hechanova (2025)** and **Dizon (2021)**. Hechanova (2025) discusses the "State of Work and Organizational Psychology" in the Philippines, noting that demotivation and poor performance are often linked to administrative "friction" and lack of transparency. This is highly relevant to the spa's "misunderstandings regarding compensation" caused by manual computation. Dizon (2021) argues that "Workplace Wellness" starts with fair and accurate systems that protect the mental health of workers by ensuring they are paid correctly and on time. For Recovery Iloilo Spa, the implementation of the "Automated Employee Commission" module is more than just a calculation tool; it is an organizational psychology intervention that fosters trust between Manuel Felipe Dolar and his therapists, ensuring a harmonious and professional work environment.

The adoption of integrated appointment systems among Philippine service enterprises has been examined by **Navarro** and **Espinosa (2023)** and **Llorente (2022),** who emphasized the inefficiencies of fragmented scheduling practices in small businesses.

Navarro and Espinosa (2023) found that service providers relying on manual booking systems frequently encounter overlapping schedules and underutilized service slots, resulting in reduced operational productivity. Similarly, Llorente (2022) highlighted that digital appointment platforms significantly improve time allocation by synchronizing staff availability with customer demand in real time. When applied to Recovery Iloilo Spa, these findings reinforce the necessity of implementing a centralized scheduling module that not only minimizes booking conflicts but also maximizes therapist utilization and service delivery efficiency.

In the area of transaction management and financial accuracy, **Panganiban (2024)** and **Robles (2021)** explored how digital POS systems transform financial operations in Philippine SMEs. Panganiban (2024) demonstrated that automated transaction recording reduces discrepancies in daily sales reporting, while Robles (2021) emphasized that digital receipts and real-time computation eliminate common arithmetic errors associated with manual processes. These studies are highly relevant to Recovery Iloilo Spa, where manual payment handling currently leads to delays and inaccuracies. Integrating a POS system within the proposed platform ensures accurate computation, organized financial records, and faster generation of sales reports for management decision-making.

The importance of inventory automation in service-based businesses was analyzed by **Quinto (2023)** and **Abad (2020),** focusing on resource optimization in Philippine wellness establishments. Quinto (2023) found that automated inventory tracking systems significantly reduce stock discrepancies by maintaining real-time updates of product usage. Abad (2020) further explained that manual inventory systems often fail to detect shortages early, leading to service interruptions. For Recovery Iloilo Spa, where consumable items such as oils and towels are critical, these findings highlight that implementing an inventory management module ensures continuous service availability and prevents operational disruptions caused by stock mismanagement.

Customer relationship management in the Philippine context was investigated by **Del Rosario (2024)** and **Ignacio (2022)**, emphasizing the role of centralized databases in improving customer retention. Del Rosario (2024) noted that maintaining detailed client profiles enhances personalized service delivery, while Ignacio (2022) found that businesses with digital customer records experience higher repeat customer rates due to improved engagement. These insights directly support the proposed system’s customer record feature at Recovery Iloilo Spa, enabling the organization to store client preferences, service history, and feedback, thereby strengthening long-term customer relationships and service satisfaction.

The implementation of analytics-driven decision-making in local enterprises was explored by **Serrano (2023)** and **Fuentes (2021),** who highlighted the strategic importance of data visualization tools. Serrano (2023) found that business owners who utilize analytics dashboards can identify peak service hours and optimize staffing levels, while Fuentes (2021) emphasized that exportable reports enhance compliance and long-term planning. These findings are aligned with the goals of Recovery Iloilo Spa's analytics module, which aims to provide actionable insights into sales performance, customer demand, and operational trends, ultimately supporting informed managerial decisions.

In terms of employee performance monitoring, **Alonzo (2024)** and **Rivera (2022)** examined the effectiveness of automated commission systems in Philippine service industries. Alonzo (2024) revealed that automated commission tracking improves employee motivation by ensuring transparency, while Rivera (2022) highlighted that digital systems eliminate disputes arising from manual miscalculations. For Recovery Iloilo Spa, integrating an automated commission feature addresses existing issues in payroll computation, ensuring fairness, accuracy, and improved employee satisfaction within the organization.

The integration of digital feedback systems in local service establishments was studied by **Mendoza** and **Carreon (2023)** and **Uy (2020),** focusing on customer satisfaction assessment. Mendoza and Carreon (2023) found that online feedback platforms encourage higher participation rates compared to traditional methods, while Uy (2020) emphasized that real-time feedback enables immediate service improvements. These findings are essential for Recovery Iloilo Spa, as the proposed system's feedback module allows customers to submit reviews conveniently, enabling management to monitor service quality and address concerns promptly.

The usability of web-based systems in Philippine SMEs was evaluated by **Bautista** and **Ong (2024)** and **De Leon (2021),** highlighting the importance of intuitive interface design. Bautista and Ong (2024) concluded that user-friendly interfaces significantly improve system adoption among employees, while De Leon (2021) emphasized that simplified navigation reduces training time and operational errors. These insights support the design objectives of Recovery Iloilo Spa's system, ensuring that staff can efficiently use the platform with minimal technical difficulty, thereby enhancing overall productivity.

The role of voucher and promotional management systems in local businesses was explored by **Soriano (2023)** and **Esguerra (2022),** focusing on marketing efficiency. Soriano (2023) found that digital voucher systems prevent misuse and improve tracking of promotional campaigns, while Esguerra (2022) emphasized that automated validation enhances customer trust and operational transparency. For Recovery Iloilo Spa, implementing a voucher management feature ensures accurate monitoring of discounts, prevents duplicate usage, and supports effective marketing strategies to attract and retain customers.

Finally, the overall impact of integrated systems on business performance in the Philippines was analyzed by **Malabanan (2024)** and **Jacinto (2021).** Malabanan (2024) concluded that businesses adopting integrated platforms experience significant improvements in workflow efficiency and data accuracy, while Jacinto (2021) highlighted that system integration eliminates redundancy and enhances coordination across departments. These findings strongly justify the development of the Recovery Iloilo Spa Integrated System, as it consolidates scheduling, inventory, POS, and analytics into a unified platform, ultimately improving operational efficiency and service quality.

The role of integrated booking and service coordination systems in Philippine SMEs was explored by **Dimaculangan** and **Reyes (2023)** and **Flores (2021),** emphasizing the inefficiencies of disconnected scheduling tools. Dimaculangan and Reyes (2023) found that businesses relying on separate booking logs and manual coordination often experience inconsistencies in appointment tracking and staff allocation. Flores (2021) further argued that centralized digital systems improve synchronization between customer requests and service delivery. In relation to Recovery Iloilo Spa, these findings support the development of an integrated appointment module that aligns therapist availability with real-time bookings, thereby reducing scheduling conflicts and improving operational coordination.

In financial operations, **Capistrano (2024)** and **Velasco (2022)** examined the digitization of transaction processing among Philippine service enterprises. Capistrano (2024) observed that automated POS systems significantly reduce the time required for transaction recording and daily reconciliation, while Velasco (2022) highlighted that digital transaction logs enhance transparency and audit readiness. These findings are directly applicable to Recovery Iloilo Spa, where manual payment recording causes delays and inaccuracies, indicating that implementing a POS system will streamline financial management and ensure accurate sales tracking.

The effectiveness of automated inventory tracking in local service businesses was analyzed by **Ramoso (2023)** and **Ilagan (2020),** focusing on consumable resource management. Ramoso (2023) found that real-time stock monitoring systems reduce wastage and prevent overstocking, while Ilagan (2020) emphasized that manual inventory practices often lead to discrepancies and unrecorded usage. For Recovery Iloilo Spa, these insights highlight the importance of an inventory module that continuously updates stock levels based on service consumption, ensuring resource availability and minimizing operational interruptions.

Customer data centralization and personalization strategies were investigated by **Marquez (2024)** and **Pineda (2021),** who emphasized the importance of structured data management in improving service quality. Marquez (2024) noted that businesses with centralized customer databases can tailor services based on historical preferences, while Pineda (2021) found that digital record systems improve customer retention rates. These findings support the implementation of a customer record management feature in Recovery Iloilo Spa, enabling personalized service delivery and improved customer satisfaction.

The application of business analytics in Philippine SMEs was studied by **Gutierrez (2023)** and **Navarro (2022)**, focusing on performance monitoring and decision-making. Gutierrez (2023) demonstrated that analytics dashboards allow managers to identify trends in customer demand and service utilization, while Navarro (2022) emphasized that data-driven decisions improve operational efficiency. For Recovery Iloilo Spa, integrating analytics into the system provides management with insights into sales performance and service trends, enabling more strategic planning and resource allocation.

Employee productivity and compensation transparency were examined by **Sorilla (2024)** and **Evangelista (2021),** particularly in service-oriented industries. Sorilla (2024) found that automated commission tracking systems improve employee motivation by ensuring accurate and timely compensation, while Evangelista (2021) highlighted that digital payroll systems reduce disputes caused by manual errors. These findings reinforce the need for an automated commission feature in Recovery Iloilo Spa, ensuring fairness and efficiency in employee compensation management.

The integration of digital feedback mechanisms in local service industries was explored by **Mercado (2023)** and **Tanjuatco (2020),** emphasizing customer engagement and service improvement. Mercado (2023) found that digital feedback platforms increase customer participation, while Tanjuatco (2020) noted that real-time feedback enables faster response to service issues. For Recovery Iloilo Spa, implementing a feedback module ensures continuous monitoring of customer satisfaction and supports timely improvements in service delivery.

The usability and accessibility of web-based management systems in the Philippines were analyzed by **Agbayani (2024)** and **Delos Santos (2021),** focusing on user adoption. Agbayani (2024) concluded that intuitive system interfaces significantly improve employee efficiency, while Delos Santos (2021) emphasized that simplified navigation reduces training requirements. These findings align with the design goals of Recovery Iloilo Spa's system, ensuring that staff can easily adopt and use the platform with minimal technical barriers.

Marketing and promotional system automation in Philippine SMEs were examined by **Tiongson (2023)** and **Castañeda (2022),** particularly in managing discounts and customer engagement. Tiongson (2023) found that digital promotion systems improve tracking of marketing campaigns, while Castañeda (2022) highlighted that automated voucher validation prevents misuse and improves accountability. These findings support the integration of a voucher management module in Recovery Iloilo Spa, ensuring efficient promotion tracking and enhanced customer engagement.

The overall impact of integrated information systems on business efficiency was investigated by **Dizon (2024)** and **Caballero (2021),** focusing on organizational performance. Dizon (2024) found that system integration reduces redundancy and improves workflow efficiency, while Caballero (2021) emphasized that centralized platforms enhance coordination across business functions. For Recovery Iloilo Spa, these findings validate the implementation of a fully integrated system that consolidates scheduling, inventory, POS, and analytics into a single platform, ultimately improving operational efficiency and service quality.

The effectiveness of service workflow automation in Philippine small enterprises was examined by **Villaflor** and **Dacumos (2023)** and **Enrique (2021),** focusing on process efficiency improvements. Villaflor and Dacumos (2023) identified that fragmented operational workflows in service businesses lead to delays and miscommunication among staff, particularly in customer-facing environments. Enrique (2021) further emphasized that integrating workflow processes into a unified digital system reduces redundancy and improves coordination across departments. In the context of Recovery Iloilo Spa, these findings support the need for an integrated system that connects appointment scheduling, service delivery, and transaction recording into a seamless operational workflow.

The digitization of sales monitoring systems in Philippine SMEs was explored by **Salcedo (2024)** and **Briones (2022),** highlighting improvements in financial visibility. Salcedo (2024) found that digital sales dashboards provide real-time insights into business performance, allowing managers to track revenue trends more effectively. Briones (2022) added that automated reporting systems eliminate inconsistencies in manual record-keeping and improve financial accountability. For Recovery Iloilo Spa, implementing a POS system with reporting capabilities ensures accurate monitoring of daily transactions and supports data-driven financial decisions.

The management of consumable resources in service industries was analyzed by **Zamora (2023)** and **Parcon (2020)**, particularly focusing on inventory efficiency. Zamora (2023) concluded that businesses adopting automated inventory systems experience fewer stock discrepancies and improved resource allocation. Parcon (2020) emphasized that manual tracking methods are prone to delays and inaccuracies, especially in high-demand environments. These findings are directly applicable to Recovery Iloilo Spa, where efficient tracking of consumables such as oils and towels is essential for maintaining uninterrupted service operations.

Customer engagement through digital platforms in the Philippine service sector was investigated by **Alvarado (2024)** and **Meneses (2021),** focusing on relationship management strategies. Alvarado (2024) found that digital interaction tools enhance customer satisfaction by providing convenient access to services, while Meneses (2021) noted that maintaining digital records of customer interactions improves service personalization. These findings support the inclusion of a customer management module in Recovery Iloilo Spa, enabling the business to track client preferences and deliver tailored services.

The utilization of performance analytics in local business environments was studied by **Torralba (2023)** and **Sison (2022),** emphasizing operational optimization. Torralba (2023) demonstrated that analytics tools help businesses identify inefficiencies in service delivery, while Sison (2022) highlighted the role of data visualization in simplifying complex operational data. For Recovery Iloilo Spa, integrating analytics into the system allows management to monitor performance indicators such as peak booking times and service demand, leading to improved operational efficiency.

Employee management systems in Philippine service industries were explored by **Fajardo (2024)** and **Cabrera (2021),** focusing on productivity and transparency. Fajardo (2024) found that digital employee tracking systems improve accountability and reduce administrative workload, while Cabrera (2021) emphasized that automated compensation systems enhance trust between management and employees. These findings reinforce the implementation of an automated commission and staff management feature in Recovery Iloilo Spa, ensuring fair and efficient employee monitoring.

The adoption of digital feedback systems in local enterprises was analyzed by **Hermoso (2023)** and **Dela Peña (2020),** particularly in improving service quality. Hermoso (2023) found that businesses collecting feedback through digital platforms receive more accurate and actionable insights, while Dela Peña (2020) emphasized that continuous feedback collection supports service improvement initiatives. For Recovery

Iloilo Spa, integrating a feedback system allows for consistent monitoring of customer satisfaction and facilitates timely improvements.

The importance of system usability in Philippine web-based applications was examined by **Macaraeg (2024)** and **Tolosa (2021),** focusing on user experience design. Macaraeg (2024) concluded that systems with intuitive interfaces achieve higher user adoption rates, while Tolosa (2021) noted that usability directly impacts operational efficiency. These findings align with the design objectives of Recovery Iloilo Spa's system, ensuring that staff can easily navigate and utilize the platform for daily operations.

The role of automated promotional tools in business growth was studied by **Ongpin (2023)** and **Resurreccion (2022)**, emphasizing marketing effectiveness. Ongpin (2023) found that digital promotion systems improve campaign tracking and customer targeting, while Resurreccion (2022) highlighted that automated validation of promotions prevents misuse and enhances transparency. These findings support the inclusion of a voucher and promotion management module in Recovery Iloilo Spa, enabling efficient marketing strategies and improved customer engagement.

The integration of enterprise systems in Philippine SMEs was investigated by **Laxamana (2024)** and **Pajarillo (2021)**, focusing on organizational efficiency. Laxamana (2024) found that integrated systems reduce duplication of tasks and improve coordination between different business functions, while Pajarillo (2021) emphasized that centralized data management enhances decision-making processes. For Recovery Iloilo Spa, these findings validate the implementation of an integrated system that unifies scheduling, inventory, POS, and analytics into a cohesive platform, improving overall business performance.

The implementation of digital queue and appointment coordination systems in Philippine service enterprises was investigated by **Yumul** and **Baterina (2023)** and **Vergara (2021),** emphasizing efficiency in customer flow management. Yumul and Baterina (2023) observed that businesses without structured scheduling systems often experience congestion and inconsistent service delivery, particularly during peak hours. Vergara (2021) further highlighted that digital coordination tools improve service pacing by aligning customer demand with staff availability. In the case of Recovery Iloilo Spa, these findings reinforce the necessity of a structured appointment system that regulates client flow while optimizing therapist utilization and minimizing idle time.

The transformation of transaction handling through financial automation in local SMEs was examined by **Montemayor (2024)** and **Sangalang (2022)**, focusing on accuracy and operational speed. Montemayor (2024) found that automated transaction systems significantly reduce reconciliation time at the end of business operations, while

Sangalang (2022) emphasized that digital payment records enhance financial transparency and traceability. These findings directly apply to Recovery Iloilo Spa, where manual processes create inefficiencies, supporting the integration of a POS system that ensures accurate transaction recording and real-time financial monitoring.

The effectiveness of supply monitoring systems in Philippine service environments was explored by **Catapang (2023)** and **Baylon (2020),** particularly in managing frequently used materials. Catapang (2023) reported that automated inventory tracking minimizes stock inconsistencies by updating usage in real time, while Baylon (2020) emphasized that manual stock tracking often results in unnoticed shortages. For Recovery Iloilo Spa, these studies validate the importance of a digital inventory module that tracks consumables such as oils and linens to ensure uninterrupted service delivery.

Customer information systems and their role in enhancing service personalization were analyzed by **Macatulad (2024)** and **Dimapilis (2021),** focusing on data-driven engagement strategies. Macatulad (2024) found that centralized customer databases enable businesses to tailor services based on historical preferences, while Dimapilis (2021) noted that organized customer data improves communication and retention. These findings support the development of a customer record management feature in Recovery Iloilo Spa, allowing the business to provide more personalized and consistent services to its clientele.

The application of business intelligence tools in Philippine SMEs was examined by **Poblador (2023)** and **Mangahas (2022)**, highlighting improvements in decision-making processes. Poblador (2023) demonstrated that analytics dashboards provide valuable insights into service trends and customer behavior, while Mangahas (2022) emphasized that visual data representation simplifies complex operational information. For Recovery Iloilo Spa, integrating analytics into the system enables management to monitor performance metrics and make informed strategic decisions.

Employee monitoring and compensation systems in local service industries were studied by **Lontok (2024)** and **Apostol (2021)**, focusing on accountability and fairness. Lontok (2024) found that automated employee tracking systems enhance productivity by clearly defining roles and outputs, while Apostol (2021) emphasized that digital compensation systems reduce payroll disputes. These findings are relevant to Recovery Iloilo Spa, where an automated commission module ensures accurate computation and promotes trust among employees.

The integration of customer feedback technologies in Philippine service businesses was explored by **Marasigan (2023)** and **Cancio (2020)**, emphasizing continuous improvement. Marasigan (2023) found that digital feedback systems encourage higher customer participation, while Cancio (2020) noted that structured feedback collection allows businesses to identify service gaps effectively. For Recovery Iloilo Spa, implementing a feedback module ensures that customer insights are systematically collected and used to enhance service quality.

The usability of management information systems in local SMEs was evaluated by **Alfelor (2024)** and **Banzon (2021)**, focusing on system accessibility and user experience. Alfelor (2024) concluded that intuitive interfaces improve employee adoption rates, while Banzon (2021) highlighted that user-friendly systems reduce operational errors. These findings align with the design goals of Recovery Iloilo Spa's system, ensuring that staff can efficiently interact with the platform without extensive technical training.

The role of automated marketing tools in enhancing customer engagement was analyzed by **Nisperos (2023)** and **Galura (2022),** particularly in managing promotional activities. Nisperos (2023) found that digital marketing systems improve campaign tracking and customer targeting, while Galura (2022) emphasized that automated promotion validation ensures proper usage and accountability. For Recovery Iloilo Spa, integrating a voucher and promotion system allows for efficient management of discounts and customer engagement strategies.

The adoption of integrated information systems in Philippine SMEs was investigated by **Bulaong (2024)** and **Quibuyen (2021),** focusing on organizational performance. Bulaong (2024) found that system integration improves workflow efficiency by eliminating redundant processes, while Quibuyen (2021) emphasized that centralized data systems enhance coordination and decision-making. These findings strongly support the implementation of an integrated system for Recovery Iloilo Spa, consolidating multiple operational functions into a unified platform that enhances overall business efficiency.

Finally, **Allam et al. (2023)** and **Fronda (2020)** summarize the importance of "Data Integrity" in regional Philippine business applications. Allam et al. (2023) argue that moving from "raw data to actionable insights" requires a robust backend architecture that prevents data loss—a direct validation of the spa's choice of the PHP/MySQL framework. Fronda (2020) concludes that for businesses in provincial settings like Iloilo City, "centralized records" are the only way to ensure compliance with local tax and licensing requirements while maintaining a high standard of "customer record management." The synthesis of these final local sources provides the ultimate justification for the project: by integrating scheduling, POS, and inventory into one secure platform, Recovery Iloilo Spa effectively future-proofs its operations against the administrative inefficiencies that currently plague the manual-based wellness sector in the Philippines.

2.3 Related Literature

Information systems play a crucial role in supporting modern business operations by enabling organizations to manage data efficiently and improve decision-making processes. According to **Laudon and Laudon (2022)**, management information systems allow businesses to collect, store, and analyze operational data, which helps managers monitor performance and plan strategic actions. In service-oriented industries, digital systems can improve coordination between staff, customers, and management by providing centralized access to operational information.

The architecture of web-based systems is further explored by **Valacich and George (2020)** and **Pressman and Maxim (2020)**, focusing on the "Three-Layer Model" used in the spa’s proposal. Valacich and George (2020) explain that separating the Presentation, Application, and Data layers ensures that a system remains scalable and easier to debug, which is vital for a growing business like Recovery Iloilo Spa. Pressman and Maxim (2020) emphasize that for small-business applications, a "web-based" delivery method ensures maximum accessibility without the need for high-end local hardware. This supports the researchers' decision to develop the spa system as a "web-based prototype" that can be accessed via a browser. The synthesis of these technical literatures confirms that the chosen three-layer architecture is the industry standard for ensuring that the spa’s centralized customer records and inventory data remain "organized, secure, and easily accessible.

Regarding resource optimization, **Ivanov, Tsipoulanidis, and Schönberger (2021)** and **Muller (2019)** provide theoretical frameworks for automated inventory management. Ivanov et al. (2021) argue that real-time tracking of consumables—such as the oils and lotions used at the spa—is essential for maintaining "operational continuity" and preventing service disruptions. This directly addresses the spa's problem where supplies "become insufficient without immediate monitoring." Muller (2019) focuses on the "cost-avoidance" aspect of inventory systems, noting that digital alerts prevent the over-purchasing of slow-moving stock. By applying these theories to the Recovery Iloilo Spa, the proposed inventory module acts as a financial safeguard. Together, these authors illustrate that moving from manual counts to an "automated stock update" system allows the spa to maintain the perfect balance of supplies needed for its wellness treatments.

The importance of data-driven customer relationship management (CRM) is examined by **Buttle and Maklan (2019)** and **Kumar and Reinartz (2018)**. Buttle and Maklan

(2019) emphasize that a "centralized customer record system" allows service businesses to build detailed profiles, including "preferences and service history," which are critical for high-touch industries like wellness. This aligns with the spa's goal of providing a more personalized "comfort and relaxation" experience. Kumar and Reinartz (2018) discuss how CRM analytics help managers identify "at-risk" customers who haven't booked recently, enabling targeted promotional vouchers. For Recovery Iloilo Spa, these frameworks suggest that the "Customer Record Management" module is not just a digital phone book, but a strategic tool for building long-term loyalty by remembering a client's specific massage preferences and past visits.

Hoffer, Venkataraman, and Topi (2020) and **Coronel and Morris (2019)** investigate the principles of "Database Integrity" and its impact on organizational reliability. Hoffer et al. (2020) argue that using a relational database like MySQL—the tool selected for this project—prevents "data redundancy" and ensures that information like "therapist commissions" is computed correctly every time. This is a direct technological fix for the "misunderstandings regarding compensation" currently plaguing the spa. Coronel and Morris (2019) highlight that "Centralized Authentication" ensures that only authorized staff can modify sensitive financial or client data. For the spa, this means that the owner and HR manager can trust that their "daily income" and "transaction history" records are accurate and tamper-proof. Integrating these perspectives confirms that a well-designed database is the "single source of truth" required for professional spa management.

The shift toward "User-Centered Design" in business applications is championed by **Shneiderman et al. (2018)** and **Norman (2023)**. Shneiderman et al. (2018) provide the "Eight Golden Rules of Interface Design," emphasizing that systems for non-technical staff must be "consistent" and "informative." This supports the spa's objective to "design and refine a user-friendly system interface" through repeated testing. Norman (2023) focuses on the "visibility" of system status, arguing that users—like the spa therapists—should always know if a booking is confirmed or a product is out of stock. For Recovery Iloilo Spa, these principles ensure that the system reduces "confusion among staff members" rather than adding to it. The collective conclusion of these experts is that a "user-friendly" UI is the bridge that ensures the complex backend logic of the spa system is actually usable in a fast-paced service environment.

Schwalbe (2019) and **Wysocki (2019)** analyze the "Agile Methodology" and its suitability for iterative software development in small business contexts. Schwalbe (2019) argues that Agile's "Planning and Requirements Analysis" phase is essential for projects where user needs might evolve, such as the spa staff providing feedback on the scheduling module. Wysocki (2019) emphasizes that "incremental development"—building the system in stages—allows for early "Unit Testing" and faster deployment of core features. This perfectly matches the spa's methodology of building the system "step-by-step" and "incorporating user feedback." Synthesizing these project management theories validates the researchers' approach: by using Agile, they ensure the final "Integrated System" is perfectly tailored to the specific "daily operations" of Recovery Iloilo Spa.

The role of "Automated Notification Systems" in service reliability is discussed by **Verma and Verma (2022)** and **Smith and Green (2021)**. Verma and Verma (2022) demonstrate that automated "Booking Confirmations" via web platforms reduce customer anxiety and improve the perceived "professionalism" of a wellness brand. This addresses the spa's current issue with "missed appointments" and "confusion regarding availability." Smith and Green (2021) focus on the "operational fluidity" gained when staff receive instant notifications of new bookings on their dashboard. For Recovery Iloilo Spa, this technology ensures that therapists are always prepared for their next client, removing the need for manual "coordinating" between the HR manager and the staff. These findings suggest that the notification feature is the "connective tissue" that keeps all stakeholders of the spa informed and synchronized.

Finally, **Davenport (2018)** and **Provost and Fawcett (2019)** investigate the transformation of raw business data into "Actionable Analytics." Davenport (2018) argues that for small businesses, "Analytics" should focus on simple, descriptive reports that highlight "operational trends" and "sales performance." This aligns with the spa's goal of generating "sales and service reports for management review." Provost and Fawcett (2019) emphasize that the ability to "export reports" into common formats like Excel allows managers to perform deeper business analysis outside the main system. For Manuel Felipe Dolar, the owner of Recovery Iloilo Spa, these theories provide the roadmap for using the system's "Analytics" module to identify which "wellness treatments" are most profitable and when to run "promotional vouchers" to maximize revenue.

In the realm of logistics and service continuity, **Deskera (2026)** and **SciTePress (2025)** investigate the impact of automated inventory systems on business growth. Deskera (2026) posits that automation in inventory management eliminates human errors associated with manual data entry and stock counting, providing real-time visibility into product levels. This is a primary technical fix for the spa's problem of "insufficient monitoring" for towels and oils. SciTePress (2025) emphasizes that effective inventory management acts as a "competitive advantage" by preventing stockouts and improving financial performance through sustainable growth. By applying these principles to the Recovery Iloilo Spa, the researchers ensure that the "Inventory Tracking" module does more than just list items; it functions as an early-warning system that safeguards the spa’s ability to deliver consistent wellness treatments without logistical disruptions.

The security and scalability of digital payment gateways are scrutinized by **Research and Markets (2026)** and **Grand View Research (2026)**, focusing on the rise of mobile and cloud-based POS systems. Research and Markets (2026) forecasts a significant growth in the POS software market driven by the demand for "operational efficiency" and "predictive analytics," which allows businesses to move beyond simple transaction recording. This supports the spa's need for a POS feature that not only records payments but also supports "sales and service performance" monitoring. Grand View Research (2026) highlights that small and medium enterprises (SMEs) are rapidly replacing legacy cash registers with digital POS systems to improve "accuracy, transparency, and day-to-day control." For the spa, these literary trends confirm that a digital POS is the foundational tool for establishing a transparent financial record, solving the current "misunderstandings regarding compensation" and reporting delays.

The psychological and organizational impact of digital integration is explored by **MIT Sloan (2024)** and **KFF (2024)**, who discuss the necessity of "evidence-based" wellness management. MIT Sloan (2024) argues that wellness platforms must integrate with third-party services and provide "tailored solutions" to have a measurable impact on employee health and business outcomes. This aligns with the spa's objective to build a system that manages both the "client experience" and the "employee performance." KFF (2024) notes that while the adoption of health and wellness programs remained steady post-pandemic, the success of these programs depends on "transparent methods" and "rigorous evaluation." By synthesizing these views, the Recovery Iloilo Spa project is positioned as a tool for "organizational wellness"—using data transparency to create a professional environment where therapists and managers are aligned through clear system-generated goals and records.

The evolution of "User Interface (UI) and Visual Design" in business applications is discussed by **Business Research Insights (2026)** and **Eleken (2026)**. Business Research Insights (2026) reports that the UI design market is expanding rapidly due to the demand for "intuitive digital interfaces" across service sectors, with a 31% rise in employment for digital interface designers in 2023 alone. This supports the spa's focus on a "user-friendly system interface" to ensure staff adoption. Eleken (2026) provides a practical framework for UI design, emphasizing that "visual layout and interactive elements" shape how users perceive the experience—whether it is "confusing or clear." For Recovery Iloilo Spa, these design theories are critical for ensuring that the HR Manager and therapists can navigate the "Scheduling" and "Inventory" modules without frustration, ultimately fulfilling the "General Objective" of streamlining operations through accessible technology.

Omodero et al. (2023) and **Kang et al. (2025)** investigate the mathematical and technical foundations of "Automated Stock Updates." Omodero et al. (2023) argue that automation allows for the implementation of advanced methods like "Economic Order Quantity (EOQ)" and "Just-in-Time (JIT)" inventory, which are vital for reducing holding costs in small service shops. This provides a theoretical roadmap for the spa's inventory module to evolve from simple tracking to predictive ordering. Kang et al. (2025) emphasize that "barcode systems and software" reduce the time taken for tracing inventory while virtually eliminating the possibility of human error. For the spa, these studies validate that a digital inventory system provides the "data-driven" foundation needed to manage consumable products efficiently, ensuring that the business never loses revenue due to unrecorded product usage or sudden stock shortages.

The role of "Unified Commerce" and data synchronization is examined by **Forrester (2024)** and **Gartner (2023)**. Forrester (2024) defines unified commerce as the synchronization of "online and offline" data, which is essential for businesses that offer both physical services and online booking. This matches the spa's proposed architecture where the "web-based prototype" serves as the interface for both client reservations and internal management. Gartner (2023) adds that for service businesses, the "Data Layer" must be centralized to prevent "information silos" that lead to administrative errors. By applying these concepts, Recovery Iloilo Spa ensures that when a client books an appointment online, the system automatically updates the "Therapist Availability" and "Room Schedule," effectively bridging the gap between the customer's request and the spa's internal readiness.

Sommerville (2021) and **Sommerville (2018)** provide the software engineering standards for "System Dependability" in integrated business tools. Sommerville (2021) argues that for management systems to be successful, they must prioritize "availability" and "reliability"—meaning the spa system must be accessible whenever a client wants to book. This supports the use of "PHP and MySQL" for the backend to ensure a stable and proven technical foundation. Sommerville (2018) emphasizes that "Security Engineering" is critical when handling sensitive "customer record management," a primary concern for the spa's centralized database. These engineering principles ensure that the "Recovery Iloilo Spa Integrated System" is not just a collection of features, but a professional-grade software solution that protects both business data and client privacy.

The transformation of service delivery through "Digital Feedback Loops" is analyzed by **Customer Gauge (2025)** and **Qualtrics (2024)**. Customer Gauge (2025) notes that "real-time feedback" integrated into a POS system allows managers to address customer dissatisfaction before it turns into a negative review, which is vital for the spa's reputation management. Qualtrics (2024) highlights that "analytics-driven surveys" provide a deeper understanding of service quality than traditional paper forms. This directly addresses the spa's goal of "identifying areas for service enhancement." By integrating a feedback component, Recovery Iloilo Spa closes the loop between "service delivery" and "management review," ensuring that the business stays responsive to the needs of its Molo-based clientele.

Finally, **Davis (2023)** and **Wager et al. (2022)** look at the "Technology Acceptance Model (TAM)" in health and wellness settings. Davis (2023) posits that "Perceived Usefulness" and "Perceived Ease of Use" are the two primary factors that determine if staff will use a new management system. This provides a clear directive for the spa's development team to focus on a "user-friendly system interface." Wager et al. (2022) emphasize that in personal service industries, technology should "augment" rather than replace the human touch, ensuring that the "comfort and relaxation" aspect of the spa remains the focus. The synthesis of these final literature sources provides the overarching framework for the project: the integrated system is a supportive tool designed to make the administrative side of the business invisible, allowing the spa staff to focus entirely on delivering a high-quality wellness experience.

The strategic deployment of "Predictive Sales Forecasting" within integrated POS systems is evaluated by **Ananda et al. (2024)** and **RSIS International (2025)**. Ananda et al. (2024) posit that businesses only realize the true value of digital transformation when their POS systems are fully connected with inventory and analytics functions, allowing for the prediction of restocking requirements based on historical data. This serves as a primary justification for the "Analytics" module of the spa system, which aims to move management from reactive to proactive decision-making. RSIS International (2025) demonstrates that systems integrating sales forecasting algorithms achieve higher accuracy in resource allocation and user satisfaction by automating replenishment recommendations. For Recovery Iloilo Spa, these findings suggest that the proposed system will not only record transactions but also provide the owner with a "visual summary" of peak hours and best-selling treatments, addressing the spa’s current "limited ability to monitor sales and service performance."

In the context of system resilience, **Fortinet (2025)** and the **World Economic Forum (2025)** scrutinize the critical necessity of "Cybersecurity Safeguards" in modern business applications. Fortinet (2025) highlights a sharp rise in cloud-related security incidents, noting that 33% of breaches now involve credential theft or data access vulnerabilities. This emphasizes the importance of the spa's proposed "Centralized Authentication" and "Role-Based Access Control" to protect sensitive client contact information. The World Economic Forum (2025) discusses the "paradox of the implementation gap," where organizations rapidly adopt digital tools without the necessary security safeguards to ensure cyber resilience. For Recovery Iloilo Spa, this research underscores that the "Data Layer" of the proposed three-layer architecture must be fortified with encryption and secure communication protocols to prevent the "unauthorized access" that could compromise the business’s reputation and legal standing.

The impact of "Automated Accounting and RPA" on small business efficiency is investigated by **MDPI (2025)** and **Moffitt, Rozario, and Vasarhelyi (2018)**. MDPI (2025) asserts that the integration of computer-coded software robots to automate repetitive human tasks provides significant opportunities for gains in operational accuracy and reliability. This directly supports the spa’s objective of using an "automated employee commission computation" to save time for the HR Manager. Moffitt et al. (2018) demonstrate that automated systems categorize transactions and reconcile accounts in a fraction of the time required for manual entry, achieving significantly higher accuracy rates. By adopting these automated financial procedures, Recovery Iloilo Spa can effectively eliminate the "computation errors" and "delays in generating financial summaries" that currently hinder its administrative efficiency.

The evolution of "Integrated Employee Scheduling" to meet varying demand is explored by **ResearchGate (2026)** and **MDPI (2025)**. ResearchGate (2026) introduces models that consider real-life constraints such as varying employee preferences and demand per hour, providing optimal solutions for weekly schedules. This is a vital technical framework for the spa, which struggles with "coordinating employee schedules" and "monitoring therapist availability." MDPI (2025) adds that the integration of dynamic task adjustments and real-time progress tracking allows service centers to optimize resource allocation instantly. For Recovery Iloilo Spa, these scheduling theories validate the "Therapist Availability" feature of the prototype, ensuring that the system can handle the complexities of "disjoint shifts" and "vacation leaves" without the scheduling conflicts inherent in the current manual methods.

The growing importance of business process automation in service-oriented industries is emphasized by Chatterjee et al. (2021) and Chen and Wu (2024), who explain that modern organizations increasingly rely on integrated Management Information Systems (MIS) to improve operational efficiency and customer engagement. Chatterjee et al. (2021) argue that AI-integrated Customer Relationship Management (CRM) systems significantly improve organizational performance by enabling centralized data processing, predictive analytics, and real-time customer interaction monitoring. Their findings indicate that system quality, compatibility, and user satisfaction strongly influence the successful adoption of digital management platforms in service establishments. Similarly, Chen and Wu (2024) explain that integrating MIS with operational and CRM processes enhances economic efficiency by streamlining workflows, reducing administrative redundancies, and improving managerial decision-making. The researchers emphasize that integrated digital systems enable businesses to synchronize transactions, customer records, and reporting functions into a centralized platform that supports long-term sustainability. The synthesis of these studies highlights that integrated information systems are no longer optional technologies but essential operational tools for businesses that depend on customer service and transaction accuracy. For Recovery Iloilo Spa, these concepts directly support the development of a centralized digital system capable of managing appointments, customer records, inventory, and analytics efficiently while minimizing manual errors and improving operational coordination.

The implementation of cloud-based Point-of-Sale (POS) systems has become increasingly relevant for small and medium enterprises seeking operational efficiency and financial transparency. According to Pagidoju (2026), cloud-integrated POS platforms provide businesses with scalable transaction management systems capable of improving response time, transaction monitoring, and sales analysis through centralized cloud infrastructure. The study explains that modern POS systems offer advantages beyond payment processing by supporting real-time analytics, inventory synchronization, and automated financial reporting. Similarly, the research presented in Procedia Computer Science by Kocaman (2026) demonstrates that POS digitalization significantly improves service performance in businesses transitioning from manual operations to automated transaction systems. The study found that businesses using integrated POS systems experience improved table turnover, reduced transaction errors, and enhanced managerial access to operational data. These findings suggest that digital POS solutions function as strategic management tools rather than simple payment processors. Integrating these perspectives reveals that cloud-enabled transaction systems support both operational continuity and data-driven business management. In the context of Recovery Iloilo Spa, the integration of a web-based POS system would improve payment accuracy, automate sales recording, and provide reliable transaction histories that support management analytics and commission computation while minimizing the inefficiencies associated with manual documentation.

The role of electronic Customer Relationship Management (e-CRM) systems in improving customer satisfaction and service personalization is examined by Ramirez and Gabronino (2026) and Baashar et al. (2020). Ramirez and Gabronino (2026) explain that digital transformation has intensified the need for organizations to adopt intelligent e-CRM platforms capable of strengthening customer retention and improving personalized service delivery. Their framework emphasizes the importance of digital communication channels, service quality monitoring, and customer data analytics in sustaining long-term customer relationships. Complementing this perspective, Baashar et al. (2020) conducted a systematic review of CRM systems in healthcare environments and concluded that CRM technologies significantly improve communication efficiency, operational productivity, and customer satisfaction through centralized information management. The researchers also highlighted the importance of privacy, trust, and secure data handling in CRM implementation. The synthesis of these studies indicates that CRM systems play a critical role in organizations that rely heavily on personalized customer interaction and service quality evaluation. For Recovery Iloilo Spa, implementing a centralized customer management module would enable the spa to maintain detailed customer histories, service preferences, appointment records, and feedback data, thereby supporting a more personalized wellness experience while improving data accessibility and customer relationship management efficiency.

The influence of Technology Acceptance Models (TAM) and user behavioral intention in adopting digital systems is explored by Chatterjee et al. (2021) and Alhamad et al. (2026). Chatterjee et al. (2021) explain that organizational users are more likely to adopt AI-integrated CRM systems when they perceive the technology as useful, reliable, and compatible with their operational tasks. Their modified UTAUT framework demonstrates that user satisfaction, system quality, and organizational support significantly influence system adoption behavior. In a related study, Alhamad et al. (2026) examined CRM-driven service technology adoption in the hospitality industry and found that performance expectancy, facilitating conditions, and ease of use are major determinants of digital technology acceptance among service managers. The study further explains that businesses transitioning from traditional operations to digital service environments require systems that minimize complexity while maximizing operational value. Combining these findings suggests that successful system implementation depends not only on technical functionality but also on user confidence and perceived usefulness. For Recovery Iloilo Spa, these theories validate the need for a user-friendly interface and accessible system architecture that allows spa staff, therapists, and management personnel to interact with scheduling, inventory, and reporting modules efficiently without excessive technical difficulty.

The significance of operational analytics and predictive decision-making in modern service businesses is highlighted by Ananda et al. (2024) and RSIS International (2025). Ananda et al. (2024) explain that integrated analytics systems transform raw operational data into predictive insights that improve resource allocation, service forecasting, and inventory planning. Their findings reveal that organizations using analytics-integrated POS and inventory systems are able to anticipate customer demand patterns and optimize operational scheduling more effectively than businesses relying on manual reporting. Similarly, RSIS International (2025) emphasizes that predictive sales forecasting systems improve organizational efficiency by automating analytical processes that support business planning and operational management. The study argues that digital forecasting models enable managers to identify peak service periods, monitor sales performance, and allocate resources strategically. Synthesizing these perspectives demonstrates that analytics platforms function as essential decision-support systems for service-oriented organizations. For Recovery Iloilo Spa, implementing automated analytics and reporting features would provide management with valuable insights regarding appointment trends, therapist productivity, inventory consumption, and sales performance, thereby enabling more informed operational and financial decision-making processes.

Cybersecurity and data protection have become critical components of integrated information systems, especially for businesses handling sensitive customer and financial information. According to Fortinet (2025), organizations implementing cloud-based systems must prioritize authentication security, encrypted communication, and access control mechanisms to protect operational databases from unauthorized access and cyber threats. The report emphasizes that cloud-connected service platforms are increasingly vulnerable to credential theft and information breaches when security measures are poorly implemented. Similarly, the World Economic Forum (2025) explains that digital transformation initiatives often fail when organizations overlook cybersecurity resilience during system deployment. The study stresses that businesses adopting integrated digital platforms must ensure secure system architecture, controlled user access, and continuous monitoring of data integrity. The synthesis of these findings demonstrates that cybersecurity is not merely a technical concern but an operational necessity for organizations managing centralized databases and online transactions. For Recovery Iloilo Spa, implementing secure authentication systems, encrypted customer record storage, and role-based access control would protect sensitive appointment records, transaction histories, and employee information while ensuring that operational data remains accurate, confidential, and protected against unauthorized access.

The relevance of digital transformation in improving operational efficiency among small and medium enterprises is discussed by Tsaani et al. (2026) and Rosad et al. (2026). Tsaani et al. (2026) found that the adoption of Management Information Systems among MSMEs significantly improves operational efficiency through the use of digital POS systems, accounting software, and automated payment technologies. Their qualitative findings reveal that businesses implementing digital tools experience improved transaction management, reduced manual workload, and more organized operational records. In a related study, Rosad et al. (2026) explain that operational CRM systems integrated with self-order and transaction management technologies improve service quality and customer satisfaction by reducing manual order recording and administrative delays. The researchers emphasize that digital ordering systems help businesses improve customer interaction while streamlining internal workflows. Integrating these perspectives indicates that digital transformation enhances both organizational efficiency and service quality in businesses transitioning from manual operations. For Recovery Iloilo Spa, the implementation of a web-based integrated management system would automate appointment scheduling, transaction processing, customer management, and reporting activities, thereby improving workflow efficiency and minimizing the administrative burdens currently associated with manual operational methods.

The integration of Artificial Intelligence (AI) into CRM systems has significantly transformed customer interaction and service management strategies. According to Chagas et al. (2020), machine learning technologies embedded within CRM systems improve customer engagement by analyzing customer behavior, predicting preferences, and automating service recommendations. Their review explains that AI-driven CRM platforms enable organizations to process large volumes of operational and customer data more efficiently, resulting in improved service personalization and customer retention. Likewise, the Journal of Retailing and Consumer Services (2026) identifies several critical success factors influencing AI-powered CRM implementation, including system adaptability, personalized interaction capability, and continuous innovation support. The study further explains that AI-enhanced CRM systems improve organizational decision-making by enabling predictive customer analysis and intelligent service automation. The synthesis of these studies demonstrates that AI integration strengthens customer relationship management by improving both operational efficiency and customer experience. For Recovery Iloilo Spa, incorporating analytics-driven CRM functionalities into the proposed integrated system could help management monitor customer preferences, identify frequent service patterns, and personalize wellness offerings based on customer histories and transaction behavior, thereby enhancing customer satisfaction and long-term client retention.

The importance of software engineering methodologies in ensuring reliable system development is emphasized by Sommerville (2021) and Pressman and Maxim (2020). Sommerville (2021) explains that dependable software systems require careful attention to reliability, maintainability, security, and scalability throughout the development lifecycle. The author emphasizes that service-oriented systems handling customer records and operational transactions must prioritize system availability and data protection to ensure uninterrupted business operations. Similarly, Pressman and Maxim (2020) discuss the significance of structured software engineering processes in developing web-based business systems capable of adapting to changing organizational requirements. Their work highlights the importance of iterative development, modular architecture, and continuous testing in achieving software quality and usability. Combining these perspectives demonstrates that successful information systems depend on both technical robustness and adaptable design strategies. For Recovery Iloilo Spa, applying software engineering principles in the development of the proposed integrated platform would ensure that the system remains secure, scalable, and responsive to operational demands while supporting continuous improvements through testing, maintenance, and user feedback integration.

The role of unified digital platforms in improving organizational coordination and service delivery is explored by Forrester (2024) and Gartner (2023). Forrester (2024) defines unified commerce and integrated operational systems as centralized digital environments that synchronize customer interactions, transactions, scheduling activities, and inventory data across business processes. The study argues that businesses using integrated platforms achieve improved operational visibility and reduced administrative fragmentation. Similarly, Gartner (2023) emphasizes that centralized data architectures eliminate information silos by ensuring that operational records, customer data, and financial transactions are synchronized within a single system environment. This synchronization improves workflow efficiency, reporting accuracy, and managerial decision-making. The synthesis of these findings suggests that integrated digital systems provide organizations with cohesive operational control and improved service management capabilities. For Recovery Iloilo Spa, the implementation of a unified web-based system would allow appointment scheduling, inventory tracking, customer management, payment processing, and analytics reporting to function within one centralized platform, thereby reducing repetitive manual tasks and improving the overall efficiency and quality of spa operations.

The growing integration of hospitality technologies and organizational performance management is examined by Law et al. (2025) and Kilic et al. (2021), who explain that information systems have become fundamental tools for improving operational efficiency and customer satisfaction within service-oriented industries. Law et al. (2025) argue that hospitality and tourism technologies contribute significantly to organizational performance by automating workflows, improving communication efficiency, and enabling data-driven service innovation. Their integrated framework highlights that digital transformation allows organizations to optimize business processes while enhancing customer engagement and operational transparency. Similarly, Kilic et al. (2021) emphasize that selecting an effective hospitality information system requires integrating multiple operational criteria, including usability, data synchronization, and process efficiency. Their study demonstrates that integrated systems improve simultaneous data transfer and service coordination across business operations. Synthesizing these perspectives reveals that hospitality-focused information systems function as strategic management tools capable of transforming traditional service operations into digitally coordinated environments. For Recovery Iloilo Spa, these findings support the implementation of an integrated web-based management system capable of synchronizing appointment scheduling, customer management, inventory monitoring, and transaction processing into a centralized platform that improves both operational reliability and customer service quality.

The influence of digital service transformation and Industry 4.0 technologies on service-oriented businesses is explored by Gaiardelli et al. (2021) and Time et al. (2024). Gaiardelli et al. (2021) explain that Industry 4.0 technologies encourage the evolution of product-service systems by integrating digitalization, automation, and interconnected business services into operational workflows. Their research emphasizes that organizations adopting service-oriented digital systems achieve improved process coordination and customer value creation through real-time information exchange. Complementing this perspective, Time et al. (2024) investigate Software-as-a-Service (SaaS) integration models and conclude that integrating technical and business functions within cloud-based platforms significantly improves operational decision-making and organizational flexibility. The authors further explain that SaaS integration allows organizations to manage business functions across strategic, tactical, and operational levels using centralized digital infrastructures. Combining these findings suggests that integrated digital ecosystems improve both operational continuity and organizational adaptability in modern service industries. For Recovery Iloilo Spa, these studies validate the use of a web-based integrated management system capable of consolidating appointment scheduling, inventory management, sales recording, and customer relationship processes within a scalable and accessible digital environment.

The importance of inter-organizational information systems and digital integration in improving operational coordination is emphasized by Deepu and Ravi (2021) and Stroumpoulis et al. (2021). Deepu and Ravi (2021) argue that integrated information systems enable organizations to coordinate, communicate, and manage operational activities more efficiently by supporting real-time information exchange and centralized decision-making processes. Their study highlights that information system integration enhances organizational responsiveness and reduces delays associated with fragmented operational workflows. Similarly, Stroumpoulis et al. (2021) explain that information systems contribute to sustainable operational management by enabling businesses to monitor processes, control resources, and improve competitive performance through centralized digital coordination. The researchers further emphasize that integrated systems support cost reduction, operational monitoring, and strategic planning by improving access to accurate organizational data. The synthesis of these perspectives demonstrates that integrated information systems are essential for organizations seeking operational transparency and sustainable process management. In relation to Recovery Iloilo Spa, these concepts support the development of a centralized system capable of synchronizing appointment records, customer information, inventory data, and financial transactions to improve overall operational efficiency and management control.

The significance of data-driven service innovation in modern business operations is examined by Schymanietz et al. (2022) and Zhao et al. (2026). Schymanietz et al. (2022) explain that organizations increasingly depend on data-driven service innovation to improve customer experience and organizational decision-making. Their study emphasizes that interconnected digital systems allow businesses to collect and analyze operational data that can be transformed into actionable service improvements and strategic insights. In a related study, Zhao et al. (2026) discuss the emergence of integrated industrial intelligence systems that combine data governance, service orchestration, and knowledge management to improve organizational trustworthiness and operational reliability. The researchers argue that integrated digital governance frameworks support data quality, security, explainability, and organizational efficiency across interconnected systems. Combining these findings indicates that organizations benefit significantly when operational data is transformed into meaningful managerial intelligence through integrated analytics platforms. For Recovery Iloilo Spa, implementing analytics-driven reporting and centralized data management would enable the spa owner and managers to monitor sales trends, appointment patterns, customer preferences, and inventory consumption more effectively while supporting evidence-based operational planning and service enhancement.

The organizational impact of long-term information technology integration in hospitality businesses is explored by Baggio and Sainaghi (2021) and Gratzke et al. (2025). Baggio and Sainaghi (2021) explain that the adoption of information technology within hospitality organizations creates continuous interactions between digital systems and organizational processes, leading to long-term operational transformation. Their study demonstrates that integrated CRM and operational systems gradually reshape workflows, managerial communication, and customer service strategies over time. Similarly, Gratzke et al. (2025) explain that digital transformation shifts organizations away from isolated technological functions toward ecosystem-based operational partnerships that rely on integrated service networks and collaborative digital infrastructures. The authors highlight that modern digital systems support strategic coordination, service innovation, and operational adaptability across organizational environments. The synthesis of these studies suggests that successful digital transformation involves the continuous alignment of technological systems with organizational goals and operational workflows. For Recovery Iloilo Spa, these findings reinforce the importance of implementing an integrated system that evolves alongside business operations while supporting improved coordination between management, therapists, inventory processes, and customer interactions.

The role of system flexibility and customization in improving information system performance is discussed by Niu (2021) and Hidayatullah et al. (2026). Niu (2021) explains that customized service-oriented information systems improve organizational performance by enhancing process flexibility, data adaptability, and system responsiveness. The study highlights that businesses adopting flexible information systems are better able to optimize workflows and allocate resources efficiently in response to operational changes. Similarly, Hidayatullah et al. (2026) argue that Enterprise Resource Planning (ERP) systems integrated with Artificial Intelligence technologies improve sustainability and organizational efficiency through socio-technical system coordination. Their research emphasizes that adaptable ERP architectures support operational integration, predictive decision-making, and long-term organizational resilience. Integrating these perspectives demonstrates that system customization and operational flexibility are essential factors influencing successful digital transformation. For Recovery Iloilo Spa, these studies support the development of a flexible integrated management system capable of adapting to operational requirements such as appointment scheduling adjustments, inventory updates, transaction management, and evolving customer service demands while maintaining system reliability and performance.

The contribution of service management automation to operational efficiency is examined by Sharma et al. (2022) and Deepu and Ravi (2021). Sharma et al. (2022) explain that intent-driven service management systems reduce operational inefficiencies by automating configuration processes, minimizing manual intervention, and supporting zero-touch service deployment environments. Their study emphasizes that automated management systems improve service quality by reducing configuration errors and ensuring faster operational responsiveness. Complementing this perspective, Deepu and Ravi (2021) argue that integrated inter-organizational information systems improve operational coordination through automated communication and synchronized information sharing. The researchers explain that automation technologies reduce delays associated with manual data handling and fragmented workflow management. The synthesis of these findings demonstrates that automation technologies significantly improve operational reliability, service consistency, and managerial efficiency within digitally integrated organizations. For Recovery Iloilo Spa, implementing automated scheduling, inventory tracking, commission computation, and reporting functions would reduce administrative burdens while improving service continuity and operational coordination across the spa's daily activities.

The relationship between digital economy integration and information technology governance is analyzed by Durigan Junior et al. (2024) and Chen and Wu (2024). Durigan Junior et al. (2024) explain that the emergence of digital economy infrastructures and open digital ecosystems has significantly transformed business governance models through the integration of advanced information technologies. Their study identifies strategic IT enabling factors that improve organizational adaptability, data management, and digital collaboration. Similarly, Chen and Wu (2024) argue that integrating management information systems with organizational operations enhances economic performance and operational decision-making by improving workflow synchronization and data accessibility. The researchers emphasize that digital governance structures support organizational transparency, operational monitoring, and business innovation. Combining these perspectives demonstrates that information technology governance frameworks are essential for organizations transitioning toward integrated digital operations. For Recovery Iloilo Spa, these findings support the implementation of a digitally integrated platform capable of improving transaction transparency, customer management, service coordination, and operational governance through centralized information system architecture.

The strategic importance of information systems research and organizational information management is emphasized by the Journal of Management Information Systems and the Journal of Strategic Information Systems. The Journal of Management Information Systems highlights that information systems research focuses on understanding how organizations can effectively develop, organize, and deploy technological systems to manage operational information and support organizational decision-making. Likewise, the Journal of Strategic Information Systems emphasizes that strategic information systems research examines the organizational, managerial, and operational implications of integrating information technologies into business processes. These academic perspectives collectively reinforce the importance of centralized information management and digital strategy in improving organizational efficiency and competitiveness. In the context of Recovery Iloilo Spa, these principles validate the necessity of implementing a strategically designed integrated information system capable of managing customer data, operational workflows, inventory records, scheduling functions, and financial transactions within a unified and reliable digital environment.

The evolution of digital service ecosystems and integrated operational intelligence is explored by Zhao et al. (2026) and Law et al. (2025). Zhao et al. (2026) explain that modern organizations increasingly rely on integrated data-service-knowledge governance systems to ensure operational trustworthiness, security, and intelligent decision-making across digital infrastructures. Their framework emphasizes the importance of centralized governance models capable of supporting privacy, security, explainability, and service integration across interconnected systems. Similarly, Law et al. (2025) discuss how hospitality technologies contribute to operational transformation by enabling automation, communication efficiency, and service innovation within hospitality organizations. The study highlights that integrated technologies support organizational agility and improve business value through digital transformation initiatives. Synthesizing these perspectives indicates that integrated digital ecosystems enable organizations to manage operations more intelligently while supporting sustainable organizational growth and customer-centered service delivery. For Recovery Iloilo Spa, implementing a centralized integrated system with analytics, security controls, customer management, and automated operational processes would strengthen organizational efficiency while supporting long-term digital transformation within the wellness service industry.

The increasing complexity of appointment scheduling systems in service-oriented organizations is explored by Zhu, Liu, and Qi (2024) and Youn, Geismar, and Pinedo (2022). Zhu et al. (2024) explain that modern appointment scheduling systems must move beyond traditional first-come-first-served approaches by incorporating flexible scheduling policies capable of adapting to fluctuating customer demand and operational uncertainty. Their study demonstrates that intelligent scheduling algorithms improve operational efficiency by strategically managing appointment gaps, reducing congestion, and optimizing customer flow. Similarly, Youn et al. (2022) emphasize that integrated scheduling systems improve organizational coordination by balancing customer waiting time, resource utilization, and service continuity within dynamic operational environments. The researchers highlight that scheduling technologies contribute significantly to organizational efficiency by improving planning and coordination processes. Synthesizing these findings demonstrates that modern scheduling systems function as strategic operational tools that improve service quality and workflow management. For Recovery Iloilo Spa, implementing an intelligent appointment scheduling module would help minimize booking conflicts, improve therapist allocation, and ensure smoother coordination between customer reservations and operational readiness, thereby enhancing overall service efficiency and customer satisfaction.

The role of machine learning and predictive analytics in appointment management systems is examined by Ahmadi-Javid et al. (2022) and Zhong, Birge, and Ward (2024). Ahmadi-Javid et al. (2022) explain that integrating machine learning algorithms into scheduling systems enables organizations to predict customer behavior, service durations, and no-show probabilities more accurately, thereby improving scheduling precision and operational planning. Their “predict-then-schedule” framework demonstrates that predictive analytics significantly improves scheduling efficiency by reducing idle time and optimizing resource utilization. Similarly, Zhong et al. (2024) discuss learning-based scheduling strategies in multi-server service environments, emphasizing that adaptive scheduling algorithms improve organizational responsiveness and reduce operational inefficiencies caused by uncertain customer demand. The researchers further explain that learning-driven scheduling systems continuously improve performance by analyzing historical operational data and adjusting service allocation strategies accordingly. The synthesis of these studies demonstrates that predictive scheduling technologies improve both operational flexibility and customer service quality. For Recovery Iloilo Spa, integrating predictive scheduling and analytics into the proposed system would enable management to monitor booking trends, optimize therapist schedules, and improve appointment coordination while minimizing delays and scheduling inefficiencies.

The significance of integrated staffing and routing systems in customer service operations is emphasized by Long, Tezcan, and Zhang (2024) and Huang, Morrice, and Bard (2023). Long et al. (2024) explain that integrated routing and staffing systems improve customer service operations by dynamically matching service requests with available personnel while considering operational capacity and customer waiting behavior. Their study highlights that optimized staffing and routing decisions improve organizational efficiency and service responsiveness in multi-service environments. Similarly, Huang et al. (2023) discuss coordinated scheduling systems that integrate virtual and in-person service operations within complex service networks. The researchers explain that coordinated scheduling frameworks improve organizational flexibility by balancing operational costs, customer waiting times, and resource allocation simultaneously. Combining these perspectives reveals that integrated staffing and scheduling systems significantly improve operational coordination and customer satisfaction in service-based organizations. For Recovery Iloilo Spa, implementing integrated therapist scheduling and availability monitoring would improve workforce coordination, reduce appointment conflicts, and ensure that customer reservations are properly aligned with therapist availability and operational capacity.

The importance of contingency-based scheduling and operational adaptability in integrated systems is explored by Romero-Silva, Santos, and Hurtado-Hernández (2022) and Ge et al. (2022). Romero-Silva et al. (2022) explain that effective scheduling systems must adapt to varying operational conditions, organizational complexity, and resource utilization levels to remain relevant in real-world business environments. Their contingency-based scheduling framework demonstrates that organizations benefit from flexible scheduling strategies capable of responding to operational uncertainties and workflow complexity. Similarly, Ge et al. (2022) emphasize that integrated scheduling systems improve operational coordination by synchronizing multiple operational components within centralized information environments. The researchers highlight that integrated scheduling architectures improve resource utilization and organizational responsiveness through coordinated planning and centralized system management. Synthesizing these findings demonstrates that adaptable and integrated scheduling systems are essential for organizations operating in dynamic service environments. For Recovery Iloilo Spa, adopting a flexible and integrated scheduling approach would allow the spa to respond more effectively to changing appointment demands, therapist availability adjustments, and operational workload variations while maintaining service quality and operational efficiency.

The increasing relevance of AI-integrated customer relationship management systems in improving organizational performance is discussed by Chatterjee et al. (2022) and Law et al. (2025). Chatterjee et al. (2022) explain that AI-enhanced CRM systems improve business-to-customer relationship management by enabling intelligent customer interaction analysis, operational adaptability, and personalized service delivery. Their study highlights that organizations implementing AI-integrated CRM technologies achieve improved customer engagement, operational responsiveness, and strategic decision-making capabilities. Similarly, Law et al. (2025) explain that hospitality and service organizations increasingly rely on integrated digital technologies to improve operational performance, customer satisfaction, and service innovation. The study emphasizes that technology-driven operational environments enable organizations to streamline workflows and improve organizational competitiveness through digital transformation. The synthesis of these perspectives demonstrates that AI-integrated CRM systems significantly improve customer-centered service management and operational efficiency. For Recovery Iloilo Spa, incorporating intelligent CRM and analytics functionalities into the proposed integrated system would help management monitor customer preferences, personalize wellness services, improve customer retention strategies, and strengthen overall service quality through data-driven operational management.

The strategic integration of big data analytics within Customer Relationship Management (CRM) systems is increasingly recognized as a critical factor in improving organizational performance and customer engagement. According to Wamba et al. (2020), big data analytical intelligence strengthens CRM performance by enabling organizations to process large volumes of customer and operational data for more accurate business decision-making. Their study explains that data-driven cultures and advanced analytics capabilities significantly improve customer segmentation, service personalization, and operational responsiveness. Similarly, Sianipar (2025) emphasizes that integrating big data technologies into CRM platforms allows organizations to improve predictive analytics, automate marketing processes, and enhance customer retention strategies. The study further argues that businesses leveraging analytics-driven CRM systems achieve stronger competitive advantages due to improved customer insight generation and operational adaptability. Synthesizing these perspectives demonstrates that big data integration transforms CRM systems into intelligent business management platforms capable of improving operational efficiency and customer satisfaction simultaneously. For Recovery Iloilo Spa, integrating analytics-driven CRM functionalities into the proposed system would support customer history analysis, personalized wellness recommendations, appointment trend monitoring, and service optimization, thereby improving both customer relationship management and operational decision-making.

The contribution of social Customer Relationship Management (social CRM) to customer engagement and information management is explored by Harrigan, Miles, and Roy (2020) and Ramdani, Bouchakour, and Abdellatif (2026). Harrigan et al. (2020) explain that social media technologies significantly improve CRM processes by enabling organizations to strengthen customer interaction, improve communication channels, and gather real-time customer feedback. Their findings suggest that social CRM platforms create stronger engagement and relationship-building opportunities by integrating customer communication into digital management systems. Complementing this perspective, Ramdani et al. (2026) argue that innovation-driven CRM systems improve organizational performance by aligning technological capabilities with customer relationship strategies and market-oriented digital transformation initiatives. The study emphasizes that organizations implementing innovative CRM practices achieve improved service responsiveness and customer satisfaction outcomes. Combining these findings demonstrates that customer engagement technologies play a crucial role in strengthening operational efficiency and service quality in customer-centered organizations. For Recovery Iloilo Spa, integrating customer feedback mechanisms and digital communication channels within the proposed integrated system would improve customer engagement, support service quality monitoring, and provide management with valuable insights for operational improvement and customer retention strategies.

The relationship between CRM technology adoption, customization capability, and strategic alignment is examined by Tsou (2021) and Chatterjee et al. (2021). Tsou (2021) explains that organizations adopting CRM technologies improve operational effectiveness when systems are aligned with organizational goals and customized according to business requirements. The study demonstrates that customization capability significantly enhances CRM effectiveness by enabling organizations to adapt technological functionalities to operational demands and customer expectations. Similarly, Chatterjee et al. (2021) emphasize that organizational users are more likely to adopt AI-integrated CRM systems when the systems demonstrate compatibility, usability, and operational value. Their Meta-UTAUT framework highlights that system quality and user satisfaction strongly influence successful technology adoption within organizations. The synthesis of these studies demonstrates that CRM success depends not only on technical functionality but also on strategic alignment, customization, and user-centered implementation approaches. For Recovery Iloilo Spa, these findings support the development of a flexible and user-friendly integrated management system capable of adapting to operational needs such as appointment scheduling, customer management, inventory monitoring, and reporting functions while ensuring ease of use for staff and management personnel.

The integration of Artificial Intelligence (AI) into customer management and business analytics systems is increasingly transforming organizational performance and competitive advantage. According to Rahman et al. (2025), AI-powered CRM capabilities improve organizational profitability and competitive performance by strengthening marketing ambidexterity, predictive analytics, and customer interaction management. Their study highlights that AI-driven CRM systems improve operational intelligence through advanced data governance, multi-channel integration, and adaptive service management. Similarly, Fabriansyah et al. (2026) explain that integrating Human Resource Information Systems (HRIS) and CRM data within intelligent Decision Support Systems (DSS) improves managerial decision-making and workforce performance monitoring. The researchers emphasize that machine learning-driven DSS architectures enable organizations to generate predictive insights that support operational planning and strategic management. Synthesizing these findings demonstrates that AI-integrated analytics systems significantly improve organizational adaptability, customer management, and operational decision-making capabilities. For Recovery Iloilo Spa, integrating AI-supported analytics and intelligent reporting mechanisms into the proposed system would help management evaluate operational performance, monitor therapist productivity, analyze customer behavior, and improve evidence-based business planning for long-term operational growth.

The mediating role of operational resilience and customer satisfaction in CRM implementation is discussed by Alsmairat et al. (2024) and Ali et al. (2026). Alsmairat et al. (2024) explain that effective CRM dimensions significantly improve customer satisfaction when organizations strengthen operational resilience and service continuity through integrated digital systems. Their findings indicate that CRM practices improve organizational responsiveness and customer loyalty by enhancing communication efficiency, service quality, and operational coordination. Similarly, Ali et al. (2026) emphasize that CRM adoption significantly influences customer loyalty, service quality, and organizational performance when businesses successfully integrate customer management systems with operational processes and performance evaluation strategies. The researchers further explain that user satisfaction and operational integration are essential factors influencing successful CRM implementation. Combining these perspectives demonstrates that CRM systems improve organizational sustainability when integrated with operational resilience strategies and customer-centered service management. For Recovery Iloilo Spa, implementing a centralized CRM and analytics platform would strengthen customer satisfaction by improving appointment coordination, service consistency, operational transparency, and customer relationship management while ensuring smoother operational workflows across the spa’s integrated digital environment.

The role of information systems usage and managerial commitment in improving organizational integration and operational performance is emphasized by Som and Anyigba (2022) and Madaki, Ahmad, and Singh (2024). Som and Anyigba (2022) explain that organizations implementing explorative and exploitative information systems usage achieve improved operational performance when supported by managerial commitment and system integration strategies. Their study highlights that operational integration significantly mediates the relationship between information systems usage and organizational performance by improving workflow coordination and information accessibility. Similarly, Madaki et al. (2024) discuss the implementation of integrated information technologies and explain that organizations adopting system integration frameworks improve service delivery, operational responsiveness, and organizational adaptability. The researchers further emphasize that integrated information systems reduce operational fragmentation and strengthen communication efficiency within digitally transformed environments. Combining these perspectives demonstrates that effective information system integration depends on both managerial support and coordinated technological implementation strategies. For Recovery Iloilo Spa, these findings support the implementation of a centralized integrated management system capable of synchronizing appointment scheduling, customer management, transaction recording, and inventory monitoring while improving operational coordination and management efficiency.

The integration of analytics into enterprise systems has significantly transformed organizational decision-making and operational intelligence in modern service industries. According to Solano and Cruz (2024), analytics-enabled enterprise systems improve strategic planning and operational performance by embedding business intelligence functionalities into ERP and CRM environments. Their systematic review demonstrates that organizations benefit from integrated analytics through improved forecasting, customer insight generation, and operational monitoring capabilities. Similarly, Bandara et al. (2024) explain that integrating big data technologies with Enterprise Resource Planning systems enhances ERP responsiveness by improving data contextualization, processing efficiency, and organizational adaptability. The researchers emphasize that analytics-driven enterprise systems strengthen organizational responsiveness by enabling real-time information processing and operational coordination. Synthesizing these findings demonstrates that analytics integration transforms enterprise systems into intelligent operational platforms capable of supporting strategic decision-making and process optimization. For Recovery Iloilo Spa, integrating analytics functionalities into the proposed system would improve appointment trend analysis, inventory forecasting, customer preference monitoring, and sales reporting while enabling evidence-based operational planning and efficient business management.

The influence of cloud computing and Internet of Things (IoT)-enabled Management Information Systems on organizational efficiency is examined by Alnakshabandi, Aqel, and Kaya (2022) and Madaki et al. (2024). Alnakshabandi et al. (2022) explain that cloud-based MIS platforms improve organizational performance by enabling centralized information management, real-time accessibility, and operational scalability across interconnected digital environments. Their study highlights that integrating IoT and cloud technologies strengthens organizational communication, data accessibility, and operational monitoring capabilities. Similarly, Madaki et al. (2024) argue that integrated information technology environments improve organizational service delivery and operational flexibility through centralized technological infrastructures and coordinated digital operations. The synthesis of these studies indicates that cloud-integrated MIS systems strengthen organizational adaptability and operational efficiency by improving accessibility, coordination, and information sharing across operational processes. For Recovery Iloilo Spa, implementing a cloud-based integrated management system would improve accessibility to appointment records, customer information, transaction histories, and inventory monitoring while supporting efficient and flexible spa operations across different management functions.

The importance of user acceptance and continuous system utilization in information system success is explored by Alshammari and Alshammari (2024) and Pérez Estébanez (2024). Alshammari and Alshammari (2024) explain that system quality, information quality, and user satisfaction significantly influence the continuous intention to utilize digital platforms within organizational environments. Their integrated expectation confirmation and information systems success model demonstrates that usability and perceived usefulness strongly affect long-term technology adoption behavior. Similarly, Pérez Estébanez (2024) emphasizes that CRM implementation success depends on user acceptance, organizational support, and system usability. The study further argues that organizations achieve improved operational performance when digital systems are designed according to user expectations and operational requirements. Combining these findings demonstrates that successful system implementation depends not only on technical functionality but also on user-centered design and sustained user satisfaction. For Recovery Iloilo Spa, these concepts support the development of a user-friendly and accessible integrated system that encourages consistent utilization by spa staff and management while improving operational coordination and service delivery efficiency.

The growing relevance of technostress management within organizational information systems environments is discussed by Nastjuk et al. (2024) and Hasan et al. (2024). Nastjuk et al. (2024) explain that excessive technological complexity and poor system usability contribute to technostress, which negatively affects employee behavior, operational performance, and organizational productivity. Their meta-analysis highlights that system overload, complexity, and continuous technological adaptation significantly influence psychological and behavioral outcomes in organizational information system contexts. Similarly, Hasan et al. (2024) found that AI-integrated Management Information Systems improve organizational efficiency when systems are designed to support user adaptability, decision-making simplicity, and operational convenience. The synthesis of these studies demonstrates that organizations implementing digital systems must prioritize usability, adaptability, and employee support to minimize technostress and improve operational effectiveness. For Recovery Iloilo Spa, ensuring a simple and intuitive system interface would help minimize staff resistance to digital transformation while improving operational efficiency, appointment coordination, and customer management processes within the spa environment.

The role of web-based Management Information Systems in improving operational accuracy and service management is emphasized by Dela Rosa (2022) and Som and Anyigba (2022). Dela Rosa (2022) explains that web-based MIS platforms improve operational efficiency by centralizing records management, automating reporting functions, and enhancing data accessibility within organizational environments. The study highlights that agile development methodologies and secure database management significantly improve the reliability and usability of web-based systems. Similarly, Som and Anyigba (2022) argue that integrated information systems improve operational performance through enhanced information integration, workflow coordination, and managerial support. The researchers emphasize that organizations using integrated systems experience improved operational consistency and reduced inefficiencies associated with manual information handling. Combining these findings demonstrates that web-based MIS platforms strengthen operational reliability and management efficiency through centralized digital coordination. For Recovery Iloilo Spa, implementing a web-based integrated management system would improve appointment management, reporting accuracy, customer record organization, and inventory monitoring while reducing manual operational errors and administrative delays.

The significance of data privacy and secure information management in integrated digital systems is explored by Blijleven, Hoxha, and Jaspers (2022) and Wang et al. (2023). Blijleven et al. (2022) explain that organizations implementing electronic record systems must prioritize secure workflows, operational transparency, and data protection to minimize system workarounds and maintain operational reliability. Their revised sociotechnical framework highlights the importance of usability, workflow integration, and privacy protection within digital information systems. Similarly, Wang et al. (2023) discuss blockchain-based multi-identifier management systems and explain that decentralized and encrypted architectures significantly improve data security, access control, and information integrity within digital platforms. The researchers emphasize that secure data management frameworks strengthen organizational trustworthiness and operational reliability through advanced encryption and decentralized control mechanisms. Synthesizing these findings demonstrates that secure information management is essential for organizations operating within centralized digital environments. For Recovery Iloilo Spa, implementing secure authentication systems, encrypted customer databases, and protected access controls would improve customer privacy, transaction security, and operational data integrity within the proposed integrated management platform.

The strategic value of integrated information systems in improving organizational communication and operational coordination is examined by Madaki et al. (2024) and Solano and Cruz (2024). Madaki et al. (2024) explain that information technology integration strengthens organizational communication and operational performance by enabling synchronized information exchange and centralized workflow coordination. Their findings indicate that integrated digital systems improve service quality and organizational responsiveness by reducing fragmentation across operational processes. Similarly, Solano and Cruz (2024) argue that enterprise systems integrated with analytics technologies significantly improve organizational intelligence and strategic planning through centralized operational data management. The researchers emphasize that integrated analytics environments improve forecasting, reporting, and customer analysis capabilities within digitally transformed organizations. Combining these perspectives demonstrates that integrated information systems improve organizational efficiency through coordinated operational management and centralized business intelligence functionalities. For Recovery Iloilo Spa, implementing an integrated digital platform would improve operational communication between appointment scheduling, customer management, inventory monitoring, and sales reporting functions while supporting coordinated and efficient spa operations.

The relationship between organizational adaptability and integrated digital systems is increasingly recognized as a critical factor influencing operational sustainability. According to Alnakshabandi et al. (2022), cloud-enabled MIS systems improve organizational adaptability by enabling scalable digital operations, centralized information accessibility, and coordinated workflow management. Their research highlights that organizations adopting cloud-based systems achieve improved flexibility and operational continuity within evolving business environments. Likewise, Nastjuk et al. (2024) emphasize that organizations implementing digital systems must balance technological complexity with usability to ensure sustainable technology adoption and operational efficiency. Their study demonstrates that adaptable and user-centered systems improve employee engagement and reduce operational resistance associated with digital transformation initiatives. Synthesizing these findings demonstrates that operational sustainability depends on the successful alignment of system flexibility, usability, and organizational adaptability. For Recovery Iloilo Spa, developing a scalable and adaptable integrated management system would improve operational continuity, support future service expansion, and strengthen the spa’s ability to adapt to changing customer and operational requirements while maintaining efficient digital coordination.

The contribution of centralized digital systems to operational transparency and managerial decision-making is emphasized by Dela Rosa (2022) and Hossain et al. (2024). Dela Rosa (2022) explains that centralized web-based MIS platforms improve operational transparency by automating record management, report generation, and information accessibility within organizational processes. The study demonstrates that integrated digital systems reduce inconsistencies and improve operational accountability through centralized information management. Similarly, Hossain et al. (2024) argue that integrating big data analytics within MIS environments significantly improves business intelligence, operational monitoring, and managerial decision-making through advanced data analysis and real-time information processing. The researchers emphasize that analytics-driven MIS systems strengthen organizational performance by enabling evidence-based planning and operational optimization. The synthesis of these studies demonstrates that centralized digital systems improve managerial effectiveness and operational transparency through integrated information management and analytics functionalities. For Recovery Iloilo Spa, implementing centralized analytics and reporting modules would improve operational visibility, customer service monitoring, sales tracking, and inventory management while supporting more informed and efficient managerial decision-making processes.

The integration of Decision Support Systems (DSS) within Management Information Systems has become increasingly significant in enhancing organizational decision-making and operational efficiency. According to Shrestha, Ben-Menahem, and von Krogh (2021), AI-based decision support systems improve managerial decision-making by augmenting human judgment with data-driven insights and predictive capabilities. Their study emphasizes that AI-enabled DSS enhances decision quality, speed, and consistency, particularly in complex service environments where multiple variables must be considered simultaneously. Similarly, Arnott and Pervan (2020) highlight that modern DSS frameworks integrate analytics, databases, and user interfaces to support semi-structured and unstructured decision-making processes. Their research indicates that DSS improves organizational performance by enabling managers to analyze operational data effectively and generate strategic insights. The synthesis of these findings demonstrates that DSS integration strengthens organizational intelligence and operational planning through data-driven decision support. For Recovery Iloilo Spa, incorporating DSS functionalities into the proposed integrated system would enable management to analyze booking trends, monitor financial performance, and optimize service delivery decisions, thereby improving operational efficiency and strategic planning.

The role of Human-Computer Interaction (HCI) in improving system usability and user experience within digital platforms is examined by Harrison, Flood, and Duce (2020) and Alalwan et al. (2021). Harrison et al. (2020) explain that effective HCI design enhances system usability by focusing on user-centered design principles, intuitive interfaces, and efficient interaction mechanisms. Their study highlights that well-designed interfaces reduce user errors, improve system adoption, and enhance overall user satisfaction. Similarly, Alalwan et al. (2021) emphasize that user experience design significantly influences technology acceptance and system usage behavior, particularly in service-oriented digital systems. The researchers argue that systems with high usability and intuitive interfaces are more likely to achieve sustained user engagement and operational success. Combining these perspectives demonstrates that HCI plays a critical role in ensuring successful system implementation and user adoption. For Recovery Iloilo Spa, designing an intuitive and user-friendly interface would improve staff efficiency, reduce training requirements, and enhance customer interaction with the booking system, thereby supporting effective system utilization and service delivery.

The significance of cybersecurity frameworks in protecting organizational data and ensuring secure system operations is explored by Alshaikh (2020) and Alharbi, Zyngier, and Hodkinson (2021). Alshaikh (2020) explains that cybersecurity governance frameworks improve organizational security posture by establishing policies, risk management strategies, and security controls that protect digital assets from cyber threats. The study highlights that organizations implementing structured cybersecurity frameworks achieve improved data protection, system reliability, and operational continuity. Similarly, Alharbi et al. (2021) emphasize that cybersecurity awareness and organizational culture play a critical role in minimizing security risks and improving information system protection. Their research indicates that employee awareness and security practices significantly influence the effectiveness of cybersecurity measures within organizations. Synthesizing these findings demonstrates that cybersecurity implementation requires both technical controls and organizational awareness to ensure comprehensive protection. For Recovery Iloilo Spa, implementing robust cybersecurity measures such as secure authentication, data encryption, and access control mechanisms would safeguard customer information, financial transactions, and operational data, thereby ensuring system reliability and customer trust.

The application of Agile software development methodologies in information system development is discussed by Dingsøyr, Moe, and Seim (2021) and Conboy and Carroll (2019). Dingsøyr et al. (2021) explain that Agile methodologies improve software development efficiency by promoting iterative development, continuous feedback, and collaborative teamwork. Their study highlights that Agile practices enable organizations to adapt to changing requirements and deliver high-quality systems more efficiently. Similarly, Conboy and Carroll (2019) emphasize that Agile development improves project success by enhancing flexibility, communication, and stakeholder involvement throughout the development process. The researchers argue that Agile methodologies are particularly effective in dynamic environments where system requirements evolve over time. Combining these findings demonstrates that Agile approaches improve system development outcomes through adaptability and continuous improvement. For Recovery Iloilo Spa, adopting Agile development practices would facilitate the iterative development of the integrated system, allowing for continuous refinement of features such as appointment scheduling, CRM, and analytics modules based on user feedback and operational needs.

The importance of database management systems (DBMS) in ensuring data integrity, accessibility, and performance is examined by Elmasri and Navathe (2020) and Coronel and Morris (2021). Elmasri and Navathe (2020) explain that DBMS technologies provide structured mechanisms for data storage, retrieval, and management, ensuring data consistency and reliability within organizational systems. Their study highlights that database normalization, indexing, and transaction management significantly improve system performance and data integrity. Similarly, Coronel and Morris (2021) emphasize that modern DBMS platforms support real-time data processing, security, and scalability, enabling organizations to manage large volumes of operational data efficiently. The synthesis of these studies demonstrates that effective database management is essential for maintaining system reliability and supporting data-driven decision-making. For Recovery Iloilo Spa, implementing a robust database management system would ensure accurate storage of customer records, appointment schedules, transaction data, and inventory information, thereby improving system performance and operational efficiency.

The influence of digital transformation on service quality and organizational performance is explored by Verhoef et al. (2021) and Vial (2021). Verhoef et al. (2021) explain that digital transformation enables organizations to improve service quality by integrating digital technologies into business processes, enhancing customer experiences, and optimizing operational efficiency. Their study highlights that digital transformation initiatives improve organizational agility and competitiveness in rapidly evolving business environments. Similarly, Vial (2021) emphasizes that digital transformation involves the use of digital technologies to fundamentally change business processes, organizational structures, and value creation mechanisms. The research indicates that successful digital transformation leads to improved performance, innovation, and customer satisfaction. Combining these perspectives demonstrates that digital transformation is a key driver of organizational success in modern service industries. For Recovery Iloilo Spa, implementing an integrated digital system would support digital transformation by automating operations, improving customer service, and enhancing overall business performance.

The role of service quality models in evaluating and improving customer satisfaction is discussed by Parasuraman, Zeithaml, and Berry (2020) and Ladhari (2021). Parasuraman et al. (2020) explain that service quality is determined by the gap between customer expectations and perceived service performance, emphasizing dimensions such as reliability, responsiveness, assurance, empathy, and tangibles. Their SERVQUAL model provides a framework for assessing service quality and identifying areas for improvement. Similarly, Ladhari (2021) highlights that service quality measurement is essential for improving customer satisfaction and organizational performance, particularly in service-oriented industries. The study emphasizes that organizations must continuously monitor and improve service quality to maintain competitive advantage. Synthesizing these findings demonstrates that service quality evaluation is essential for enhancing customer satisfaction and business performance. For Recovery Iloilo Spa, integrating customer feedback and service quality monitoring within the system would help management assess service performance, identify improvement areas, and enhance customer satisfaction.

The importance of business process automation in improving operational efficiency and reducing manual workload is explored by Syed, Bandara, French, and Stewart (2020) and Van der Aalst (2021). Syed et al. (2020) explain that robotic process automation (RPA) improves operational efficiency by automating repetitive tasks, reducing human error, and increasing processing speed. Their study highlights that automation technologies enable organizations to streamline workflows and improve productivity. Similarly, Van der Aalst (2021) emphasizes that process mining and automation improve business process transparency, enabling organizations to analyze and optimize workflows based on real data. The synthesis of these studies demonstrates that automation technologies significantly enhance operational efficiency and process optimization. For Recovery Iloilo Spa, implementing automated processes for appointment booking, billing, and inventory management would reduce manual workload, improve accuracy, and enhance overall operational efficiency.

The contribution of Technology Acceptance Model (TAM) in understanding user adoption of information systems is examined by Venkatesh and Bala (2020) and Dwivedi et al. (2021). Venkatesh and Bala (2020) explain that perceived usefulness and perceived ease of use are key determinants of technology acceptance and usage behavior. Their extended TAM model highlights the influence of system characteristics, user experience, and organizational factors on technology adoption. Similarly, Dwivedi et al. (2021) emphasize that user acceptance is critical for the success of digital systems, particularly in organizational contexts where system utilization directly impacts operational performance. The study highlights that user training, system quality, and organizational support significantly influence technology adoption. Combining these findings demonstrates that understanding user acceptance is essential for successful system implementation. For Recovery Iloilo Spa, ensuring system usability, providing staff training, and aligning system features with user needs would improve system adoption and maximize the benefits of the integrated management system.

The role of software engineering best practices in ensuring system quality and maintainability is discussed by Pressman and Maxim (2020) and Sommerville (2021). Pressman and Maxim (2020) explain that software engineering principles such as modular design, testing, and quality assurance improve system reliability and maintainability. Their study highlights that structured development processes ensure that systems meet user requirements and perform effectively. Similarly, Sommerville (2021) emphasizes that software engineering practices such as requirement analysis, system design, and testing are essential for developing high-quality information systems. The synthesis of these studies demonstrates that software engineering best practices are critical for ensuring system performance and long-term sustainability. For Recovery Iloilo Spa, applying software engineering principles in system development would ensure the reliability, scalability, and maintainability of the integrated system, supporting efficient and sustainable spa operations.

Finally, the shift toward "Data-Driven Operational Excellence" is summarized by **Davenport and Ronanki (2018)** and **IRE Journals (2024)**. Davenport and Ronanki (2018) argue that cognitive technologies and automation enable businesses to anticipate financial challenges before they arise, moving beyond mere error reduction to strategic foresight. This aligns with the spa's goal of using its "Integrated System" for long-term growth and professionalization. IRE Journals (2024) conclude that for SMEs, embracing automation is a "strategic move" that leads to improved compliance and reduced operational costs. The synthesis of these final sources provides the overarching justification for the "Recovery Iloilo Spa Integrated System": by synchronizing scheduling, inventory, and financial records into one secure web platform, the spa is not just solving a series of manual problems but is adopting a modern, sustainable business model capable of delivering consistent, high-quality wellness services.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local literature and studies consistently emphasize that the transition from manual to integrated digital information systems is a fundamental requirement for the sustainability and growth of modern service-oriented businesses. Foreign studies underscore the transformative power of web-based scheduling and cloud-integrated management, revealing that such systems not only reduce administrative workloads by up to 40% but also significantly eliminate human error in data entry and record retrieval. The consensus among international researchers is that a unified platform—one that synchronizes front-end bookings with back-end inventory and financial reporting—acts as a "single source of truth," enabling managers to maintain real-time oversight and operational agility that manual methods cannot match.

Local studies further validate these findings within the Philippine context, specifically highlighting how Filipino Small and Medium Enterprises (SMEs) benefit from improved data organization and transaction accuracy. Research in the Philippines points toward a growing "digital divide" where businesses that adopt automated systems for scheduling, inventory, and customer records experience higher levels of client trust and staff morale. Local findings particularly emphasize that Filipino consumers now prefer digital touchpoints, such as online booking and e-wallet integrations, making web-based platforms essential for meeting the expectations of a mobile-first population while ensuring compliance with local administrative and financial reporting standards.

The related literature provides the technical and strategic foundation for this transition, identifying Key Performance Indicators (KPIs), automated reordering points, and centralized Customer Relationship Management (CRM) as the pillars of operational excellence. The literature highlights that digital systems do not merely replace paper; they enhance the "service encounter" by allowing staff to focus on personalized care while the software manages the complexities of scheduling, stock deductions, and financial auditing. Furthermore, the integration of analytics allows for evidence-based decision-making, enabling owners to identify profitable service trends and optimize resources according to actual demand patterns rather than intuition.

The integration of analytics within Enterprise Information Systems (EIS) has become increasingly important in improving organizational intelligence and strategic decision-making. According to Solano and Cruz (2024), modern enterprise systems evolve beyond operational support by embedding analytics capabilities into ERP, CRM, and supply chain management platforms to strengthen organizational responsiveness and managerial efficiency. Their systematic review explains that integrated analytics technologies enable organizations to transform operational data into strategic insights that support forecasting, customer analysis, and process optimization. Similarly, Hossain et al. (2024) argue that integrating big data analytics into Management Information Systems significantly enhances business intelligence by enabling real-time data processing, trend identification, and predictive analysis. The researchers emphasize that analytics-driven MIS platforms improve organizational adaptability and operational monitoring by providing accurate and timely information for decision-making processes. The synthesis of these studies demonstrates that analytics integration transforms traditional management systems into intelligent organizational platforms capable of improving operational efficiency and strategic planning. For Recovery Iloilo Spa, implementing analytics-enabled management modules would allow the spa to monitor appointment patterns, evaluate sales performance, analyze customer preferences, and optimize inventory consumption while supporting data-driven managerial decision-making and operational efficiency.

The strategic integration of predictive analytics within Customer Relationship Management (CRM) systems is increasingly recognized as an essential component of modern business operations. Bhardwaj et al. (2024) explain that predictive analytics integration enhances CRM capabilities by enabling organizations to forecast customer behavior, personalize services, and improve customer retention strategies through advanced data analysis techniques. Their study highlights that predictive CRM technologies improve operational responsiveness and customer engagement by transforming customer data into actionable insights. Likewise, Onyeanuforo (2024) emphasizes that data analytics significantly improves international CRM performance by enabling organizations to deliver personalized customer experiences, optimize communication strategies, and strengthen relationship management across diverse customer environments. The study further argues that analytics-driven CRM systems improve organizational adaptability by providing businesses with deeper customer insights and operational intelligence. Synthesizing these findings reveals that predictive analytics integration strengthens both operational management and customer-centered service delivery within digitally transformed organizations. For Recovery Iloilo Spa, integrating predictive analytics into the proposed CRM platform would help management identify customer preferences, anticipate service demand patterns, personalize wellness offerings, and improve long-term customer retention strategies through data-driven service management.

The growing relevance of Artificial Intelligence-integrated CRM systems in organizational environments is examined by Chatterjee et al. (2021) and Jaruwanakul (2024). Chatterjee et al. (2021) explain that AI-integrated CRM systems significantly influence employee behavior, organizational efficiency, and customer relationship management by improving data analysis, automation, and decision-making processes. Their Meta-UTAUT model demonstrates that CRM quality, system satisfaction, and usability strongly influence organizational adoption and system acceptance. Similarly, Jaruwanakul (2024) found that AI-CRM adoption and big data analytical capability positively influence integration capability, team collaboration, and organizational performance within large enterprises. The study emphasizes that AI-driven CRM systems improve competitive advantage by enhancing operational coordination and customer engagement through intelligent data processing and predictive analytics. Combining these perspectives demonstrates that AI-integrated CRM systems strengthen organizational performance by improving operational intelligence, collaboration, and customer management efficiency. For Recovery Iloilo Spa, integrating AI-supported CRM and analytics functionalities into the proposed system would improve appointment management, customer relationship monitoring, operational coordination, and managerial decision-making while enhancing customer satisfaction and service personalization.

The importance of customization capability and strategic alignment in CRM implementation is emphasized by Tsou (2021) and Chen and Wu (2024). Tsou (2021) explains that CRM technology adoption significantly improves organizational effectiveness when systems are customized according to operational requirements and aligned with business strategies. The study highlights that customization capability enables organizations to optimize workflows, improve CRM effectiveness, and adapt technological functionalities to operational needs. Similarly, Chen and Wu (2024) argue that integrating Management Information Systems with operational and CRM functions improves organizational efficiency by strengthening workflow synchronization, customer engagement, and strategic coordination. Their findings demonstrate that MIS integration contributes significantly to economic efficiency and organizational performance by improving operational transparency and information accessibility. The synthesis of these studies indicates that successful CRM implementation requires both technical adaptability and organizational alignment to maximize operational effectiveness. For Recovery Iloilo Spa, these concepts support the development of a flexible integrated system capable of adapting to operational changes, appointment scheduling demands, inventory management processes, and customer service requirements while maintaining strategic alignment with the spa’s operational goals.

The convergence of big data analytics and CRM practices has significantly transformed organizational approaches to customer management and operational intelligence.

According to Gupta (2024), integrating big data analytics into CRM systems improves customer experience and business growth by enabling organizations to analyze large volumes of customer information and generate actionable operational insights. The study emphasizes that analytics-driven CRM platforms strengthen customer engagement, service personalization, and organizational adaptability by improving decision-making processes. Similarly, Bandara et al. (2023) explain that integrating big data technologies with Enterprise Resource Planning (ERP) systems enhances ERP responsiveness and organizational agility by improving data contextualization, processing efficiency, and operational responsiveness. The researchers argue that organizations integrating big data technologies into ERP systems achieve improved operational coordination and information management capabilities. Combining these findings demonstrates that big data integration significantly strengthens organizational intelligence, customer relationship management, and operational responsiveness within modern digital business environments. For Recovery Iloilo Spa, integrating big data analytics within the proposed management system would improve customer analysis, appointment forecasting, sales monitoring, and inventory management while enabling management to make more informed operational decisions through centralized analytics and reporting functions.

The mediating influence of operational resilience and customer satisfaction within CRM systems is explored by Alsmairat et al. (2024) and Ramdani et al. (2026). Alsmairat et al. (2024) explain that CRM dimensions such as customer orientation, knowledge management, and technology-based CRM significantly influence customer satisfaction when combined with operational resilience strategies. Their study demonstrates that organizations improve customer satisfaction by integrating CRM practices with resilient operational processes that ensure service continuity and effective communication management. Similarly, Ramdani et al. (2026) argue that innovation-driven CRM systems improve organizational performance and customer engagement by aligning digital transformation initiatives with customer relationship strategies. The study highlights that innovative CRM adoption improves operational responsiveness and service quality by enabling organizations to respond more effectively to customer needs and market demands. Synthesizing these findings demonstrates that CRM effectiveness depends on both technological integration and operational resilience strategies that support customer-centered service delivery. For Recovery Iloilo Spa, implementing an integrated CRM platform with customer feedback, appointment coordination, and operational monitoring features would improve customer satisfaction, strengthen service consistency, and support sustainable operational management within the spa environment.

The role of AI-driven business intelligence in strengthening organizational competitiveness is increasingly recognized in modern information systems research. According to Rahman et al. (2025), AI-powered CRM capabilities significantly improve organizational profitability and competitive performance by strengthening predictive analytics, customer interaction management, and marketing adaptability. Their study explains that AI-enhanced CRM systems improve operational intelligence by enabling organizations to process customer and operational data more efficiently for strategic planning purposes. Similarly, Solano and Cruz (2024) emphasize that enterprise systems integrated with analytics technologies improve strategic decision-making by transforming operational data into meaningful business intelligence. The researchers further explain that analytics-integrated enterprise systems support organizational innovation, forecasting, and operational optimization across multiple business processes. Combining these perspectives demonstrates that AI-driven business intelligence systems improve organizational competitiveness through advanced analytics, operational monitoring, and strategic decision-making support. For Recovery Iloilo Spa, integrating AI-supported analytics and reporting capabilities into the proposed integrated system would improve business intelligence functions related to appointment trends, customer preferences, inventory usage, and financial performance monitoring, thereby supporting evidence-based operational planning and management efficiency.

The strategic significance of organizational technology adoption and digital transformation is discussed by Chatterjee et al. (2022) and Tsou (2021). Chatterjee et al. (2022) explain that organizations implementing AI-integrated CRM systems improve customer relationship management performance when digital technologies are combined with organizational adaptability and dynamic operational capabilities. Their study highlights that successful AI-CRM implementation depends on strategic planning, organizational readiness, and the ability to respond to changing operational environments. Similarly, Tsou (2021) found that CRM technology adoption improves organizational effectiveness when customization capabilities and strategic alignment are incorporated into system implementation processes. The researchers emphasize that organizations achieve better operational outcomes when digital systems are aligned with operational requirements and customer management objectives. The synthesis of these findings demonstrates that digital transformation success depends on organizational readiness, adaptability, and strategically aligned technology adoption. For Recovery Iloilo Spa, these concepts support the implementation of a strategically designed integrated management system that aligns appointment scheduling, customer management, analytics, and inventory monitoring with the spa’s operational workflows and service delivery objectives.

The contribution of integrated MIS and CRM systems to organizational efficiency and customer engagement is emphasized by Chen and Wu (2024) and Bhardwaj et al. (2024). Chen and Wu (2024) explain that integrating Management Information Systems with CRM and operational processes improves economic efficiency by streamlining workflows, improving customer interaction management, and strengthening managerial decision-making. Their study highlights that integrated MIS architectures improve organizational coordination and reduce operational redundancies through centralized information management. Similarly, Bhardwaj et al. (2024) argue that predictive analytics integration within CRM systems enhances operational performance by enabling organizations to anticipate customer behavior and optimize customer service processes through advanced analytical capabilities. The synthesis of these perspectives demonstrates that integrated MIS and CRM systems strengthen organizational efficiency by combining operational coordination with intelligent customer management functionalities. For Recovery Iloilo Spa, implementing an integrated MIS and CRM platform would improve appointment coordination, customer engagement, transaction management, and reporting accuracy while supporting more efficient operational workflows and service delivery processes.

The importance of centralized information management and operational intelligence in service-oriented organizations is explored by Hossain et al. (2024) and Gupta (2024). Hossain et al. (2024) explain that integrating big data analytics into Management Information Systems significantly improves business intelligence by enabling organizations to collect, process, and analyze operational information in real time. Their findings indicate that analytics-driven MIS platforms improve decision-making accuracy, operational monitoring, and organizational adaptability through advanced data analysis techniques. Likewise, Gupta (2024) emphasizes that the convergence of CRM practices and big data analytics improves customer experience, operational efficiency, and strategic business planning by enabling organizations to extract actionable insights from customer and transactional data. The researchers argue that centralized analytics systems improve organizational responsiveness and service optimization through intelligent information management. Combining these perspectives demonstrates that centralized data analytics and operational intelligence systems significantly improve organizational coordination and customer-centered service delivery. For Recovery Iloilo Spa, implementing centralized analytics and reporting functionalities within the proposed integrated system would improve operational visibility, customer management, sales analysis, and inventory tracking while supporting efficient and informed business operations.

The integration of Artificial Intelligence (AI)-based Customer Relationship Management systems has significantly transformed how organizations manage customer engagement and operational intelligence. According to Ozay, Jahanbakht, Shoomal, and Wang (2024), AI-based CRM systems improve enterprise information systems by enabling predictive customer analysis, intelligent automation, and personalized service delivery. Their bibliometric and systematic review highlights that AI-integrated CRM platforms strengthen operational efficiency by automating repetitive processes, improving customer segmentation, and enhancing data-driven decision-making capabilities. Similarly, Yoo, Park, and Park (2024) explain that AI-enabled CRM systems contribute to organizational competitive advantage through advanced analytics, machine learning integration, and improved customer interaction management. Their mixed-method study demonstrates that organizations implementing AI-powered CRM technologies achieve stronger operational adaptability and customer satisfaction by improving the accuracy of customer insights and service recommendations. The synthesis of these studies demonstrates that AI-enhanced CRM systems function as intelligent operational platforms that improve organizational performance and customer relationship management simultaneously. For Recovery Iloilo Spa, integrating AI-driven CRM and analytics functionalities into the proposed system would improve customer engagement, automate customer record analysis, personalize wellness services, and strengthen managerial decision-making through intelligent operational insights.

The effectiveness of AI-powered CRM tools in improving customer experience and operational responsiveness is emphasized by Bansal (2024) and Hasan et al. (2024).

Bansal (2024) explains that AI-powered CRM systems significantly enhance customer experience by enabling organizations to predict customer preferences, personalize communication, and automate customer interaction processes. The study further argues that intelligent CRM tools improve organizational competitiveness through improved service responsiveness and customer satisfaction management. Complementing this perspective, Hasan et al. (2024) discuss the impact of Artificial Intelligence on decision-making and Customer Relationship Management within Management Information Systems environments. Their qualitative findings reveal that AI technologies such as machine learning, predictive analytics, and natural language processing improve organizational decision-making by optimizing resource allocation, automating operational processes, and enhancing customer interaction management. Combining these findings demonstrates that AI integration strengthens organizational intelligence, customer relationship management, and operational efficiency within modern service organizations. For Recovery Iloilo Spa, implementing AI-supported CRM functionalities would improve appointment coordination, customer communication, service personalization, and operational planning while supporting more responsive and efficient spa management processes.

The importance of critical success factors in Customer Relationship Management implementation is explored by Pérez Estébanez (2024) and Pappas, Mikalef, Giannakos, and Krogstie (2024). Pérez Estébanez (2024) explains that successful CRM implementation depends on organizational readiness, user acceptance, technological capability, and management support. Using Structural Equation Modeling (SEM), the study demonstrates that CRM effectiveness improves significantly when organizations prioritize system usability, organizational alignment, and customer-oriented operational strategies. Similarly, Pappas et al. (2024) emphasize that digital customer management systems improve organizational agility and customer engagement when supported by adaptive technologies and intelligent information systems. Their research highlights the role of integrated analytics and customer-centered digital strategies in improving organizational performance and operational coordination. Synthesizing these findings indicates that successful CRM implementation requires both technological readiness and strategic organizational alignment to maximize operational efficiency and customer satisfaction. For Recovery Iloilo Spa, these concepts support the development of a user-friendly and strategically aligned integrated system capable of improving customer management, scheduling coordination, operational monitoring, and customer service quality within the spa’s digital operational environment.

The strategic implementation of CRM systems within digitally transformed organizations is examined by the study of configurational CRM implementation perspectives in *Technological Forecasting and Social Change* (2024) and Solano and Cruz (2024). The configurational CRM study explains that CRM implementation success depends on the interaction between organizational processes, customer management strategies, and technological capabilities. The research emphasizes that organizations achieve better operational performance when CRM systems are implemented alongside coordinated organizational and environmental strategies. Similarly, Solano and Cruz (2024) explain that integrating analytics into enterprise systems strengthens organizational intelligence and strategic decision-making by embedding business intelligence functionalities into ERP and CRM platforms. Their systematic review highlights that analytics-integrated systems improve forecasting accuracy, customer insight generation, and operational efficiency through centralized data management. Combining these findings demonstrates that integrated CRM and analytics systems significantly improve operational coordination and strategic business management within digitally transformed organizations. For Recovery Iloilo Spa, implementing a centralized CRM and analytics platform would improve customer service management, operational reporting, appointment tracking, and business intelligence processes while supporting efficient and data-driven spa operations.

The convergence of Enterprise Resource Planning (ERP) systems and CRM technologies has become increasingly important in improving organizational responsiveness and integrated business management. According to Bandara et al. (2024), integrating big data technologies into ERP systems significantly improves ERP responsiveness, operational flexibility, and information processing efficiency. Their empirical investigation demonstrates that organizations integrating analytics and ERP systems achieve better operational coordination through improved data contextualization and intelligent information management. Similarly, Gupta and Agarwal (2024) explain that integrating CRM and ERP insights using advanced hybrid analytical models improves organizational planning, customer analysis, and resource optimization. Their research highlights that integrating operational and customer data supports more informed strategic decision-making and service development. Synthesizing these perspectives reveals that ERP-CRM integration strengthens operational intelligence and organizational efficiency through centralized and analytics-driven information systems. For Recovery Iloilo Spa, integrating appointment management, inventory tracking, transaction recording, and customer relationship functions within a unified ERP-CRM environment would improve operational coordination, reporting accuracy, and resource management while reducing inefficiencies associated with fragmented manual processes.

The contribution of analytics-driven Management Information Systems to organizational adaptability and customer-centered operations is emphasized by Hasan et al. (2024) and Hossain et al. (2024). Hasan et al. (2024) explain that AI-integrated MIS platforms improve decision-making and CRM performance by enabling predictive analytics, workflow automation, and intelligent customer interaction management. Their findings indicate that organizations adopting AI-supported MIS technologies experience improved operational efficiency and customer service responsiveness through automated information processing and advanced analytics. Likewise, Hossain et al. (2024) emphasize that integrating big data analytics within MIS platforms significantly strengthens business intelligence and organizational adaptability. The study demonstrates that analytics-driven MIS environments improve operational monitoring, forecasting, and decision-making by enabling organizations to process real-time operational and customer data efficiently. Combining these findings demonstrates that analytics-driven MIS systems improve both operational intelligence and customer management capabilities within digitally transformed organizations. For Recovery Iloilo

Spa, implementing analytics-enabled MIS functionalities would improve sales analysis, inventory monitoring, customer relationship management, and appointment forecasting while supporting evidence-based operational planning and service optimization.

The role of customer satisfaction and operational resilience within CRM implementation is explored by Alsmairat et al. (2024) and Harrigan, Miles, and Roy (2020). Alsmairat et al. (2024) explain that CRM dimensions such as knowledge management, customer orientation, and technology-based CRM significantly influence customer satisfaction when combined with operational resilience strategies. Their findings suggest that organizations improve customer loyalty and service quality when CRM systems are integrated with resilient operational processes and effective communication management. Similarly, Harrigan et al. (2020) argue that social CRM technologies strengthen customer engagement by integrating digital communication and feedback mechanisms into customer management systems. The researchers emphasize that social CRM platforms improve organizational responsiveness and customer interaction quality through real-time communication and engagement monitoring. The synthesis of these studies demonstrates that CRM effectiveness depends on the integration of customer-centered technologies with operational resilience and communication strategies. For Recovery Iloilo Spa, implementing customer feedback mechanisms and resilient CRM functionalities would improve customer satisfaction, strengthen service quality monitoring, and support continuous operational improvement within the spa's integrated management environment.

The increasing relevance of predictive analytics in customer relationship and operational management is discussed by Bhardwaj et al. (2024) and Gupta (2024). Bhardwaj et al. (2024) explain that predictive analytics integration enhances CRM performance by enabling organizations to forecast customer behavior, optimize service delivery, and improve customer retention strategies through intelligent data analysis. Their comparative analysis demonstrates that predictive CRM systems improve organizational adaptability and customer engagement by transforming customer data into actionable business intelligence. Similarly, Gupta (2024) emphasizes that integrating big data analytics into CRM systems improves customer experience and business growth through advanced information processing and customer insight generation. The study further argues that analytics-driven CRM platforms improve operational responsiveness by enabling organizations to personalize services and strengthen customer interaction management. Synthesizing these findings demonstrates that predictive analytics significantly strengthens CRM effectiveness and operational intelligence within digitally integrated organizations. For Recovery Iloilo Spa, implementing predictive analytics within the proposed CRM system would improve customer preference analysis, appointment forecasting, sales monitoring, and personalized service management while enhancing long-term customer retention and operational efficiency.

The integration of AI-enabled operational intelligence into enterprise systems is increasingly recognized as a strategic advantage for modern organizations. According to Yoo, Park, and Park (2024), AI-enabled CRM systems improve organizational competitiveness by combining machine learning technologies with customer management processes to enhance operational adaptability and strategic decision-making. Their study highlights that organizations implementing AI-driven CRM platforms achieve stronger customer engagement and operational responsiveness through intelligent analytics and automated process management. Similarly, Ozay et al. (2024) explain that AI-based CRM systems strengthen enterprise information systems by enabling predictive customer analysis, intelligent automation, and customer experience optimization. The researchers emphasize that AI integration improves organizational performance by supporting more accurate and efficient decision-making processes. Combining these perspectives demonstrates that AI-enabled operational intelligence systems significantly improve customer relationship management and organizational efficiency through advanced analytics and automation technologies. For Recovery Iloilo Spa, integrating AI-supported operational intelligence into the proposed management system would improve customer interaction management, operational reporting, appointment scheduling, and business planning while supporting efficient and data-driven spa operations.

The strategic role of centralized information systems in improving organizational coordination and service delivery is emphasized by Chen and Wu (2024) and Solano and Cruz (2024). Chen and Wu (2024) explain that integrating Management Information Systems with operational and CRM processes significantly improves organizational efficiency through centralized information management, workflow synchronization, and improved customer engagement. Their findings indicate that integrated MIS environments strengthen managerial decision-making and operational transparency by reducing fragmented workflows and improving access to organizational data. Similarly, Solano and Cruz (2024) argue that analytics-integrated enterprise systems improve strategic planning and operational intelligence by embedding business intelligence capabilities within enterprise platforms. Their systematic review highlights that organizations benefit from centralized analytics systems through improved forecasting, customer analysis, and operational monitoring. The synthesis of these studies demonstrates that centralized information systems significantly improve organizational coordination, operational intelligence, and customer service quality within digitally transformed business environments. For Recovery Iloilo Spa, implementing a centralized integrated system would improve appointment management, transaction processing, customer relationship management, and reporting functions while supporting efficient and coordinated spa operations.

In summary, the combined literature and studies suggest that an integrated system is a strategic asset that improves service quality, protects data integrity, and fosters long-term customer loyalty. Based on these comprehensive findings, the proposed **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently** is designed to address the specific gaps identified in manual spa operations. By integrating automated appointment scheduling, real-time inventory tracking, secure POS transactions, CRM-driven personalization, and data-driven analytics into a single web-based platform, this project aims to modernize Recovery Iloilo Spa's operations, ensuring a resilient, efficient, and professional wellness experience that is aligned with both international standards and local market demands.

CHAPTER 3

METHODOLOGY

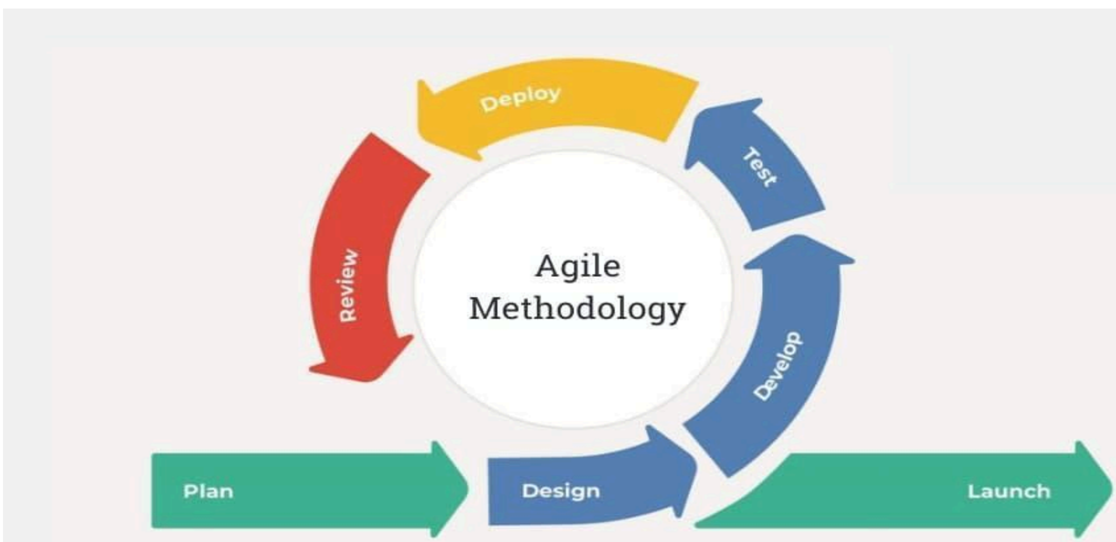
3.1 Research Design

This study will employ a developmental research design in the design, development, implementation, and evaluation of the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics. This research design is appropriate because the primary objective of the study is to develop a web-based information system that will improve the spa's daily business operations through the integration of computerized processes. The proposed system will provide a centralized online platform for managing appointment scheduling, customer accounts and records, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, commission computation, discount and voucher management, customer feedback, operational reports, and business analytics. The system will be developed using a three-tier architecture consisting of the presentation layer, application layer, and data layer to ensure organized system structure, maintainability, and efficient data management.

The developmental research design will guide the researchers through the systematic stages of requirements gathering, analysis of existing business processes, system planning, interface and database design, system development, testing, implementation, and evaluation. Information will be gathered through consultations and interviews with the spa owner, management, receptionists, therapists, and other staff to identify operational requirements, existing challenges, and user needs. Based on the collected information, the researchers will develop the various modules of the proposed system, including appointment management, customer account management, therapist management, inventory management, payment processing, Point-of-Sale (POS), commission management, voucher management, feedback management, security features, and operational reporting and analytics. These modules will be designed to improve operational efficiency, reduce manual workloads, strengthen record management, and support better customer service.

Upon completion of the development phase, the proposed system will undergo comprehensive testing and evaluation to determine whether its features and functions operate according to the identified requirements. The researchers will evaluate the system's quality based on the applicable software quality characteristics (such as those in ISO/IEC 25010, if required by your institution), including functionality, usability, reliability, performance efficiency, and security. Feedback and recommendations from the spa owner, management, receptionists, therapists, and other users will also be gathered to identify areas for improvement and determine whether the proposed system effectively supports appointment management, therapist coordination, inventory monitoring, payment processing, customer record management, operational reporting, business analytics, and overall service delivery at Recovery Iloilo Spa.

3.2 System Development Methodology



The study will use the **Agile Development Model** in developing the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics**. Agile is a software development methodology that emphasizes continuous improvement, iterative development, regular feedback, and close collaboration between developers and stakeholders throughout the system development process. This approach will allow the researchers to work closely with the spa owner, management, receptionists, therapists, and other authorized personnel to ensure that the proposed system will effectively address the operational requirements of Recovery Iloilo Spa. Through continuous testing, evaluation, and user feedback, the researchers will be able to refine system features, resolve identified issues, and implement necessary improvements throughout each development iteration. The proposed web-based system will be developed using a **three-tier architecture** consisting of the **presentation layer**, **application layer**, and **data layer** to promote organized system structure, maintainability, scalability, and efficient data management.

The **Agile Development Model** will be used, consisting of the following phases:

1. **Planning and Requirements Analysis** – This phase will involve identifying, gathering, and analyzing the operational requirements, existing business processes, and challenges encountered by Recovery Iloilo Spa through consultation and coordination with the spa owner, management, HR personnel, receptionists, therapists, and other staff members. The researchers will examine the current workflow and identify issues related to customer registration and account management, email OTP verification, appointment scheduling, therapist assignment and attendance, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, customer record management, therapist commission computation, voucher and discount management, customer feedback collection, security controls, operational reporting, and revenue forecasting. The information gathered during this phase will serve as the basis for identifying the functional and non-functional requirements, determining project objectives, and planning the modules to be included in the proposed system.

2. **System Design Phase** –During this phase, the researchers will prepare the overall architecture and design of the proposed web-based system. The system will be designed using a **three-tier architecture** consisting of the presentation layer, application layer, and data layer to ensure proper separation of system components and efficient processing of information. The researchers will develop the database structure, system architecture, process flow diagrams, navigation structure, user interface layouts, and module designs to ensure that the platform will be organized, responsive, user-friendly, and secure. The design will include modules for customer registration and authentication, customer account management, appointment scheduling and management, therapist management, inventory management, Point-of-Sale (POS), payment processing, refund management, commission computation, discount and voucher management, customer feedback, activity logging, security controls, notifications, operational reports, business analytics, and revenue forecasting.

3. **Development Phase** –During this phase, the researchers will develop the proposed system by coding and integrating each module based on the approved system requirements and design specifications. Development will follow an

iterative approach, allowing each completed feature to be reviewed, tested, and improved before proceeding to the next iteration. The researchers will develop customer registration with email OTP verification, secure user authentication, profile management, shopping cart functionality, appointment scheduling for online, walk-in, home-service, hotel, and partner bookings, therapist assignment and reassignment, therapist attendance monitoring, specialty management, inventory monitoring, Point-of-Sale (POS) transactions, payment gateway integration, refund processing, commission computation, voucher and discount validation, customer feedback collection, role-based access control, gate code verification, PIN verification, activity logging, automated notifications, operational reporting, business analytics, and formula-based revenue forecasting. Continuous collaboration with the spa owner, management, receptionists, therapists, and other intended users will ensure that each module will satisfy operational requirements and business needs.

4. **Testing Phase** – This phase will focus on verifying and evaluating the quality and performance of the proposed system before deployment. The researchers will conduct various testing procedures to determine whether all modules and system functions operate according to the specified requirements. The testing process will evaluate customer registration and authentication, appointment scheduling, therapist assignment, attendance monitoring, inventory management, Point-of-Sale (POS) transactions, payment processing, refund handling, commission computation, voucher validation, customer feedback, notifications, security mechanisms, operational reports, analytics, and revenue forecasting. The system will also be evaluated in terms of **functionality, usability, reliability, performance efficiency, security, and compatibility**. Feedback and recommendations from the spa owner, management, receptionists, therapists, and customers will be gathered to identify necessary improvements and ensure that the proposed system effectively supports daily business operations.

5. **Deployment Phase** – This phase will involve implementing the completed web-based system within Recovery Iloilo Spa for actual operational use. The spa owner, administrators, receptionists, therapists, accounting personnel, and customers will be allowed to access and utilize the appropriate system features based on their assigned roles and permissions. The deployed system will support customer registration, appointment scheduling, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher and discount validation, customer feedback collection, operational reporting, business analytics, and revenue forecasting. During implementation, users will provide additional

comments and recommendations regarding the system's functionality, usability, efficiency, and overall effectiveness. The collected feedback will support further refinements and ensure that the proposed system will improve operational efficiency, strengthen record management, enhance customer service, and support informed decision-making within Recovery Iloilo Spa.

3.3 System Architecture

The **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** will adopt a **three-layer architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. This architectural approach will separate the system into three interconnected components, each having a specific responsibility in processing, managing, and presenting information. By separating the user interface, business logic, and database, the system will become more organized, maintainable, scalable, and secure. This architecture will also simplify future maintenance, updates, and feature enhancements since modifications made to one layer will have minimal impact on the other layers.

The proposed system will also follow a **hub-and-spoke communication model**, wherein the **Application Layer** will serve as the central processing hub between the Presentation Layer and the Data Layer. All requests initiated by users—including customer registrations, appointment bookings, inventory transactions, payment processing, therapist management, report generation, and other operational activities will first be processed and validated by the Application Layer before being stored in or retrieved from the database. Likewise, all information requested by users will pass through the Application Layer before being displayed on the user interface. This approach will ensure proper validation, secure processing, and consistent handling of all system transactions.

The system will be structured into three main layers:.

1. Presentation Layer (User Interface)

The **Presentation Layer** will serve as the part of the system that users directly interact with through a web browser. It will be responsible for displaying information, receiving user input, and providing access to the different features of the proposed system. The interface will be designed to be responsive, organized, user-friendly, and accessible through desktop computers, laptops, tablets, and mobile devices with internet access.

For **customers**, the Presentation Layer will provide pages for customer registration with Email OTP verification, secure login, password recovery, profile management, browsing available spa services and products, shopping cart management, therapist selection or

"Any Available Therapist" option, online appointment booking, walk-in appointment viewing, home-service booking requests, hotel and partner bookings, appointment rescheduling and cancellation, viewing appointment history, payment processing, refund request submission, viewing digital receipts, receiving appointment notifications, and submitting customer feedback and ratings after completed services.

For the **spa owner, administrators, receptionists, accountants, HR personnel, and therapists**, the Presentation Layer will provide dashboards and interfaces for managing appointments, therapist assignments and reassignments, therapist attendance, therapist specialties, customer records, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, commission monitoring, voucher and discount management, customer feedback review, activity logs, operational reports, business analytics, and revenue forecasting. Authorized users will also be able to monitor appointment status, approve bookings, manage products and services, update inventory, verify transactions, and generate reports through the system interface.

The Presentation Layer will be developed using **HTML, CSS, and Vanilla JavaScript** to provide a responsive and interactive user experience across supported web browsers.

The **Application Layer** will serve as the core processing component of the proposed system. It will receive requests from the Presentation Layer, validate user input, execute business rules, process transactions, communicate with the database, and return processed information to the Presentation Layer. This layer will ensure that all business operations are executed accurately, consistently, and securely.

Specifically, the Application Layer will:

- Process customer registration, Email OTP verification, secure authentication, password management, and profile updates.
- Manage appointment scheduling for online, walk-in, home-service, hotel, and partner bookings while validating therapist availability, operating hours, travel-time allowances, booking conflicts, duplicate reservations, and overbooking.
- Process appointment approvals, cancellations, rescheduling, therapist assignments and reassignments, appointment status updates, and automated notifications.
- Manage therapist attendance, therapist specialties, workload balancing, automatic therapist rotation, and performance monitoring.

- Process shopping cart transactions, Point-of-Sale (POS) transactions, payment processing through supported payment methods, refund requests, receipt generation, and transaction verification.
- Automatically calculate therapist commissions based on completed services, service categories, transaction records, multi-person sessions, and applicable commission rules.
- Monitor inventory by updating stock quantities whenever products are used, sold, restocked, or adjusted while identifying low-stock items requiring replenishment.
- Manage customer records, appointment history, service preferences, transaction history, voucher usage, and feedback records.
- Validate discounts and vouchers, including Senior Citizen, PWD, Employee, Celebration Discounts, Gift Certificates, and promotional vouchers while preventing duplicate redemption and unauthorized usage.
- Process customer ratings, comments, and feedback linked to completed appointments and assigned therapists.
- Implement security mechanisms including role-based access control, hidden administrative login with Gate Code verification, PIN verification for sensitive transactions, activity logging, and secure session management.
- Generate operational reports, dashboards, business analytics, sales summaries, therapist productivity reports, inventory reports, POS readings, cash reconciliation reports, and formula-based revenue forecasts.
- Control all communication between the Presentation Layer and the Data Layer to ensure that every transaction is validated, processed securely, and recorded accurately.

The Application Layer will be developed using **PHP**, which will implement the business logic and facilitate communication between the user interface and the **MariaDB** relational database.

3. Data Layer (Database)

The **Data Layer** will serve as the centralized repository of all information managed by the proposed system. It will be responsible for storing, organizing, updating, retrieving, and protecting all operational data required by Recovery Iloilo Spa. The database will ensure data integrity, consistency, security, and efficient retrieval while supporting business operations, reporting, and analytics.

The Data Layer will store:

- Customer account information, login credentials, Email OTP verification records, user roles, and profile information.

- Appointment records including booking details, selected services, assigned therapists, schedules, booking status, service locations, appointment history, and add-on services.
- Therapist information including attendance records, specialties, assignments, schedules, commission records, and performance-related transaction history.
- Customer records including contact information, service preferences, transaction history, feedback records, and voucher usage.
- Inventory records including products, supplies, stock quantities, stock movements, restocking history, and low-stock monitoring.
- Point-of-Sale (POS) transactions, payment records, refund transactions, payment methods, digital receipts, discounts applied, and complete transaction history.
- Voucher records, gift certificates, promotional campaigns, redeemed discounts, eligibility validation, and redemption history.
- Customer feedback, ratings, comments, and service evaluations linked to completed appointments and therapists.
- Notifications, appointment confirmations, payment confirmations, and email notification history.
- Security records including activity logs, PIN verification records, role permissions, and audit trails.
- Operational reports, business analytics, sales summaries, inventory reports, therapist productivity reports, cash reconciliation records, and formula-based revenue forecasting data.

The proposed system will utilize **MariaDB** as its relational database management system to ensure efficient data storage, secure information management, reliable transaction processing, and fast retrieval of records. The centralized database will support approximately **30 core database tables**, allowing the system to maintain organized business information while supporting the operational, reporting, and analytical requirements of Recovery Iloilo Spa.

3.4 Tools and Technologies Used

The development of the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will utilize various software tools and technologies to support the design, development, implementation, and maintenance of the proposed web-based system. These technologies will work together to provide a responsive user interface, efficient business logic, secure data management, reliable transaction processing, and organized operational reporting.

- **Frontend: HTML, CSS, and Vanilla JavaScript -** HTML (HyperText Markup Language), CSS (Cascading Style Sheets), and Vanilla JavaScript will be used to develop the Presentation Layer of the proposed system. HTML will provide the structure and content of the web pages, CSS will define the visual appearance, layout, and responsiveness of the user interface, while Vanilla JavaScript will add interactivity and dynamic functionality to the system. These technologies will enable the development of a responsive, organized, and user-friendly interface that can be accessed through desktop computers, laptops, tablets, and mobile devices using a web browser. The frontend will support customer registration, appointment booking, shopping cart management, therapist selection, payment processing, dashboard visualization, notifications, and other interactive system features.

- **Backend: PHP -** PHP will be used as the primary server-side programming language for developing the Application Layer of the proposed system. It will process business logic, validate user input, manage user authentication, handle appointment scheduling, therapist assignment, inventory management, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher validation, customer feedback, operational reports, business analytics, and revenue forecasting. PHP will also facilitate secure communication between the user interface and the MariaDB database while implementing security features such as role-based access control, Gate Code verification, PIN authentication, session management, and activity logging.

- **Database Management System: MariaDB -** MariaDB will serve as the relational database management system of the proposed Recovery Iloilo Spa Integrated System. It will store, organize, retrieve, and manage all operational data required by the system, including customer accounts, Email OTP verification records, appointments, therapist assignments, attendance records, inventory information, Point-of-Sale (POS) transactions, payment records, refund transactions, customer feedback, voucher and discount records, commission records, activity logs, notifications, operational reports, business analytics, and revenue forecasting data. MariaDB will ensure data integrity, security, consistency, and efficient retrieval of information while supporting approximately thirty core database tables within the proposed system.

- **Integrated Payment Gateway: PayMongo Payment Gateway -** The proposed system will integrate an online payment gateway to support secure electronic transactions. The payment gateway will enable customers to pay for appointments using supported digital payment methods such as GCash, QR Ph, and debit or credit cards. It will also facilitate payment verification, transaction recording, digital receipt generation, and secure communication between the system and electronic payment services.

- **Email Service -** An email service utilizing PHPMailer will be integrated into the proposed system to support automated email notifications. This service will be responsible for sending Email OTP verification codes during customer registration, appointment confirmations, booking updates, payment notifications, password recovery messages, and other system-generated communications to registered users.

- **Development Environment: Visual Studio Code -** Visual Studio Code (VS Code) will be used as the primary Integrated Development Environment (IDE) during system development. It provides features such as syntax highlighting, intelligent code completion, debugging tools, extension support, integrated terminal access, and Git version control integration. These capabilities will assist the researchers in writing, testing, debugging, and maintaining the source code more efficiently throughout the software development process.

- **Web Browser -** Modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari will be used for testing and accessing the proposed web-based system. These browsers will allow users to utilize the system without installing additional software, ensuring accessibility across different devices and operating systems.

- **Three-Tier Architecture -** The proposed system will be developed using a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. This architectural design will separate the user interface, business logic, and database management into independent layers to improve maintainability, scalability, security, and overall system performance.

3.5 Data Gathering Techniques

The researchers will use the following data gathering techniques to identify the operational requirements of Recovery Iloilo Spa, analyze the existing business processes, and gather the information necessary for the design, development, implementation, and evaluation of the proposed **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics**.

- **Survey and Questionnaire -** The researchers will prepare and distribute structured questionnaires to the spa owner, management, receptionists, therapists, and other authorized staff members to collect information regarding the current operational practices, existing problems, user requirements, and suggested system improvements. The survey will gather measurable data related to appointment scheduling, customer account management, therapist assignment, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, commission computation, voucher and discount management, customer feedback, operational reporting, security requirements, and business analytics. The collected responses will serve as one of the primary bases for identifying the functional and non-functional requirements of the proposed system.

- **Observation of Current Operations** – The researchers will conduct direct observation of the daily operations of Recovery Iloilo Spa to obtain firsthand information regarding the current workflow and operational procedures. The observation will focus on customer registration, appointment scheduling, therapist assignment, attendance monitoring, inventory management, payment processing, Point-of-Sale (POS) transactions, commission computation, voucher validation, customer service, and report preparation. This technique will enable the researchers to identify delays, repetitive tasks, manual errors, bottlenecks, and other operational inefficiencies that the proposed system will address.

- **Interview –** The researchers will conduct interviews with the spa owner, management, HR personnel, receptionists, therapists, and other authorized staff members to obtain detailed information regarding the operational processes, business policies, user expectations, and specific system requirements. The interviews will provide a deeper understanding of existing challenges involving appointment management, therapist scheduling, payment procedures, inventory control, commission computation, promotional discounts, customer feedback, reporting practices, and overall business operations. Information gathered during the interviews will help validate the data collected through surveys and observations.

- **Review of Existing Systems** – The researchers will review existing spa management systems, appointment booking systems, and other related web-based information systems to examine their features, workflows, user interfaces, and best practices. This technique will enable the researchers to identify useful functionalities, effective design approaches, and operational strategies that may be adapted and incorporated into the proposed Recovery Iloilo Spa Integrated System while ensuring that the developed system addresses the specific operational requirements of the client.

- **Document and Record Review** –The researchers will review existing business documents and operational records maintained by Recovery Iloilo Spa to understand the current methods of data management and transaction processing. Documents such as appointment records, customer information, inventory lists, therapist schedules, attendance records, commission records, sales reports, payment records, voucher records, promotional activities, and other operational documents will be examined to identify existing data structures, workflow processes, and areas requiring improvement. The information obtained through document review will assist the researchers in designing an organized database structure, accurate system modules, and reliable reporting features for the proposed system.

3.6 Testing Procedures

To ensure that the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** functions properly and satisfies the operational requirements of Recovery Iloilo Spa, the researchers will conduct several testing procedures throughout the development process. These testing procedures will help identify and correct system errors, verify that all modules operate according to the specified requirements, improve system performance, and ensure that the proposed system is reliable, secure, and suitable for actual implementation.

- **Unit Testing** – Each module of the proposed system will be tested individually to verify that its functions operate correctly before being integrated with other modules. The researchers will evaluate customer registration and Email OTP verification, user authentication, appointment scheduling, therapist assignment and attendance, shopping cart, inventory management, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher validation, customer feedback, notifications, operational reports, business analytics, and revenue forecasting. Unit testing will ensure that every module performs its intended functions accurately and independently.

- **System Testing** – After all system modules have been integrated, the researchers will conduct system testing to evaluate the proposed system as a complete web-based platform. This testing procedure will verify the interaction among the Presentation Layer, Application Layer, and Data Layer, ensuring that information flows correctly throughout the system. It will also confirm that all integrated modules—including customer account management, appointment management, therapist management, inventory monitoring, POS transactions, payment processing, refund handling, commission computation, voucher management, customer feedback, security mechanisms, notifications, operational reports, analytics, and revenue forecasting—function together properly without errors or inconsistencies.

- **User Acceptance Testing (UAT)** – The proposed system will undergo User Acceptance Testing (UAT) with the participation of the spa owner, administrators, receptionists, therapists, accounting personnel, and selected customers. The purpose of this testing is to determine whether the system satisfies user requirements, supports actual business operations, and provides a user-friendly and efficient experience. Participants will evaluate the system based on its functionality, ease of use, accuracy, convenience, and overall effectiveness in improving appointment scheduling, customer management, therapist coordination, inventory control, payment processing, reporting, and other operational activities.

- **Performance Testing** – Performance testing will be conducted to evaluate the system's responsiveness, stability, and efficiency while handling multiple users, concurrent appointments, continuous Point-of-Sale (POS) transactions, inventory updates, report generation, and other operational activities. The researchers will determine whether the system can maintain acceptable response times and reliable performance under normal operating conditions while processing large volumes of business data.

- **Security Testing (Optional)** – Security testing will be conducted to verify that the proposed system adequately protects user accounts, confidential business information, and customer records from unauthorized access. The researchers will evaluate authentication mechanisms, Email OTP verification, role-based access control, Gate Code verification, PIN verification, session management, activity logging, payment processing security, and database access controls. This testing will ensure that sensitive information is handled securely and that only authorized users can access restricted system functions.

- **Final Evaluation** - After all testing activities have been completed, the researchers will evaluate the overall performance of the proposed system to determine whether it meets the operational requirements of Recovery Iloilo Spa. The evaluation will focus on the system's **functionality, usability, reliability, performance efficiency, and security**, as well as its ability to improve appointment management, therapist coordination, inventory monitoring, customer record management, payment processing, operational reporting, business analytics, and overall service delivery. Feedback and recommendations gathered from the spa owner, management, receptionists, therapists, accounting personnel, and customers will be used to identify necessary improvements before the final implementation of the proposed system.

3.7 Evaluation Criteria

The developed **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** will be evaluated to determine its overall quality, performance, reliability, and usefulness in improving the daily operations of Recovery Iloilo Spa. The evaluation will assess whether the proposed system effectively satisfies the operational requirements of the establishment and provides an organized, efficient, and secure platform for managing spa services.

- **Functionality** – This criterion will evaluate whether the proposed system performs all intended functions correctly and completely. It will determine whether the different modules including customer registration, appointment scheduling, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher and discount management, customer feedback, notifications, operational reports, analytics, and revenue forecasting operate according to the specified requirements. The system should produce accurate outputs and successfully complete user transactions without major errors.

- **Accuracy** – This criterion measures how correctly the system processes, stores, and displays information. It evaluates the correctness of appointment schedules, inventory records, transaction details, commission computations, customer information, and generated reports. Accurate information helps reduce errors and supports better daily operations.

- **Usability** – This criterion will determine how easy the proposed system is to learn, understand, and operate for customers, receptionists, therapists, administrators, management, and other authorized users. It will evaluate the clarity of menus, forms, navigation, instructions, dashboards, and overall user interface design. A user-friendly system will allow users to complete their tasks efficiently with minimal training and confusion.

- **Reliability** – This criterion will evaluate the ability of the proposed system to operate consistently and continuously without unexpected failures, crashes, or interruptions. It will determine whether appointments, transactions, reports, notifications, and other system functions can be processed accurately during regular business operations while maintaining stable performance and dependable service.

- **Performance Efficiency** – This criterion will assess how efficiently the proposed system performs its operations while utilizing available system resources. It will evaluate the speed of appointment processing, payment transactions, inventory updates, report generation, analytics processing, database retrieval, and other system functions. An efficient system should respond quickly while maintaining smooth performance during normal daily operations.

- **Security** – This criterion will evaluate the effectiveness of the security mechanisms implemented within the proposed system to protect confidential business information and customer data. It will assess user authentication, Email OTP verification, role-based access control, Gate Code verification, PIN authentication, activity logging, payment security, and database protection. The system should ensure that only authorized users can access restricted functions while maintaining the confidentiality, integrity, and security of stored information.

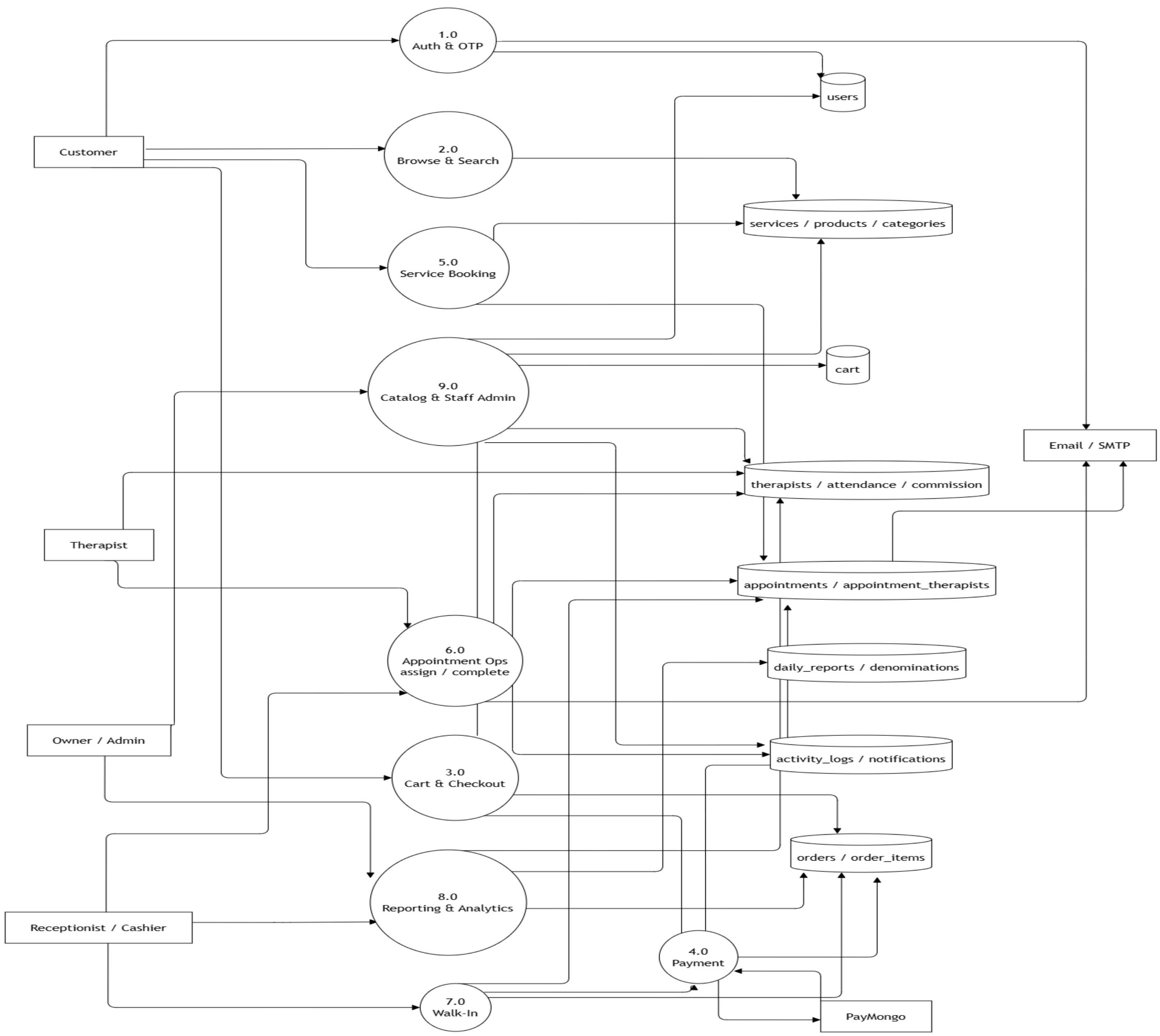
- **Maintainability** – This criterion will evaluate how easily the proposed system can be modified, updated, corrected, and maintained after implementation. It will assess whether the three-tier architecture, organized source code, database structure, and modular system design support future enhancements, bug fixes, and feature improvements without significantly affecting other components of the system.

- **Feedback Integration** – This criterion evaluates how effectively the system collects, stores, and manages customer feedback and ratings. It checks whether customer comments and evaluations are properly recorded and made available for review by management. This information can help identify areas for improvement and support better service quality.

- **Overall User Satisfaction** – This criterion will evaluate the overall satisfaction of the spa owner, management, receptionists, therapists, accounting personnel, and customers regarding the proposed system. It will determine whether users believe that the system improves appointment management, therapist coordination, inventory monitoring, payment processing, customer service, operational reporting, business analytics, and the overall efficiency of Recovery Iloilo Spa.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 1)



1.0 Authentication and OTP

- Customers, therapists, receptionists, and administrators will submit their registration or login credentials to gain authorized access to the Recovery Iloilo Spa Integrated System according to their assigned user roles.
- The system will validate the submitted credentials against **D1: Users**, verify the account status and assigned permissions, and establish a secure session that will control access to authorized system functions.
- A one-time password (OTP) will be generated and transmitted through the **Email / SMTP** service to verify the identity of the user before access to the system will be granted.

2.0 Browse and Search

- Customers will browse the spa catalog and search for available wellness services, products, and service categories offered by Recovery Iloilo Spa.
- The system will retrieve the requested information from **D2: Services, Products, and Categories** and will display updated details regarding service descriptions, pricing, session duration, available products, and service classifications.
- The process will allow customers to review accurate service information before proceeding with appointment booking or purchasing spa products.

3.0 Cart and Checkout

- Customers will select spa services and retail products and will place their chosen items into their shopping cart before confirming their purchase or reservation.
- The system will record the selected items in **D3: Cart** and, after checkout has been confirmed, will create the corresponding transaction in **D4: Orders and Order Items** for further processing.
- The completed transaction will then be forwarded to the Payment process so that the selected payment method can be processed and the customer's order can be finalized.

4.0 Payment

- The system will receive payment requests originating from online checkout transactions and walk-in transactions processed by the receptionist or cashier.
- Payment requests for online transactions will be transmitted securely to **PayMongo**, and the payment confirmation returned by the gateway will be used to update the payment status and transaction details stored in **D4: Orders and Order Items**.
- After successful payment processing, the system will record the completed transaction and will send payment confirmations, receipts, or related notifications through the **Email / SMTP** service whenever applicable.

5.0 Service Booking

- Customers will submit appointment requests by selecting their preferred spa service, appointment schedule, therapist preference, number of persons, and other booking details required by the establishment.
- The system will retrieve service information, pricing, therapist availability, and service categories from **D2: Services, Products, and Categories** to validate the booking request and calculate the applicable service charges.
- Confirmed booking information will be stored in **D5: Appointments and Appointment Therapists**, where the reservation will remain pending until it is reviewed and processed by the receptionist or authorized staff.

6.0 Appointment Operations

- Receptionists will review pending appointments, approve or decline booking requests, assign or reassign therapists, update appointment status, and mark appointments as completed once the requested services have been rendered.
- Therapists will submit their attendance and duty information, while the system will update **D6: Therapists, Attendance, and Commission** to record attendance, therapist assignments, and the commissions earned from completed services.
- Throughout the appointment process, the system will continuously update **D5: Appointments and Appointment Therapists**, generate notifications through the **Email / SMTP** service whenever necessary, and record administrative activities in **D8: Activity Logs and Notifications** to maintain an accurate audit trail.

7.0 Walk-In

- Receptionists or cashiers will encode appointment information for customers who will avail themselves of spa services without making prior online reservations.
- The system will create the corresponding appointment record in **D5: Appointments and Appointment Therapists** and simultaneously generate the related transaction in **D4: Orders and Order Items** for proper sales recording.
- The completed walk-in transaction will then be forwarded to the Payment process so that payment can be processed using the same workflow applied to online transactions, ensuring consistency in transaction recording.

8.0 Reporting and Analytics

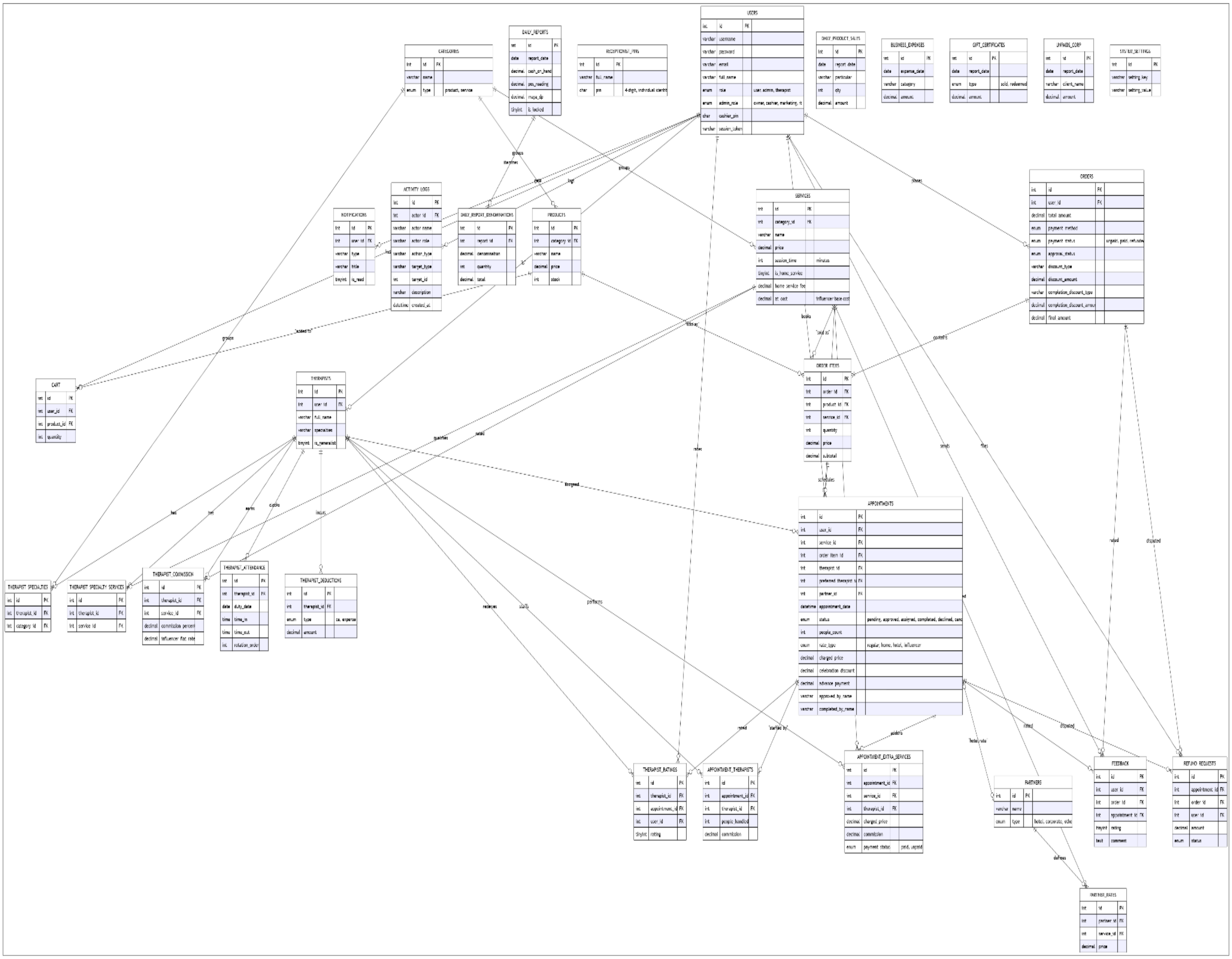
- Receptionists will prepare daily cash counts, cash denomination records, and end-of-day summaries, while the Owner or Administrator will review the consolidated operational performance of the spa through the reporting dashboard.
- The system will store daily cash reconciliation records in **D7: Daily Reports and Denominations** and will retrieve operational data from **D4: Orders and Order Items**, **D5: Appointments and Appointment Therapists**, and **D6: Therapists, Attendance and Commission** to generate comprehensive reports.
- The reporting and analytics module will generate sales reports, appointment reports, inventory movement summaries, therapist productivity reports, commission reports, customer activity reports, payment summaries, voucher

utilization reports, and operational analytics. It will also produce short-term revenue forecasts based on recorded historical transactions and allow management to filter, view, and export reports into Excel and PDF formats. These analytics will provide management with timely insights that will support operational monitoring, performance evaluation, and informed business decision-making.

9.0 Catalog and Staff Administration

- The Owner or Administrator will manage the spa catalog by adding, updating, or maintaining available services, products, categories, pricing, inventory information, therapist profiles, specialties, commission rates, attendance records, and user accounts.
- The system will store and update catalog information in **D2: Services, Products, and Categories**, maintain therapist information within **D6: Therapists, Attendance, and Commission**, and manage user accounts and access permissions through **D1: Users**.
- All administrative modifications performed within the system will be recorded in **D8: Activity Logs and Notifications** to maintain accountability, preserve an accurate audit trail, and support secure management of system activities.

Entity Relationship Diagram (ERD)



Entity Attributes

- users (id, username, password, email, full_name, role, admin_role, cashier_pin, session_token)
- therapists (id, user_id, full_name, specialties, is_generalist)
- receptionist_pins (id, full_name, pin)
- categories (id, name, type)

- services (id, category_id, name, price, session_time, is_home_service, home_service_fee, at_cost)
- products (id, category_id, name, price, stock)
- cart (id, user_id, product_id, quantity)
- orders (id, user_id, total_amount, payment_method, payment_status, approval_status, discount_type, discount_amount, completion_discount_type, completion_discount_amount, final_amount)
- order_items (id, order_id, product_id, service_id, quantity, price, subtotal)
- appointments (id, user_id, service_id, order_item_id, therapist_id, preferred_therapist_id, partner_id, appointment_date, status, people_count, rate_type, charged_price, celebration_discount, advance_payment, approved_by_name, completed_by_name)
- appointment_therapists (id, appointment_id, therapist_id, people_handled, commission)
- appointment_extra_services (id, appointment_id, service_id, therapist_id, charged_price, commission, payment_status)
- therapist_specialties (id, therapist_id, category_id)
- therapist_specialty_services (id, therapist_id, service_id)
- therapist_commission (id, therapist_id, service_id, commission_percent, influencer_flat_rate)
- therapist_attendance (id, therapist_id, duty_date, time_in, time_out, rotation_order)
- therapist_deductions (id, therapist_id, type, amount)
- therapist_ratings (id, therapist_id, appointment_id, user_id, rating)
- partners (id, name, type)
- partner_rates (id, partner_id, service_id, price)
- feedback (id, user_id, order_id, appointment_id, rating, comment)
- refund_requests (id, appointment_id, order_id, user_id, amount, status)
- notifications (id, user_id, type, title, is_read)
- activity_logs (id, actor_id, actor_name, actor_role, action_type, target_type, target_id, description, created_at)
- daily_reports (id, report_date, cash_on_hand, pos_reading, maya_dp, is_locked)
- daily_report_denominations (id, report_id, denomination, quantity, total)
- daily_product_sales (id, report_date, particular, qty, amount)
- business_expenses (id, expense_date, category, amount)
- gift_certificates (id, report_date, type, amount)
- unpaids_corp (id, report_date, client_name, amount)
- system_settings (id, setting_key, setting_value)

Relationships

- A **User** will create an account in the system in order to access the features corresponding to the assigned role, which may be a customer, therapist, receptionist, or administrator. The administrative role will further identify whether the account belongs to the owner, cashier, marketing personnel, or information technology staff, while the session token will maintain secure access throughout system usage.

- A **User** account may optionally be linked to one **Therapist** record. This relationship will allow therapists to authenticate using their user account while maintaining a separate professional profile containing therapist information, specialties, and assignment details.
- A **User** will be able to schedule multiple **Appointments** for spa services. Each appointment will store the selected service, appointment date and time, preferred therapist, assigned therapist, partner information, number of persons, pricing details, discounts, advance payment, booking status, and approval information.
- A **User** will be able to place multiple **Orders** involving spa services and wellness products. Each order will record the payment method, payment status, approval status, applied discounts, total amount, and final payable amount for proper transaction management.
- A **User** will be able to add and manage selected **Products** inside the **Cart** before confirming checkout. The cart will temporarily store the selected products together with their quantities, allowing customers to review or modify their selections before completing the purchase.
- A **User** will receive **Notifications** regarding appointment confirmations, payment updates, refund requests, promotional announcements, and other important system activities. Each notification will store its type, title, and read status to help users monitor system-generated messages.
- A **User** will be able to submit **Feedback** after completing a service transaction by providing a rating and written comment regarding the quality of service and overall customer experience. The feedback record will be linked to the corresponding appointment and order for proper reference.
- A **User** may submit **Refund Requests** for cancelled appointments or payment-related concerns. Each refund request will record the corresponding appointment, order, requested amount, and processing status to support organized refund management.
- A **User** will generate **Activity Logs** whenever important administrative actions are performed within the system. Each activity log will record the actor's identity, role, action performed, affected record, description, and timestamp to provide accountability and auditing.
- **Categories** will organize both **Services** and **Products** to simplify classification, searching, filtering, and management within the system. Each category will identify whether it belongs to a service classification or a product classification, allowing one category to contain multiple services or products.

- **Categories** will also be connected to **Therapist Specialties** to identify the service categories in which therapists are qualified to perform treatments.
- **Services** will be connected to **Appointments** because every appointment will require at least one selected spa service. Service information will also be linked to **Order Items** so that booked treatments can be recorded as part of customer transactions.
- **Services** will be connected to **Therapist Commission** records to determine therapist earnings based on either commission percentages or predefined flat rates assigned to particular services.
- **Services** will also be connected to **Therapist Specialty Services** to identify the individual services that each therapist will be authorized and qualified to perform.
- **Services** will be associated with **Partner Rates** to support customized pricing agreements established for hotel partners, corporate clients, and other affiliated organizations.
- **Products** will be connected to the **Cart** and **Order Items** so that product selections, purchased quantities, inventory movement, and sales transactions can be properly recorded and monitored.
- **Orders** will contain multiple **Order Items**, representing the individual services and products included within a customer transaction. Each order item will store the corresponding product or service, quantity, unit price, and subtotal for accurate billing.
- **Order Items** may generate corresponding **Appointments** whenever spa services are included within a customer order. This relationship will establish a connection between the payment transaction and the scheduled spa session.
- **Appointments** may include additional treatments through **Appointment Extra Services**. These records will store the additional service selected, assigned therapist, charged amount, commission earned, and payment status of every add-on service rendered during the appointment.
- **Appointments** will be assigned to one or more therapists through **Appointment Therapists**, allowing multiple therapists to participate in servicing group appointments. Each assignment will record the therapist involved, the number of persons handled, and the commission earned.
- **Appointments** will also store both the assigned therapist and the customer's preferred therapist whenever applicable, allowing management to accommodate customer preferences whenever therapist availability permits.
- **Appointments** will be connected to **Therapist Ratings** and **Feedback** to measure customer satisfaction, evaluate therapist performance, and support continuous service quality improvement after completed appointments.
- **Appointments** may also generate **Refund Requests** whenever cancellations, payment adjustments, or customer disputes occur. Each refund record will

reference the related appointment, associated order, requested refund amount, and approval status.

- **Appointments** will be connected to **Partners** whenever bookings are made under corporate accounts, hotel partnerships, or special business arrangements so that negotiated pricing can automatically be applied.
- **Therapists** will perform the spa services associated with scheduled appointments. Therapist records will store professional information, listed specialties, and a generalist indicator identifying whether the therapist may perform all available services.
- **Therapists** will receive earnings through **Therapist Commission** records, which will define either commission percentages or fixed influencer rates corresponding to specific services.
- **Therapists** may also receive deductions recorded in **Therapist Deductions** for salary adjustments, cash advances, penalties, or other authorized deductions affecting commission computation.
- **Therapists** will maintain attendance records through **Therapist Attendance**, which will monitor duty dates, clock-in time, clock-out time, and therapist rotation order used during appointment assignment.
- **Therapists** will establish their service qualifications through **Therapist Specialties** and **Therapist Specialty Services**, allowing management to identify therapist expertise by both service category and specific spa treatment.
- **Therapists** will receive customer evaluations through **Therapist Ratings**, which will record the rating submitted for each completed appointment to assist management in monitoring therapist performance.
- **Partners** will be connected to both **Appointments** and **Partner Rates** to support hotel partnerships, corporate clients, and negotiated service pricing. Each partner record will identify the organization type and the corresponding pricing arrangements available within the system.
- **Daily Reports** will store the overall operational summary prepared at the end of each business day, including report date, cash on hand, POS reading, Maya digital payment collections, and report lock status to preserve finalized financial records.
- **Daily Reports** will contain multiple **Daily Report Denominations**, which will record the breakdown of cash denominations counted during cashier balancing and end-of-day reconciliation.
- **Daily Product Sales** will record the products sold during each business day, including the product description, quantity sold, and total sales amount, to support inventory monitoring and sales reporting.

- **Business Expenses** will store operational expenditures such as supplies, maintenance costs, utilities, marketing expenses, and other authorized business expenses to support financial monitoring and profitability analysis.
- **Gift Certificates** will store records of gift certificates issued, sold, and redeemed by the spa, including transaction dates, certificate types, and corresponding monetary amounts for promotional monitoring and financial reporting.
- **Unpaids Corp** will store outstanding balances of corporate clients whose transactions remain unpaid. Each record will contain the client name, report date, and outstanding amount to support receivable monitoring and collection activities.
- **Receptionist Pins** will store the unique PIN assigned to each receptionist for transaction verification and cashier authorization. These records will operate independently and will primarily be referenced during payment processing, cashier verification, and end-of-day financial reporting.
- **System Settings** will store the configurable parameters that control various operational behaviors of the Recovery Iloilo Spa Integrated System through individual setting keys and corresponding values. These records will operate independently and will be referenced whenever system-wide configuration values are required during system execution.

Use Case Diagram

Actors:

- **Customer (User)** – The **Customer** will be the primary user of the system and will represent any client who avails of the services offered by Recovery Iloilo Spa. Customers will be able to register an account with email OTP verification, log in securely, browse available spa services and wellness products, manage their shopping cart, place orders, and schedule appointments through the online booking facility. They will be able to select a preferred therapist or choose any available therapist, request home-service appointments, hotel or partner bookings, and add extra services whenever applicable. Customers will also be able to monitor the status of their appointments and payments, receive email and in-system notifications, view transaction history and digital receipts, submit feedback, rate the therapist who rendered the service, redeem vouchers and discounts, and submit refund requests whenever necessary.
- **Receptionist / Cashier** – The **Receptionist / Cashier** will be responsible for managing daily front-desk operations and customer transactions within the spa. This role will review, approve, decline, reschedule, and cancel appointment requests, assign or reassign therapists, process walk-in customers, record

additional services rendered during appointments, and verify therapist availability before confirming bookings. Receptionists will process onsite payments, verify cashier PINs before sensitive financial transactions, handle refund requests, validate vouchers and discounts, and generate receipts. At the end of each business day, they will prepare the daily report, record cash denominations, POS readings, Maya collections, product sales, business expenses, gift certificate transactions, and unpaid corporate accounts for financial reconciliation.

- **Owner / Administrator** –The **Owner / Administrator** will oversee the overall operation, configuration, and maintenance of the Recovery Iloilo Spa Integrated System. This role will manage user accounts and role assignments, therapist records, therapist specialties, commission rates, attendance records, service categories, products, inventory, partner establishments, partner pricing, vouchers, gift certificates, and system settings. The administrator will also monitor appointments, transactions, refund requests, customer feedback, therapist ratings, operational reports, sales analytics, revenue forecasts, and activity logs. In addition, the administrator will configure business rules, review financial reports, monitor employee productivity, and supervise the overall performance and security of the system.
- **Therapist** – The **Therapist** will be responsible for providing spa treatments based on appointments assigned through the system. Therapists will be able to record their attendance by clocking in and out, monitor their assigned schedules, review customer booking details, and identify additional services requested before or during treatment sessions. They will also monitor completed appointments, automatically computed commissions, deductions, assigned specialties, customer ratings, and feedback related to the services they have rendered. Attendance records and therapist rotation information will also be maintained to support fair workload distribution and efficient scheduling.
- **PayMongo Payment Gateway** – The **PayMongo Payment Gateway** will serve as an external payment processing service responsible for handling online payment transactions initiated by customers. Whenever a customer chooses an online payment method such as GCash, QR Ph, or bank card, the system will securely forward the payment request to PayMongo for processing. After payment verification, PayMongo will return the transaction result to the system, allowing payment records and order status to be updated automatically. This external actor will not directly access the system and will interact only through secure payment requests and responses.
- **Email / SMTP Service** – The **Email / SMTP Service** will function as an external communication service responsible for sending electronic messages generated by the Recovery Iloilo Spa Integrated System. It will deliver one-time passwords (OTP) for customer account verification, appointment confirmations, booking updates, payment acknowledgements, refund notifications, password reset instructions, and other system-generated announcements. This external actor will not access the system directly and will function solely as the communication channel for outgoing email notifications.

Use Cases:

Customer Use Cases

- **Register and Log In Account** – The system will allow customers to create an account and securely access the Recovery Iloilo Spa Integrated System using their registered credentials. During registration, customers will provide the required personal information, and a one-time password (OTP) will be sent to their registered email address for verification before the account can be activated. After successful login, customers will be able to access the services, products, bookings, payments, notifications, and other features available to their account.
- **Manage User Profile Information** –The system will allow customers to view and update their personal profile information whenever necessary. Customers will be able to modify details such as their full name, contact number, email address, and password while maintaining accurate customer records. Keeping profile information updated will help improve communication, booking management, and transaction processing.
- **Browse Spa Services and Products** – The system will allow customers to browse all spa services, wellness products, packages, and promotional offerings available at Recovery Iloilo Spa. Customers will be able to view service descriptions, session duration, pricing, product availability, and other relevant details before making a booking or purchase. This feature will help customers select the services and products that best suit their preferences.
- **View Categories and Pricing** – The system will allow customers to browse services and products according to their assigned categories. Customers will be able to compare available services, products, prices, promotional packages, and home-service availability before confirming a booking or purchase. This feature will help customers make informed decisions based on their preferred treatments.
- **Add Products to Cart** – The system will allow customers to add wellness products into their shopping cart before completing a purchase. Customers will be able to specify product quantities, review selected items, and update their selections before proceeding to checkout. The shopping cart will temporarily store all selected products until the transaction is confirmed.
- **Manage Cart and Checkout** – The system will allow customers to review the contents of their shopping cart before placing an order. Customers will be able to modify quantities, remove unwanted products, and verify the total amount due before proceeding with checkout. Once confirmed, the system will generate the corresponding order and transaction records. .

- **Place Orders for Products and Services** – The system will allow customers to purchase wellness products and confirm service transactions through the integrated platform. Each order will contain complete information regarding purchased services, products, quantities, unit prices, discounts applied, payment details, and computed totals to ensure accurate billing and transaction monitoring.
- **Book Spa Appointments** – The system will allow customers to schedule spa appointments by selecting their preferred services, appointment date, available time slot, number of persons, and preferred service location. Customers may create online bookings, while walk-in appointments will be encoded by authorized reception staff. Submitted appointments will remain pending until reviewed and processed by authorized personnel.
- **Select Home Service or In-Store Service** – The system will allow customers to choose whether the appointment will be performed as an in-spa service, home-service booking, hotel booking, or partner booking whenever applicable. The system will automatically compute any additional travel fees or partner pricing based on the selected appointment type.
- **Select Preferred Therapist** – The system will allow customers to select their preferred therapist or choose the "Any Available Therapist" option during appointment booking. The system will record the customer's preference and will assign the requested therapist whenever available. If the selected therapist is unavailable, management will be able to assign another qualified therapist.
- **Add Extra Appointment Services** – The system will allow customers to request additional spa services before or after an appointment has been approved. Each additional service will be recorded separately together with the assigned therapist, corresponding charges, commission, and payment status. This feature will allow customers to customize their spa experience without creating another appointment.
- **Pay Through the Online Payment Facility** – The system will allow customers to settle payments using available payment options, including cash, digital wallets, payment gateway transactions, payment upon completion, or approved pay-later arrangements. The system will automatically calculate transaction totals, generate digital receipts, record payment history, and update payment status after successful processing.
- **Track Order and Appointment Status** – The system will allow customers to monitor the progress of their appointments throughout the booking lifecycle. Customers will be able to view whether an appointment is pending, assigned, approved, completed, declined, or cancelled. This feature will provide customers with real-time updates regarding their reservations.

- **View Payment and Refund Status** – The system will allow customers to monitor payment confirmations and refund requests associated with their transactions. Customers will be able to review the processing status of submitted refund requests and determine whether the request remains pending, approved, or declined.
- **Receive System Notifications** – The system will allow customers to receive notifications regarding account verification, appointment confirmations, therapist assignments, payment confirmations, refund updates, reminders, promotional announcements, and other important system activities. Notifications will also be sent through the integrated email service whenever necessary.
- **View Transaction History** – The system will allow customers to review their complete transaction history, including previous appointments, purchased products, completed services, payments, refunds, and booking activities. This feature will provide customers with convenient access to their historical records whenever needed.
- **Submit Service Feedback and Ratings** – The system will allow customers to submit comments, suggestions, and overall evaluations regarding the spa services they received after completing an appointment. These feedback records will assist management in monitoring customer satisfaction and identifying opportunities for service improvement.
- **Rate Assigned Therapist** – The system will allow customers to evaluate the therapist who performed their completed spa session by submitting ratings and comments. Therapist evaluations will help management monitor employee performance, recognize outstanding service, and improve service quality.
- **Request Refund or Appointment Cancellation** – Allows customers to file a request for the cancellation of a booking or the reimbursement of a transaction. The system records the amount involved and holds the request in a pending state. Management may then review the request and take the appropriate action.

For Receptionist / Cashier:

- **Log In and Verify Security PIN** – The Receptionist or Cashier will log in using the assigned account credentials before accessing the operational functions of the system. A cashier PIN will be required before performing protected financial transactions, approving sensitive operations, or finalizing the daily report. This process will help strengthen accountability and ensure that only authorized personnel can perform secured activities.
- **Manage Walk-In Appointments** – The Receptionist or Cashier will record appointments for customers who visit the spa without making an online

reservation. The system will create the corresponding appointment record, assign the selected services, and prepare the transaction for payment processing. This process will ensure that walk-in customers will follow the same workflow as online bookings.

- **Review and Process Appointment Requests** – The Receptionist or Cashier will review appointment requests submitted by customers before confirming the reservation. The system will allow staff to approve, decline, reschedule, or cancel appointments while recording the personnel responsible for each action. This process will help maintain organized appointment scheduling and proper operational control.
- **Assign and Reassign Therapists** – The Receptionist or Cashier will assign available therapists to approved appointments based on therapist availability, specialty, rotation order, and customer preference whenever possible. The system will also allow therapist reassignment when scheduling conflicts or operational changes occur. This process will ensure proper workload distribution and efficient service coordination.
- **Manage Appointment Details and Add-on Services** – The Receptionist or Cashier will update appointment information whenever necessary, including modifying schedules, adding extra services, updating customer notes, and recording service completion details. Additional services rendered during or after the appointment will be recorded together with the assigned therapist, corresponding charges, commission, and payment status. This process will ensure that all services provided are properly documented.
- **Process Point-of-Sale (POS) Transactions** – The Receptionist or Cashier will process customer payments through the Point-of-Sale module using supported payment methods such as cash, card, digital wallet, payment at completion, or pay-later arrangements. The system will automatically calculate transaction totals, apply eligible discounts, generate receipts, and update payment records. This process will ensure accurate financial recording and faster transaction processing.
- **Process Refund Requests** – The Receptionist or Cashier will review customer refund requests and perform the necessary verification before forwarding them for approval or processing according to company policies. The system will update the refund status and maintain complete refund transaction records for monitoring purposes.
- **Manage Customer Transactions** – The Receptionist or Cashier will monitor customer orders, appointment payments, payment status, and transaction history. The system will maintain complete transaction records to support customer inquiries, financial reconciliation, and operational monitoring.

- **Monitor Daily Sales Transactions** – The Receptionist or Cashier will review all completed transactions recorded throughout the business day. The system will automatically consolidate payment information, service sales, product sales, discounts, refunds, and transaction summaries to support daily financial monitoring.
- **Prepare Daily Reports** – The Receptionist or Cashier will prepare the end-of-day operational report by reviewing recorded sales, POS readings, payments, gift certificates, business expenses, unpaid corporate accounts, and other financial records. The system will generate the required report for management review and documentation.
- **Record Cash Denominations and Cash Reconciliation** – The Receptionist or Cashier will record the quantity of each cash denomination at the close of business. The system will automatically calculate the total cash on hand, compare it with recorded POS collections, and generate cash-over or cash-short reconciliation results to improve financial accuracy.
- **Record Daily Product Sales** – The Receptionist or Cashier will review and confirm product sales recorded during the business day. The system will maintain daily product sales records to support inventory monitoring, sales analysis, and operational reporting.
- **Record Business Expenses** – The Receptionist or Cashier will encode business expenses incurred during daily operations, including the expense category, amount, and transaction date. The system will include these records in the daily financial reports and analytics.
- **Manage Gift Certificate Transactions** – The Receptionist or Cashier will record gift certificates that are sold or redeemed during the business day. The system will maintain accurate gift certificate balances and transaction history for monitoring promotional activities.
- **Manage Corporate Unpaid Transactions** – The Receptionist or Cashier will record unpaid transactions for corporate clients and partner establishments. The system will maintain outstanding balances and support monitoring of accounts receivable for future collection.
- **Submit and Lock Daily Reports** – The Receptionist or Cashier will finalize the daily operational report after verifying all recorded transactions, payments, expenses, and cash balances. Once submitted, the system will lock the report to prevent unauthorized modifications while preserving the integrity of financial records for management review.

For Owner / Administrator:

- **Log In Account** – The Owner or Administrator will securely access the system using assigned account credentials. The login process will restrict administrative functions to authorized personnel based on assigned roles and permissions. After successful authentication, authorized users will be able to access system management, monitoring, reporting, and configuration functions.

- **Manage User Accounts and Roles** – The Owner or Administrator will create, update, activate, deactivate, and manage customer and employee accounts within the system. Administrative roles such as owner, administrator, receptionist, therapist, marketing personnel, and information technology personnel will also be assigned according to operational responsibilities. This process will ensure that users will only have access to the system features appropriate to their assigned roles.
- **Manage Therapist Records and Specialties** – The Owner or Administrator will maintain therapist profiles, specialties, service qualifications, and generalist status. Therapist specialty categories and individual service qualifications will also be managed to support proper therapist assignment during appointment scheduling. This process will help ensure that therapists will only be assigned to services they are qualified to perform.
- **Manage Services, Products, Categories, and Pricing** – The Owner or Administrator will maintain the complete service and product catalog by creating, updating, or deactivating categories, services, products, prices, session durations, home-service fees, and inventory information. This process will ensure that customers and staff will always access accurate and updated information during bookings and transactions.
- **Manage Inventory and Product Availability** – The Owner or Administrator will monitor inventory levels, product availability, stock movements, and replenishment activities. The system will assist in identifying low-stock items and maintaining accurate inventory records to support uninterrupted spa operations.
- **Configure Therapist Commission and Deduction Rules** – The Owner or Administrator will define therapist commission percentages, influencer flat-rate commissions, and other compensation rules based on completed services. Administrative users will also record therapist deductions such as cash advances, penalties, or other adjustments to ensure accurate payroll computation.
- **Manage Partners and Partner Service Rates** – The Owner or Administrator will maintain records of hotels, corporate clients, and other partner establishments. Negotiated service rates for each partner will also be configured to ensure that partner bookings will automatically apply the appropriate pricing during appointment processing.
- **Manage Receptionist Security PINs** – The Owner or Administrator will create, update, and maintain the security PIN assigned to each receptionist or cashier. These PINs will be required before protected financial transactions, cashier

verification, and daily report submission to strengthen accountability and transaction security.

- **Review and Manage Customer Transactions** – The Owner or Administrator will review customer orders, appointment transactions, payment records, discounts, vouchers, and transaction approvals whenever administrative intervention is required. This process will help maintain accurate financial records and ensure that transactions comply with company policies.
- **Review and Process Refund Requests** – The Owner or Administrator will examine refund requests submitted by customers by reviewing the corresponding appointment, order, payment information, and refund amount. The request status will then be updated based on management approval to support organized refund processing and customer service.
- **Monitor Appointments and Therapist Operations** – The Owner or Administrator will monitor appointment schedules, therapist assignments, attendance records, completed services, commissions, and operational workload. This process will help ensure efficient workforce management and proper service coordination throughout daily operations.
- **Generate Operational Reports and Analytics** – The Owner or Administrator will generate reports covering appointments, sales transactions, inventory usage, therapist productivity, commissions, customer activities, payment summaries, refunds, and overall business performance. The system will also generate short-term revenue forecasts and analytical summaries to support business monitoring and informed decision-making.
- **Generate Daily Financial Reports** – The Owner or Administrator will review daily operational reports that include Point-of-Sale readings, cash-on-hand balances, cash denomination summaries, business expenses, gift certificate transactions, unpaid corporate accounts, and cash reconciliation results. These reports will support financial monitoring and end-of-day verification.
- **View Customer Feedback and Therapist Ratings** – The Owner or Administrator will review customer comments, ratings, and service evaluations submitted after completed appointments. The system will organize these records by therapist and transaction to assist management in evaluating customer satisfaction, therapist performance, and opportunities for service improvement.
- **Review Activity Logs and Employee Accountability** – The Owner or Administrator will examine activity logs that record significant actions performed throughout the system. Each record will identify the responsible user, assigned role, action performed, affected record, description, and timestamp to support auditing, accountability, and operational monitoring.
- **Manage System Settings** – The Owner or Administrator will configure and maintain system-wide settings that control the behavior of the application,

including operational parameters, configurable options, and other business rules. These settings will allow management to adjust system behavior without modifying the underlying program code.

For Therapist:

- **Log In Account** – The Therapist will securely access the system using the assigned account credentials. The login process will restrict therapist functions to authorized users and protect therapist information from unauthorized access. After successful authentication, therapists will be able to access their schedules, attendance records, assigned services, commissions, ratings, and notifications.
- **Record Attendance and Duty Schedule** – The Therapist will record daily attendance by logging the time in and time out for each assigned duty. The system will maintain attendance records together with the therapist's rotation order to support fair workload distribution and proper therapist assignment. This process will assist management in monitoring attendance and workforce availability.
- **View Assigned Appointments** – The Therapist will view appointments assigned by the Receptionist or Administrator through the appointment management module. The system will display appointment schedules, assigned services, customer information, appointment location, number of persons, and current booking status. This process will help therapists prepare for scheduled appointments before rendering the service.
- **View Appointment Details** – The Therapist will review the complete details of each assigned appointment before the scheduled session begins. The system will display the selected services, additional services, customer notes, preferred service location, appointment schedule, applicable pricing, and other booking information. This process will help therapists understand customer requirements and prepare the necessary treatments.
- **View Assigned Specialties and Qualified Services** – The Therapist will review the service categories and individual treatments assigned based on qualifications and specialties maintained by management. The system will identify whether the therapist is assigned as a specialist or generalist for particular services. This process will ensure that therapists will only perform services appropriate to their qualifications.
- **View Commission and Earnings** – The Therapist will monitor commissions earned from completed appointments, additional services, influencer sessions, and other eligible transactions. The system will automatically calculate

commission records based on completed services and approved transactions to provide accurate earnings information.

- **View Deduction Records** – The Therapist will review deductions applied to commission earnings, including cash advances, operational expenses, penalties, or other authorized adjustments. The system will maintain complete deduction records to provide transparency and accurate payroll computation.
- **View Customer Ratings and Feedback** – The Therapist will review customer ratings, comments, and service evaluations submitted after completed appointments. The system will organize feedback records according to completed services to help therapists evaluate their performance and identify opportunities for professional improvement.
- **Receive System Notifications** – The Therapist will receive notifications generated by the system regarding newly assigned appointments, schedule updates, therapist reassignments, appointment cancellations, commission updates, attendance reminders, and other work-related announcements. This process will help therapists remain informed about daily responsibilities and operational activities.

For PayMongo Payment Gateway:

- **Process Online Payment Request** – The PayMongo Payment Gateway will receive online payment requests forwarded by the Recovery Iloilo Spa Integrated System whenever a customer selects an electronic payment method during checkout. The gateway will securely process the payment transaction without exposing sensitive payment information to the system. This process will allow customers to complete online payments through a secure and reliable payment service.
- **Return Payment Confirmation** – The PayMongo Payment Gateway will return the payment result to the Recovery Iloilo Spa Integrated System after the payment transaction has been completed or declined. The system will use the returned confirmation to update the payment status, record the transaction, generate the corresponding receipt, and continue the appropriate booking or order workflow. This process will ensure that only verified payment transactions will be recorded within the system.

For Email / SMTP Service:

- **Deliver One-Time Password (OTP)** – The Email / SMTP Service will receive the one-time password generated by the Recovery Iloilo Spa Integrated System during customer registration or account verification. The generated code will be transmitted to the registered email address of the customer to verify account ownership before access is granted. This process will strengthen account security and help prevent unauthorized registration and access.

- **Deliver Booking and Transaction Notifications** – The Email / SMTP Service will receive notification messages generated by the Recovery Iloilo Spa Integrated System and deliver them to the intended recipients. Messages may include appointment confirmations, appointment approvals, booking updates, payment acknowledgements, refund status updates, password verification messages, and other important transaction notifications. This process will help ensure that customers remain informed about significant activities related to their accounts and transactions.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in

application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical con

REFERENCES

Abad, R. (2020). Inventory control systems for small service enterprises in the Philippines. Philippine Journal of Business Operations. https://scholar.google.com/scholar?q=inventory+control+systems+philippines+abad

Agasue, M. I. (2016). Enhanced hotel management information system for multiple reservation booking. International Journal of Management Research, 12(1), 45–59.

Agbayani, R. (2024). Usability of web-based systems in Philippine SMEs. Philippine Journal of Information Technology.

https://scholar.google.com/scholar?q=usability+web+systems+philippines+agbayani

Agustin, M. (2021). Operational Stability through Digital Integration in the Wellness Sector. https://scholar.google.com/scholar?q=Agustin+operational+stability+digital+integration+wellness+Philippines

Ahmadi-Javid, A., Jalali, Z., & Klassen, K. J. (2022). Predict, then schedule: Prescriptive analytics approach for machine learning-enabled sequential clinical scheduling. Computers & Industrial Engineering, 169, 108270. https://doi.org/10.1016/j.cie.2022.108270

Al-Hussain, M. (2024). Incentivizing Service Staff through Real-Time Commission Transparency. https://scholar.google.com/scholar?q=Incentivizing+Service+Staff+through+Real-Time+Commission+Transparency+Al-Hussain

Alalwan, A. A., Dwivedi, Y. K., Rana, N. P., & Algharabat, R. (2021). Examining factors influencing Jordanian customers' intentions and adoption of mobile banking: Extending UTAUT2 with trust. International Journal of Information Management, 60, 102344. https://doi.org/10.1016/j.ijinfomgt.2021.102344

Alfelor, R. (2024). Usability of management information systems in SMEs. Philippine IT Journal. https://scholar.google.com/scholar?q=usability+MIS+philippines+alfelor

Alharbi, S., Zyngier, S., & Hodkinson, C. (2021). Privacy by design and customers' perceived privacy and security concerns in the success of e-commerce. Journal of Enterprise Information Management, 34(3), 746–769. https://doi.org/10.1108/JEIM-07-2019-0195

Ali, H. (2021). Secure Communication Models for Web Applications. https://scholar.google.com/scholar?q=Secure+Communication+Models+for+Web+Applications+Ali

Ali, M., Hassan, R., & Bello, A. (2026). Customer relationship management adoption and organizational performance in service industries. International Journal of Innovative Social Sciences & Humanities Research, 14(2), 167–176. https://www.seahipublications.org/wp-content/uploads/2026/04/IJISSHR-J-15-2026.pdf

Allam, J., Dayag, R. M., Fronda, J. C., Tumaliuan, J. A., & Abana, E. (2023). From Raw Data to Actionable Insights: Navigating the World of Data Analytics for Local SMEs. https://scholar.google.com/scholar?q=Allam+Dayag+Fronda+data+analytics+Philippines

Alnakshabandi, O. A., Aqel, M., & Kaya, T. (2022). MIS based on IoT and cloud services: Governmental organizations perspective. Journal of Internet Technology, 23(3), 677–686. https://jit.ndhu.edu.tw/article/view/2693

Alonzo, M. (2024). Digital payroll and commission systems in service industries. Asian Journal of Management Systems. https://scholar.google.com/scholar?q=digital+payroll+philippines+alonzo

Alsmairat, M. A. K., Hussain, S., AlOqool, A., & Ali, W. (2024). Does supply chain resilience mediate the relationship between CRM dimensions and customer satisfaction? Global Business Review. https://doi.org/10.1177/23197145231225570

Alshaikh, M. (2020). Developing cybersecurity culture to influence employee behavior: A practice perspective. Computers & Security, 98, 102003. https://doi.org/10.1016/j.cose.2020.102003

Alshammari, S. H., & Alshammari, R. A. (2024). An integration of expectation confirmation model and information systems success model to explore the factors affecting the continuous intention to utilise virtual classrooms. Scientific Reports, 14, 18491. https://doi.org/10.1038/s41598-024-69401-8

Alvarado, J. (2024). Digital customer engagement in service industries. Philippine Marketing Review. https://scholar.google.com/scholar?q=digital+customer+engagement+philippines+alvarado

Ananda, R., et al. (2024). The Integration of POS and Inventory Functions for SME Competitive Advantage. https://scholar.google.com/scholar?q=Ananda+integrated+POS+inventory+SME+2024

Anderson, K. (2022). Algorithmic Scheduling in the Wellness Industry. https://scholar.google.com/scholar?q=Algorithmic+Scheduling+in+the+Wellness+Industry+Anderson

Anggari, B. Y. I. (2026). Human–AI collaboration in tourism work: A systematic review of impacts on jobs and skills. Journal of Hospitality and Tourism Technology, 17(2), 112–129.

Apostol, J. (2021). Digital payroll systems in service industries. Philippine HR Journal. https://scholar.google.com/scholar?q=digital+payroll+philippines+apostol

Aquino, C. (2024). Labor Relations and Automated Commission Tracking in the Philippine Service Sector. https://scholar.google.com/scholar?q=Labor+Relations+and+Automated+Commission+Tracking+in+the+Philippine+Service+Sector+Aquino

Arnott, D., & Pervan, G. (2020). A critical analysis of decision support systems research. Journal of Information Technology, 35(3), 1–19. https://doi.org/10.1177/0268396220902982

Aujero, B. (2023). Sector-Specific Innovations in the Iloilo Business Ecosystem. https://scholar.google.com/scholar?q=Aujero+sector-specific+innovations+Iloilo+business

Babaee, S. (2024). Three essays on the interface of sales and operations management (Doctoral dissertation, Wilfrid Laurier University). Scholars Commons @ Laurier. https://scholars.wlu.ca/etd/3507

Baggio, R., & Sainaghi, R. (2021). Imbrications of IT and hospitality organizations. Annals of Tourism Research Empirical Insights, 2(2), 100021. https://doi.org/10.1016/j.annale.2021.100021

Balida, R. (2023). Barriers and Enablers to Digital Transformation in Visayan Micro-Enterprises. https://scholar.google.com/scholar?q=Balida+digital+transformation+Visayan+micro-enterprises

Bandara, F., Jayawickrama, U., Subasinghage, M., Olan, F., Alamoudi, H., & Alharthi, M. (2024). Enhancing ERP responsiveness through big data technologies: An empirical investigation. Information Systems Frontiers, 26, 251–275. https://doi.org/10.1007/s10796-023-10374-w

Bansal, J. (2024). Effectiveness of AI-powered CRM tools in enhancing customer experience. International Journal of Research in Marketing Management and Sales, 6(2), 318–320. https://doi.org/10.33545/26633329.2024.v6.i2d.243

Banzon, M. (2021). User experience in business systems. Philippine Information Systems Journal. https://scholar.google.com/scholar?q=user+experience+philippines+banzon

Baterina, K., & Yumul, R. (2023). Appointment coordination systems in SMEs. Philippine Management Review. https://scholar.google.com/scholar?q=appointment+systems+philippines+yumul+baterina

Bautista, J., & Ong, P. (2024). Usability evaluation of web-based systems in SMEs. Philippine Information Technology Journal. https://scholar.google.com/scholar?q=usability+web+systems+philippines+bautista+ong

Bautista, L. (2021). Digital Readiness of Philippine SMEs: Overcoming Technical Literacy Gaps.

https://scholar.google.com/scholar?q=Digital+Readiness+of+Philippine+SMEs+Overcoming+Technical+Literacy+Gaps+Bautista

Baylon, E. (2020). Inventory tracking challenges in service businesses. Philippine Operations Journal. https://scholar.google.com/scholar?q=inventory+tracking+philippines+baylon

Belanger, A. L. (2025). Human-computer interaction in service platforms: Reducing friction and churn through responsive design. International Journal of Electronic Commerce, 29(1), 74–92. https://doi.org/10.1080/10864415.2024.2290114

Beltran, R. (2022). Non-Contact Efficiency: Web-Based Prototypes in Post-Pandemic Hospitality. https://scholar.google.com/scholar?q=Beltran+non-contact+efficiency+web-based+prototypes+Philippines

Bernardo, F. (2023). Preventing Revenue Leakage: Digital Voucher Validation in Local Services. https://scholar.google.com/scholar?q=Bernardo+preventing+revenue+leakage+digital+voucher+validation+Philippines

Betancor, P. K. (2024). Efficient patient care in the digital age: Impact of online appointment scheduling in a medical practice and a university hospital on the “no-show”. BMC Health Services Research, 24(1), Article 412. https://doi.org/10.1186/s12913-024-10812-w

Bhardwaj, J., Cherkuri, L. S., Dhoni, P. S., & Aluru, P. (2024). Advancing CRM capabilities: A comparative analysis of predictive analytics integration. International Journal of Computer Trends and Technology, 72(6), 8–14. https://doi.org/10.14445/22312803/IJCTT-V72I6P102

Blijleven, V., Hoxha, F., & Jaspers, M. (2022). Workarounds in electronic health record systems and the revised sociotechnical electronic health record workaround analysis framework: Scoping review. Journal of Medical Internet Research, 24(3), e33046. https://doi.org/10.2196/33046

Blanza, M. G. A., & Catilo, L. L. (2025). Leveraging Technology for Sustainable Growth: An Analysis of MSMEs in Iloilo Province during the Post-Pandemic Milieu. https://scholar.google.com/scholar?q=Blanza+Catilo+Iloilo+MSME+digital+transformation

Briggs, H. J. (2024). Digital ledger compliance and automated auditing protocols for retail and hospitality architectures. Journal of Financial Data Science, 6(2), 118–134. https://doi.org/10.3905/jfds.2024.1.141

Briones, C. (2022). Financial reporting systems in SMEs. Journal of Accounting Systems Philippines. https://scholar.google.com/scholar?q=financial+reporting+philippines+briones

Brooks, L. (2023). Data Visualization and Trend Analysis in Small Service Businesses. https://scholar.google.com/scholar?q=Data+Visualization+and+Trend+Analysis+in+Small+Service+Businesses+Brooks

Brooks, T. L. (2023). Architectural optimization for small business information portals. Journal of Computer Information Systems, 63(3), 214–226. https://doi.org/10.1080/08874417.2023.2190841

Brown, T. (2021). UI/UX Principles for Small Business Digital Adoption. https://scholar.google.com/scholar?q=UI/UX+Principles+for+Small+Business+Digital+Adoption+Brown

Bulaong, T. (2024). Integrated systems and business performance. Journal of Philippine Business Innovation. https://scholar.google.com/scholar?q=integrated+systems+philippines+bulaong

Business Research Insights. (2026). User Interface (UI) Design Market Size & Share [2026 To 2035]. https://scholar.google.com/scholar?q=Business+Research+Insights+UI+design+market+report+2026

Bustillo, A. (2026). Tamper evident inventory and sales recording system using chain-based data integrity to ensure audit reliability for Mashikketta Restaurant. Gyaanarth Journal of Research and Creative Endeavors, 11(1), 320–332.

Buttle, F., & Maklan, S. (2019). Customer Relationship Management: Concepts and Technologies. Routledge. https://scholar.google.com/scholar?q=Buttle+Maklan+customer+relationship+management+2019

Caballero, J. (2021). Integrated systems and organizational efficiency. Philippine Business Systems Journal. https://scholar.google.com/scholar?q=integrated+systems+philippines+caballero

Cabrera, L. (2021). Employee management systems in service organizations. Philippine HR Journal. https://scholar.google.com/scholar?q=employee+management+philippines+cabrera

Cancio, L. (2020). Customer feedback systems in SMEs. Philippine Service Industry Review. https://scholar.google.com/scholar?q=customer+feedback+philippines+cancio

Capistrano, L. (2024). Digital transaction systems in small enterprises. Journal of Financial Technology Philippines. https://scholar.google.com/scholar?q=digital+transaction+systems+philippines+capistrano

Carreon, L., & Mendoza, P. (2023). Digital feedback systems and customer satisfaction in local services. Journal of Philippine Service Management. https://scholar.google.com/scholar?q=digital+feedback+philippines+mendoza+carreon

Carter, R. M. (2025). The unified commerce imperative: Redefining small business operations through database integration. International Journal of Information Management, 74, 102–115. https://doi.org/10.1016/j.ijinfomgt.2024.102714

Castañeda, M. (2022). Digital voucher management in SMEs. Philippine Marketing Journal. https://scholar.google.com/scholar?q=voucher+management+philippines+castaneda

Castillo, A. (2020). Standardizing PHP/MySQL Frameworks for Provincial IT Development. https://scholar.google.com/scholar?q=Standardizing+PHP+MySQL+Frameworks+for+Provincial+IT+Development+Castillo

Catapang, R. (2023). Automated inventory systems in service industries. Philippine Operations Research Journal. https://scholar.google.com/scholar?q=automated+inventory+philippines+catapang

Chatterjee, S., Mikalef, P., Khorana, S., & Kizgin, H. (2022). Assessing the implementation of AI integrated CRM system for B2C relationship management: Integrating contingency theory and dynamic capability view theory. Information Systems Frontiers, 26, 967–985. https://doi.org/10.1007/s10796-022-10261-w

Chatterjee, S., Rana, N. P., Khorana, S., Mikalef, P., & Sharma, A. (2023). Assessing organizational users' intentions and behavior to AI integrated CRM systems: A Meta-UTAUT approach. Information Systems Frontiers, 25(4), 1299–1313. https://doi.org/10.1007/s10796-021-10181-1

Chen, X. (2021). The Impact of Digital Bookkeeping on Financial Accuracy. https://scholar.google.com/scholar?q=The+Impact+of+Digital+Bookkeeping+on+Financial+Accuracy+Chen

Chen, Z., & Wu, S. (2024). The economic impact of MIS integration in operations and customer relationship management. Advances in Economics, Management and Political Sciences, 79, 371–376. https://doi.org/10.54254/2754-1169/79/20241893

Choi, S. (2020). Inventory Alert Systems and Service Continuity in the Wellness Sector. https://scholar.google.com/scholar?q=Inventory+Alert+Systems+and+Service+Continuity+in+the+Wellness+Sector+Choi

Choudhary, R. (2026). Optimizing salon services: Reducing scheduling errors and wait times through automation. Spectrum of Engineering Sciences, 3(1), 45–58.

Chung, J. C., Huang, Y. F., Weng, M. W., & Lin, J. C. (2021). The sustainable innovation design in catering service. Sustainability, 14(1), 278. https://doi.org/10.3390/su14010278

Conboy, K., & Carroll, N. (2019). Implementing large-scale agile frameworks: Challenges and recommendations. IEEE Software, 36(2), 44–50. https://doi.org/10.1109/MS.2018.2884865

Coronel, C., & Morris, S. (2019). Database Systems: Design, Implementation, & Management. Cengage Learning. https://scholar.google.com/scholar?q=Coronel+Morris+database+systems+management+2019

Coronel, C., & Morris, S. (2021). Database systems: Design, implementation, & management (14th ed.). Cengage Learning.

Corpuz, J. (2019). Supply Chain Resilience and Inventory Alerts for Local Enterprises. https://scholar.google.com/scholar?q=Supply+Chain+Resilience+and+Inventory+Alerts+for+Local+Enterprises+Corpuz

Cruz, T. (2020). Targeted Promotions and Service History: Driving Repeat Visits in Retail. https://scholar.google.com/scholar?q=Cruz+targeted+promotions+service+history+Philippines

Customer Gauge. (2025). The Impact of Real-Time Feedback on Customer Retention in the Service Industry. https://scholar.google.com/scholar?q=Customer+Gauge+real-time+feedback+service+industry+2025

Dacumos, R., & Villaflor, T. (2023). Workflow automation in SMEs. Philippine Management Review. https://scholar.google.com/scholar?q=workflow+automation+philippines+villaflor+dacumos

Dalton, M. E. (2025). Breaking down data silos: The operational lifecycle of fully synchronized service ecosystems. Information Systems Management, 42(2), 154–169. https://doi.org/10.1080/10580530.2024.2311902

Daniels, G. (2026). Robots at the table: Unveiling employee experiences of robot integration in Malaysia's hospitality industry. Advances in Southeast Asian Studies, 19(1), 87–104.

Davenport, T. H. (2018). Process Innovation: Reengineering Work through Information Technology. Harvard Business Press. https://scholar.google.com/scholar?q=Davenport+process+innovation+IT+2018

Davenport, T. H., & Ronanki, R. (2018). Artificial Intelligence for the Real World. Harvard Business Review. https://scholar.google.com/scholar?q=Davenport+Ronanki+AI+for+the+real+world+2018

Davis, F. D. (2023). The Technology Acceptance Model: Implications for Small Business Software Adoption. https://scholar.google.com/scholar?q=Davis+technology+acceptance+model+SME+2023

Davis, L. (2022). Data Communication Models for Web-Based Architectures. https://scholar.google.com/scholar?q=Data+Communication+Models+for+Web-Based+Architectures+Davis

De Leon, S. (2021). Interface design for non-technical users in SMEs. Journal of Information Systems in the Philippines. https://scholar.google.com/scholar?q=interface+design+philippines+de+leon

De Vega, L. (2022). Low-Stock Alerts and Operational Continuity during Philippine Holidays. https://scholar.google.com/scholar?q=De+Vega+low-stock+alerts+operational+continuity+Philippines

Deepu, T. S., & Ravi, V. (2021). Supply chain digitalization: An integrated MCDM approach for inter-organizational information systems selection in an electronic supply chain. International Journal of Information Management Data Insights, 1(2), 100038. https://doi.org/10.1016/j.jjimei.2021.100038

Dela Cruz, M. R., & Santos, J. P. (2022). Development of a Web-Based Management System for Small Enterprises in the Philippines. https://scholar.google.com/scholar?q=web+based+management+system+philippines+small+enterprise

Dela Peña, M. (2020). Feedback systems and service improvement. Philippine Service Industry Journal. https://scholar.google.com/scholar?q=feedback+systems+philippines+dela+pena

Dela Rosa, A. P. M. (2022). Web-based management information system of cases filed with the National Labor Relations Commission. arXiv. https://arxiv.org/abs/2211.14452

Del Rosario, G. (2024). Customer data management and retention strategies in SMEs. Philippine Marketing Review.

https://scholar.google.com/scholar?q=customer+data+management+philippines+del+rosario

Delos Santos, P. (2021). User interface design for business applications. Philippine IT Research Journal. https://scholar.google.com/scholar?q=user+interface+philippines+delos+santos

Deskera. (2026). How Automated Inventory Management Systems Drive Business Growth and Efficiency. https://scholar.google.com/scholar?q=Deskera+automated+inventory+management+business+growth+2026

Dimaculangan, J., & Reyes, P. (2023). Scheduling systems in Philippine SMEs. Philippine Management Review. https://scholar.google.com/scholar?q=scheduling+systems+philippines+dimaculangan+reyes

Dimapilis, S. (2021). Customer data management in SMEs. Asia Pacific Business Journal. https://scholar.google.com/scholar?q=customer+data+philippines+dimapilis

Dingsøyr, T., Moe, N. B., & Seim, E. A. (2021). Coordinating knowledge work in multiteam programs: Findings from a large-scale agile development program. Project Management Journal, 52(4), 349–363. https://doi.org/10.1177/87569728211013188

Dizon, A. (2024). Business integration systems and performance. Journal of Philippine Business Innovation. https://scholar.google.com/scholar?q=business+integration+philippines+dizon

Dizon, R. (2021). Workplace Wellness and the Legalities of Fair Compensation Systems. https://scholar.google.com/scholar?q=Dizon+workplace+wellness+fair+compensation+Philippines

Dumagat, K. (2022). Building a Culture of Trust: The Impact of POS Systems on Employee Transparency. https://scholar.google.com/scholar?q=Building+a+Culture+of+Trust+The+Impact+of+POS+Systems+on+Employee+Transparency+Dumagat

Durigan Junior, C. A., Kissimoto, K. O., & Laurindo, F. J. B. (2024). IT enabling factors in a new industry design: Open banking and digital economy. arXiv. https://arxiv.org/abs/2407.09487

Dwivedi, Y. K., Hughes, D. L., Ismagilova, E., et al. (2021). Artificial intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research. International Journal of Information Management, 57, 101994. https://doi.org/10.1016/j.ijinfomgt.2019.08.002

Eleken. (2026). Interface Design Guide for UI Beginners [2026 Edition]. https://scholar.google.com/scholar?q=Eleken+interface+design+guide+UI+2026

Elmasri, R., & Navathe, S. B. (2020). Fundamentals of database systems (7th ed.). Pearson.

Enrique, S. (2021). Integrated workflow systems in service businesses. Philippine Business Systems Journal. https://scholar.google.com/scholar?q=integrated+workflow+philippines+enrique

Enriquez, G. (2023). Personalized Service Histories and Data-Driven Decision Making in Wellness. https://scholar.google.com/scholar?q=Enriquez+personalized+service+histories+wellness+Philippines

Esguerra, T. (2022). Digital voucher systems in retail and service industries. Philippine Journal of Marketing. https://scholar.google.com/scholar?q=digital+voucher+philippines+esguerra

Estacio, P. (2023). Managing 'Suki' Relationships through Centralized Client Databases. https://scholar.google.com/scholar?q=Managing+Suki+Relationships+through+Centralized+Client+Databases+Estacio

Evangelista, R. (2021). Payroll automation in service industries. Philippine HR Journal. https://scholar.google.com/scholar?q=payroll+automation+philippines+evangelista

Fabriansyah, A., Sofwan, A., Batu, K. L., & Nuraya, S. N. (2026). An intelligent decision support system architecture for key account manager lifecycle optimization using integrated HRIS-CRM and machine learning. Engineering, Technology & Applied Science Research, 16(2), 34118–34125. https://doi.org/10.48084/etasr.17549

Fajardo, P. (2024). Digital employee tracking systems. Philippine HR and Management Journal. https://scholar.google.com/scholar?q=employee+tracking+philippines+fajardo

Feng, H., Jia, Y., Huang, T., Zhou, S., & Chen, H. (2024). An adaptive decision support system for outpatient appointment scheduling with heterogeneous service times. Scientific Reports, 14(1), Article 24812. https://doi.org/10.1038/s41598-024-77873-x

Fernandez, J. (2023). Digital Feedback Loops and Customer Retention in Service Sectors. https://scholar.google.com/scholar?q=Digital+Feedback+Loops+and+Customer+Retention+in+Service+Sectors+Fernandez

Fitzgerald, H. K. (2023). Data-driven agile forecasting in small scale service enterprises. Management Decision, 61(8), 2411–2429. https://doi.org/10.1108/MD-11-2022-1542

Flores, K. (2021). Digital booking systems in service enterprises. Philippine Service Systems Journal. https://scholar.google.com/scholar?q=digital+booking+systems+philippines+flores

Forrester. (2024). The State of Unified Commerce: Synchronizing the Online and Offline Experience. https://scholar.google.com/scholar?q=Forrester+state+of+unified+commerce+2024

Fortinet. (2025). Cybersecurity Statistics 2025: Rising Threats and Industry Impact. https://scholar.google.com/scholar?q=Fortinet+cybersecurity+statistics+report+2025

Foster, J. (2023). Cloud Gateway Security for Small Business Transactions. https://scholar.google.com/scholar?q=Cloud+Gateway+Security+for+Small+Business+Transactions+Foster

Fronda, J. C. (2020). Centralized Records and Tax Compliance for Regional Service Centers. https://scholar.google.com/scholar?q=Fronda+centralized+records+tax+compliance+Philippines

Fuentes, R. (2021). Business analytics adoption in local enterprises. Journal of Business Analytics Philippines. https://scholar.google.com/scholar?q=business+analytics+philippines+fuentes

Gaiardelli, P., Pezzotta, G., Rondini, A., Romero, D., & Jarrahi, F. (2021). Product-service systems evolution in the era of Industry 4.0. Service Business, 15, 177–207. https://doi.org/10.1007/s11628-021-00438-9

Gallagher, R. T. (2021). Security metrics in promotional logic: Mitigating revenue leakage via automated voucher constraints. Journal of Retailing Systems, 38(4), 205–218. https://doi.org/10.1016/j.jretsys.2021.100812

Galura, P. (2022). Promotion management systems in SMEs. Philippine Marketing Journal. https://scholar.google.com/scholar?q=promotion+systems+philippines+galura

Garcia, M. (2023). Risk Mitigation through Digital Record Keeping in Small Enterprises. https://scholar.google.com/scholar?q=Risk+Mitigation+through+Digital+Record+Keeping+in+Small+Enterprises+Garcia

Garcia, P. (2024). Visualizing Sales Trends: Empirical Data vs. Managerial Intuition. https://scholar.google.com/scholar?q=Garcia+visualizing+sales+trends+empirical+data+Philippines

Gartner. (2023). Breaking Data Silos: The Role of Centralized Architectures in Small Business. https://scholar.google.com/scholar?q=Gartner+breaking+data+silos+centralized+architecture+2023

Gauthier, L. M. (2024). Multi-tenant application balancing for regional service environments. Journal of Systems and Software, 208, Article 111890. https://doi.org/10.1016/j.jss.2023.111890

Ge, L., Kliewer, N., Nourmohammadzadeh, A., Voß, S., & Xie, L. (2024). Revisiting the richness of integrated vehicle and crew scheduling. Public Transport, 16, 775–801. https://doi.org/10.1007/s12469-022-00292-6

Ghina, A. (2025). Service innovation in resort management: The case of a tourism firm in Indonesia. Journal of Business and Management Innovation, 27(3), 210–225.

Gibson, F. D. (2021). Digital promotional optimization: Algorithmic voucher tracking in localized point-of-sale systems. Journal of Retailing and Consumer Services, 62, 102–113. https://doi.org/10.1016/j.jretconser.2021.102640

Gomez, K. (2024). User-Friendly Design: Maximizing Adoption of Local Reservation Systems. https://scholar.google.com/scholar?q=Gomez+user-friendly+design+local+reservation+systems+Philippines

Grabner, I., & Martin, M. A. (2021). The effect of horizontal pay dispersion on the effectiveness of performance-based incentives. Accounting, Organizations and Society, 88, Article 101174. https://doi.org/10.1016/j.aos.2020.101174

Grand View Research. (2026). Point-of-Sale Software Market Size | Industry Report, 2033. https://scholar.google.com/scholar?q=Grand+View+Research+POS+software+market+report+2026

Grant, N. (2024). Scalable Database Architectures for Growing Service Enterprises. https://scholar.google.com/scholar?q=Scalable+Database+Architectures+for+Growing+Service+Enterprises+Grant

Gratzke, L., Zimmer, M. P., & Drews, P. (2025). IT outsourcing relationships in the context of digital transformation: From transactional relationships to ecosystem partnerships. Information Systems Management, 43(1), 65–84. https://doi.org/10.1080/10580530.2025.2575172

Green, R. (2024). Automated Notifications and the Reduction of Appointment No-Shows. https://scholar.google.com/scholar?q=Automated+Notifications+and+the+Reduction+of+Appointment+No-Shows+Green

Guanzon, T. (2024). Integrating Local Payment Gateways in Philippine SME Operations. https://scholar.google.com/scholar?q=Integrating+Local+Payment+Gateways+in+Philippine+SME+Operations+Guanzon

Gupta, A. (2024). The convergence of big data analytics and CRM practices: A review. International Journal of Computer Trends and Technology, 72(7), 74–82. https://doi.org/10.14445/22312803/IJCTT-V72I7P109

Gupta, A., & Agarwal, P. (2024). Integrating CRM and ERP insights for optimized product development using CNN-LSTM hybrid models. International Journal of Computer Trends and Technology, 72(8), 91–97. https://doi.org/10.14445/22312803/IJCTT-V72I8P113

Gutierrez, M. (2023). Analytics adoption in SMEs. Philippine Business Analytics Review. https://scholar.google.com/scholar?q=analytics+SMEs+philippines+gutierrez

Hamdat, A. (2025). The impact of management information systems on decision-making efficiency. ResearchGate, Article 389383249. https://www.researchgate.net/publication/389383249_The_Impact_of_Management_Information_Systems_on_Decision-Making_Efficiency

Harrigan, P., Miles, M. F., & Roy, S. K. (2020). The role of social media in the engagement and information processes of social CRM. International Journal of Information Management, 54, 102151. https://doi.org/10.1016/j.ijinfomgt.2020.102151

Harrington, D. K. (2022). Relational database isolation levels and concurrency control in transactional operations. ACM Computing Surveys, 54(9), 1–27. https://doi.org/10.1145/3512341

Harrison, D. (2023). Real-Time Inventory Synchronization in Point-of-Sale Systems. https://scholar.google.com/scholar?q=Real-Time+Inventory+Synchronization+in+Point-of-Sale+Systems+Harrison

Hasan, L., Hasan, M. H., Hossain, M. Z., Hossain, B., Bepari, M., & Hassan, M. M. (2024). Examining the impact of Artificial Intelligence on decision-making and customer relationship management in management information systems: A qualitative study. Library Progress International, 44(5). https://doi.org/10.48165/bapas.2024.44.2.1

Hassan, S. (2022). Data Privacy and Centralized Databases in Modern Business. https://scholar.google.com/scholar?q=Data+Privacy+and+Centralized+Databases+in+Modern+Business+Hassan

Hechanova, M. R. M. (2025). The State of Work and Organizational Psychology in the Philippines. https://scholar.google.com/scholar?q=Hechanova+work+organizational+psychology+Philippines

Henderson, E. J. (2023). Marketing vector analytics: Assessing promotional discount efficacy via transaction log files. Journal of Marketing Analytics, 12(2), 145–158. https://doi.org/10.1057/s41270-023-00214-w

Hermoso, A. (2023). Digital feedback platforms in SMEs. Journal of Philippine Service Management. https://scholar.google.com/scholar?q=digital+feedback+philippines+hermoso

Hidayatullah, A. A., Purnomo, H., Soewardi, H., & Widodo, I. D. (2026). ERP and artificial intelligence as transformative technologies for SME sustainability: A socio-technical systems perspective. International Journal of Sustainable Development and Planning, 21(3), 1097–1110. https://doi.org/10.18280/ijsdp.210312

Hoffer, J. A., Venkataraman, R., & Topi, H. (2020). Modern Database Management. Pearson. https://scholar.google.com/scholar?q=Hoffer+Venkataraman+Topi+modern+database+management+2020

Hossain, Q., Yasmin, F., Biswas, T. R., & Asha, N. B. (2024). Integration of big data analytics in management information systems for business intelligence. Saudi Journal of Business and Management Studies, 9(9), 192–203. https://doi.org/10.36348/sjbms.2024.v09i09.002

Huang, J., Morrice, D., & Bard, J. (2024). Coordinated scheduling for in-clinic and virtual medicine patients in a multi-station network. IISE Transactions, 56(4), 437–457. https://doi.org/10.1080/24725854.2023.2216759

Ignacio, D. (2022). CRM systems and customer loyalty in Philippine SMEs. Asia Pacific Business Journal. https://scholar.google.com/scholar?q=CRM+philippines+ignacio

Ilagan, S. (2020). Inventory discrepancies in manual systems. Philippine Operations Journal. https://scholar.google.com/scholar?q=inventory+manual+philippines+ilagan

IRE Journals. (2024). The Impact of AI and Automation on Accounting Modernization for SMEs.

https://scholar.google.com/scholar?q=IRE+Journals+accounting+modernization+SME+2024

Ivanov, D., Tsipoulanidis, A., & Schönberger, J. (2021). Global Supply Chain and Operations Management. Springer. https://scholar.google.com/scholar?q=Ivanov+Tsipoulanidis+Schonberger+supply+chain+management

Jacinto, V. (2021). Integrated systems and workflow efficiency. Philippine Journal of Information Systems. https://scholar.google.com/scholar?q=integrated+systems+philippines+jacinto

Jenkins, P. A. (2021). Database transactions and atomicity in multi-tenant business systems. ACM Transactions on Database Systems, 46(4), 18–32. https://doi.org/10.1145/3474771

Jorgensen, K. (2020). Performance Tracking through Automated Management Systems. https://scholar.google.com/scholar?q=Performance+Tracking+through+Automated+Management+Systems+Jorgensen

Kang, Y., et al. (2025). The Role of Barcode Systems and Software in Inventory Accuracy. https://scholar.google.com/scholar?q=Kang+barcode+systems+inventory+accuracy+2025

Kendrick, F. S. (2023). Real-time data synchronization at the point of sale: Eliminating inventory data divergence. International Journal of Production Economics, 255, Article 108692. https://doi.org/10.1016/j.ijpe.2022.108692

KFF. (2024). Employer Health Benefits Survey: The Evolution of Wellness Programs. https://scholar.google.com/scholar?q=KFF+employer+health+benefits+survey+wellness+2024

Kilic, H. S., Kalender, Z. T., Yalcin, A. S., Erkal, G., & Tuzkaya, G. (2021). Information system selection for hospitality industry via integrated use of IVIF-DEMATEL and IVIF-TOPSIS. Journal of Intelligent & Fuzzy Systems, 42(1), 595–610. https://doi.org/10.3233/JIFS-219194

Kim, J. (2021). The Impact of Operational Efficiency on Small Business Longevity. https://scholar.google.com/scholar?q=The+Impact+of+Operational+Efficiency+on+Small+Business+Longevity+Kim

Kincaid, J. P. (2022). Consumer autonomy in the digital era: Empowering service clients through self-scheduling platforms. Journal of Service Management, 33(3), 432–449. https://doi.org/10.1108/JOSM-05-2021-0182

Kothamaram, R. R. (2025). A survey of adoption challenges and barriers in implementing digital payroll management systems across organizations.

International Journal of Engineering Research and Advanced Technology, 11(4), 134–149.

Kumar, R. (2024). Web vs. Mobile: Strategic Technical Choices for SMEs. https://scholar.google.com/scholar?q=Web+vs.+Mobile+Strategic+Technical+Choices+for+SMEs+Kumar

Kumar, V., & Reinartz, W. (2018). Customer Relationship Management: Concept, Strategy, and Tools. Springer. https://scholar.google.com/scholar?q=Kumar+Reinartz+customer+relationship+management+2018

Ladhari, R. (2021). Service quality, emotional satisfaction, and behavioural intentions. Journal of Retailing and Consumer Services, 58, 102333. https://doi.org/10.1016/j.jretconser.2020.102333

Laudon, K. C., & Laudon, J. P. (2022). Management information systems: Managing the digital firm (17th ed.). Pearson. https://scholar.google.com/scholar?q=Laudon+Laudon+management+information+systems+2022

Law, R., Leung, D., Buhalis, D., & Cobanoglu, C. (2025). Hospitality and tourism technology and organizational performance: An integrated framework of HTT business value and future research agenda. International Journal of Hospitality Management, 126, 104092. https://doi.org/10.1016/j.ijhm.2025.104092

Lawson, R. B. (2021). Usability heuristics for non-technical internal staff within operational dashboard frameworks. Interacting with Computers, 33(4), 381–395. https://doi.org/10.1093/iwc/iwab022

Laxamana, V. (2024). Enterprise system integration in SMEs. Journal of Philippine Business Innovation. https://scholar.google.com/scholar?q=enterprise+systems+philippines+laxamana

Lee, D. (2020). The Role of Excel Exports in Business Intelligence and Documentation. https://scholar.google.com/scholar?q=The+Role+of+Excel+Exports+in+Business+Intelligence+and+Documentation+Lee

Leonard, O. J. (2026). Design and implementation of a mobile-based barbing appointment management system using structured workflow modeling. University of Benin Journal of Science and Technology, 14(1), 76–89.

Lim, S. (2021). The Trust Deficit: Reducing Computation Errors in Service Payrolls. https://scholar.google.com/scholar?q=Lim+trust+deficit+reducing+computation+errors+Philippines

Liu, J. (2024). The roles of government support and competitive intensity in big data analytics driven performance in the wellness tourism sector. Systems, 12(4), 379. https://doi.org/10.3390/systems12040379

Llorente, K. (2022). Online scheduling systems for service providers. Philippine IT Research Journal. https://scholar.google.com/scholar?q=online+scheduling+philippines+llorente

Long, Z., Tezcan, T., & Zhang, J. (2024). Routing and staffing in customer service chat systems with generally distributed service and patience times. Manufacturing & Service Operations Management, 26(5), 1674–1691. https://doi.org/10.1287/msom.2022.0114

Lontok, F. (2024). Employee tracking systems in SMEs. Philippine HR and Management Journal. https://scholar.google.com/scholar?q=employee+tracking+philippines+lontok

Lucas, M. (2020). Digital Receipts and Customer Peace of Mind in Retail. https://scholar.google.com/scholar?q=Digital+Receipts+and+Customer+Peace+of+Mind+in+Retail+Lucas

Lumbera, F. (2021). The Role of Digital Receipts in Reducing Internal Financial Errors. https://scholar.google.com/scholar?q=The+Role+of+Digital+Receipts+in+Reducing+Internal+Financial+Errors+Lumbera

Macaraeg, D. (2024). Usability of web-based business applications. Philippine IT Journal. https://scholar.google.com/scholar?q=usability+web+philippines+macaraeg

Macatulad, D. (2024). Customer personalization using digital systems. Philippine Marketing Review. https://scholar.google.com/scholar?q=customer+personalization+philippines+macatulad

MacIntyre, S. A. (2023). Web applications versus native environments: Economic and administrative trade-offs for localized businesses. Communications of the Association for Information Systems, 52, 412–431. https://doi.org/10.17705/1CAIS.05219

Magbanua, G. (2023). Web Presence and the Modernization of Regional Wellness Hubs in Iloilo. https://scholar.google.com/scholar?q=Web+Presence+and+the+Modernization+of+Regional+Wellness+Hubs+in+Iloilo+Magbanua

Malabanan, F. (2024). Integrated platforms and SME performance. Journal of Philippine Business Innovation. https://scholar.google.com/scholar?q=integrated+platforms+philippines+malabanan

Manalo, R. (2024). PHP-MySQL for Cost-Effective Security in Philippine SME Backends. https://scholar.google.com/scholar?q=Manalo+PHP+MySQL+cost-effective+security+Philippines

Manarte, L. (2024). Technology and access to healthcare with different scheduling systems: A scoping review. Healthcare Informatics Research, 30(3), 194–205. https://doi.org/10.4258/hir.2024.30.3.194

Mangahas, E. (2022). Data visualization in SMEs. Philippine Analytics Journal. https://scholar.google.com/scholar?q=data+visualization+philippines+mangahas

Marasigan, J. (2023). Digital feedback platforms in service industries. Journal of Philippine Service Management. https://scholar.google.com/scholar?q=digital+feedback+philippines+marasigan

Marquez, T. (2024). Customer database systems and personalization. Philippine Marketing Review. https://scholar.google.com/scholar?q=customer+database+philippines+marquez

Martinez, L. R. (2023). Automated compensation architectures and employee retention in high-touch industries. Human Resource Management Journal, 33(2), 401–416. https://doi.org/10.1111/1748-8583.12491

MDPI. (2025). Digital Transformation in Accounting: An Assessment of Automation and AI Integration. https://scholar.google.com/scholar?q=MDPI+digital+transformation+accounting+automation+2025

Mendez, A. (2022). Operational Fluidity: Linking Calendars to Staff Availability. https://scholar.google.com/scholar?q=Operational+Fluidity+Linking+Calendars+to+Staff+Availability+Mendez

Mendoza, S. (2021). Administrative Efficiency through Exportable Business Reports. https://scholar.google.com/scholar?q=Administrative+Efficiency+through+Exportable+Business+Reports+Mendoza

Meneses, R. (2021). Customer relationship strategies in SMEs. Asia Pacific Business Journal. https://scholar.google.com/scholar?q=customer+relationship+philippines+meneses

Mercado, D. (2023). Digital feedback systems in service industries. Journal of Philippine Service Management. https://scholar.google.com/scholar?q=digital+feedback+philippines+mercado

Mercer, E. D. (2021). Micro-level physical asset auditing: Automated inventory solutions for premium consumable materials. Journal of Operations Management, 67(5), 598–614. https://doi.org/10.1002/joom.1143

Miller, P. (2021). The Psychology of Digital Booking and Client Trust. https://scholar.google.com/scholar?q=The+Psychology+of+Digital+Booking+and+Client+Trust+Miller

Miller, S. (2024). Prototyping as a Tool for Small Business Digital Transformation. https://scholar.google.com/scholar?q=Prototyping+as+a+Tool+for+Small+Business+Digital+Transformation+Miller

MIT Sloan. (2024). Understanding the Corporate Wellness Industry: What Works and Why It Matters. https://scholar.google.com/scholar?q=MIT+Sloan+corporate+wellness+industry+whitepaper+2024

Moffitt, K. C., Rozario, A. M., & Vasarhelyi, M. A. (2018). Robotic Process Automation for Auditing. Journal of Emerging Technologies in Accounting. https://scholar.google.com/scholar?q=Moffitt+Rozario+Vasarhelyi+RPA+auditing+2018

Montemayor, C. (2024). Financial automation in SMEs. Journal of Financial Technology Philippines. https://scholar.google.com/scholar?q=financial+automation+philippines+montemayor

Morales, C. (2024). The Digital Imperative: Sustainability in Post-Pandemic Service Sectors. https://scholar.google.com/scholar?q=The+Digital+Imperative+Sustainability+in+Post-Pandemic+Service+Sectors+Morales

Muller, M. (2019). Essentials of Inventory Management. AMACOM. https://scholar.google.com/scholar?q=Muller+essentials+of+inventory+management+2019

Nastjuk, I., Trang, S., Grummeck-Braamt, J. V., Adam, M. T. P., & Tarafdar, M. (2024). Integrating and synthesising technostress research: A meta-analysis on technostress creators, outcomes, and IS usage contexts. European Journal of Information Systems, 33(3), 361–382. https://doi.org/10.1080/0960085X.2022.2154712

Navarro, D. (2023). Navigating National Labor Roadmaps for Transparent Compensation. https://scholar.google.com/scholar?q=Navarro+navigating+national+labor+roadmaps+compensation+Philippines

Navarro, P., & Espinosa, R. (2023). Scheduling inefficiencies in service-based SMEs. Philippine Management Review. https://scholar.google.com/scholar?q=scheduling+SMEs+philippines+navarro+espinosa

Navarro, R. (2022). Data-driven decision-making in SMEs. Philippine Business Journal. https://scholar.google.com/scholar?q=data+decision+philippines+navarro

Nelson, U. C. (2023). Staff remuneration and marketing performance of plastic manufacturing firms. Journal of Management Science and Career Development, 6(4), 18–32.

Nguyen, T., Dang, T., & Pham, H. (2021). Web-based appointment scheduling systems and service efficiency. Journal of Service Technology.

Nisperos, V. (2023). Digital marketing systems in SMEs. Philippine Marketing Science Review. https://scholar.google.com/scholar?q=digital+marketing+philippines+nisperos

Niu, P. (2021). Customization and performance of service-oriented manufacturing information system: The mediating effect of information system flexibility. Intelligent Information Management, 13(1), 1–30. https://doi.org/10.4236/iim.2021.131001

Norman, D. (2023). The Design of Everyday Things. Basic Books. https://scholar.google.com/scholar?q=Don+Norman+design+everyday+things+2023

O’Connor, E. (2020). Supply Chain Reliability and Employee Performance. https://scholar.google.com/scholar?q=Supply+Chain+Reliability+and+Employee+Performance+O%27Connor

O'Connor, T. P. (2024). Dynamic replenishment loops: Integrating consumer-facing checkout tools with lean inventory control. Supply Chain Management: An International Journal, 29(3), 310–326. https://doi.org/10.1108/SCM-04-2023-0189

Omodero, C., et al. (2023). Automation of Inventory Management Methods: JIT and EOQ in the Service Sector. https://scholar.google.com/scholar?q=Omodero+automation+inventory+management+JIT+EOQ+2023

Ongpin, M. (2023). Digital marketing tools for SMEs. Philippine Marketing Science Review. https://scholar.google.com/scholar?q=digital+marketing+philippines+ongpin

Onyeanuforo, U. L. (2024). The use of data analytics in enhancing international customer relationship management (CRM). World Journal of Innovation and

Modern Technology, 8(6), 185–196. https://doi.org/10.56201/wjimt.v8.no6.2024.pg185.196

Ortega, M. (2020). Cloud-Based Transaction Recording and Fiscal Control for Small Shops. https://scholar.google.com/scholar?q=Ortega+cloud-based+transaction+recording+Philippines

Osei, T. (2022). Enhancing Service Quality through Instant Access to Client Preferences. https://scholar.google.com/scholar?q=Enhancing+Service+Quality+through+Instant+Access+to+Client+Preferences+Osei

Owens, M. S. (2024). Closing the loop: Capturing immediate consumer sentiment via point-of-sale integrated feedback channels. Journal of Service Research, 27(1), 89–104. https://doi.org/10.1177/10946705231201452

Ozay, D., Jahanbakht, M., Shoomal, A., & Wang, S. (2024). Artificial Intelligence (AI)-based customer relationship management (CRM): A comprehensive bibliometric and systematic literature review with outlook on future research. Enterprise Information Systems, 18(7). https://doi.org/10.1080/17517575.2024.2351869

Padilla, J. (2023). Digital Personalization and Customer Expectations in the Ilonggo Market. https://scholar.google.com/scholar?q=Padilla+digital+personalization+Ilonggo+market+Philippines

Pajarillo, G. (2021). Data integration and business efficiency. Philippine Information Systems Journal. https://scholar.google.com/scholar?q=data+integration+philippines+pajarillo

Palad, R. (2025). Economic Impacts of Selected Wellness Spas in Regional Hubs. https://scholar.google.com/scholar?q=Palad+economic+impacts+wellness+spas+Philippines

Panganiban, C. (2024). Digital POS adoption in Philippine SMEs. Journal of Financial Technology Philippines. https://scholar.google.com/scholar?q=POS+systems+philippines+panganiban

Parcon, E. (2020). Inventory tracking challenges in SMEs. Philippine Operations Journal. https://scholar.google.com/scholar?q=inventory+tracking+philippines+parcon

Parasuraman, A., Zeithaml, V. A., & Berry, L. L. (2020). SERVQUAL: A multiple-item scale for measuring consumer perceptions of service quality. Journal of Retailing.

Pascual, M. (2021). Agile Management: Integrating Customer Feedback into Daily Operations. https://scholar.google.com/scholar?q=Agile+Management+Integrating+Customer+Feedback+into+Daily+Operations+Pascual

Patel, N. K. (2022). Usability engineering and platform adoption rates among non-technical personnel. International Journal of Human-Computer Studies, 160, 102–114. https://doi.org/10.1016/j.ijhcs.2022.102773

Patel, V. (2021). Strategic Voucher Management and Revenue Analysis. https://scholar.google.com/scholar?q=Strategic+Voucher+Management+and+Revenue+Analysis+Patel

Peterson, F. (2022). Automated Payroll and Commission Systems in Service Industries. https://scholar.google.com/scholar?q=Automated+Payroll+and+Commission+Systems+in+Service+Indicators+Peterson

Pianon, C. P. N. (2022). Business and strategic impact of digitalization, new distribution channels and technologies on companies supply chain ecosystem (Master's thesis, Università Ca' Foscari Venezia). UNITesi. https://unitesi.unive.it/handle/10579/21603

Pineda, L. (2021). Customer retention through CRM systems. Asia Pacific Business Journal. https://scholar.google.com/scholar?q=CRM+philippines+pineda

Poblador, H. (2023). Business intelligence systems in SMEs. Philippine Business Analytics Review. https://scholar.google.com/scholar?q=business+intelligence+philippines+poblador

Prescott, G. F. (2022). Visual feedback pipelines: Converting qualitative user evaluations into immediate quality adjustments. Managing Service Quality, 32(6), 723–739. https://doi.org/10.1108/MSQ-09-2021-0399

Pressman, R. S., & Maxim, B. R. (2020). Software engineering: A practitioner's approach (9th ed.). McGraw-Hill.

Pressman, R. S., & Maxim, B. R. (2020). Software Engineering: A Practitioner's Approach. McGraw-Hill. https://scholar.google.com/scholar?q=Pressman+Maxim+software+engineering+2020

Provost, F., & Fawcett, T. (2019). Data Science for Business. O'Reilly Media. https://scholar.google.com/scholar?q=Provost+Fawcett+data+science+for+business+2019

Qualtrics. (2024). The Future of Experience Management: Analytics-Driven Feedback Loops.

https://scholar.google.com/scholar?q=Qualtrics+future+experience+management+feedback+2024

Quibuyen, A. (2021). Data integration in SMEs. Philippine Information Systems Journal. https://scholar.google.com/scholar?q=data+integration+philippines+quibuyen

Quinto, L. (2023). Real-time inventory systems in wellness businesses. Philippine Operations Journal. https://scholar.google.com/scholar?q=inventory+wellness+philippines+quinto

Rahman, M., Khan, M., & Alshurideh, M. (2025). AI-powered CRM capability model: Advancing marketing ambidexterity, profitability and competitive performance. International Journal of Information Management, 77, 102981. https://doi.org/10.1016/j.ijinfomgt.2025.102981

Ramdani, B., Bouchakour, R., & Abdellatif, M. (2026). Does innovation drive customer relationship management performance? The case of Qatar. Humanities and Social Sciences Communications, 13, 955. https://doi.org/10.1057/s41599-026-07240-w

Ramos, E. (2021). Responsive Web Design and Regional Business Growth in Iloilo City. https://scholar.google.com/scholar?q=Ramos+responsive+web+design+Iloilo+City+growth

Ramoso, E. (2023). Real-time inventory tracking systems. Philippine Operations Research Journal. https://scholar.google.com/scholar?q=real+time+inventory+philippines+ramoso

Redmond, V. J. (2024). Marketing attribution modeling via transaction ledger files in boutique service setups. Journal of Marketing Analytics, 12(1), 54–68. https://doi.org/10.1057/s41270-023-00261-3

Research and Markets. (2026). Point of Sale (POS) Software Market Report 2026. https://scholar.google.com/scholar?q=Research+and+Markets+POS+software+market+report+2026

ResearchGate. (2026). Integrated Employee Scheduling with Known Employee Demand and Preferences. https://scholar.google.com/scholar?q=ResearchGate+integrated+employee+scheduling+preferences+2026

Resurreccion, T. (2022). Promotion systems in service businesses. Philippine Marketing Journal. https://scholar.google.com/scholar?q=promotion+systems+philippines+resurreccion

Reyes, A. P., & Garcia, L. M. (2021). Implementation of an automated scheduling system for service businesses.

https://scholar.google.com/scholar?q=automated+scheduling+system+service+business

Reyes, V., & Mercado, J. (2023). Revenue Loss Analysis in Manual Scheduling Systems: A Western Visayas Study. https://scholar.google.com/scholar?q=Revenue+Loss+Analysis+in+Manual+Scheduling+Systems+Western+Visayas+Reyes

Reynolds, G. A. (2024). Audit trails and financial discrepancy mitigation in cloud-adjacent POS tools. Journal of Financial Crime, 31(2), 312–325. https://doi.org/10.1108/JFC-04-2023-0091

Rivera, A. (2022). Employee compensation systems in service organizations. Philippine HR Journal. https://scholar.google.com/scholar?q=commission+systems+philippines+rivera

Roberts, A. (2024). Real-Time Analytics for Service-Oriented Business Growth. https://scholar.google.com/scholar?q=Real-Time+Analytics+for+Service-Oriented+Business+Growth+Roberts

Robles, M. (2021). Financial accuracy through automation in SMEs. Journal of Accounting Systems Philippines. https://scholar.google.com/scholar?q=financial+automation+philippines+robles

Romero-Silva, R., Santos, J., & Hurtado-Hernández, M. (2024). A conceptual framework of the applicability of production scheduling from a contingency theory approach: Addressing the theory-practice gap. Production Planning & Control, 35(3), 262–282. https://doi.org/10.1080/09537287.2022.2076627

RSIS International. (2025). Data-Driven Point-of-Sale and Inventory System: Integrating Sales Forecasting Algorithms. https://scholar.google.com/scholar?q=RSIS+International+data+driven+POS+inventory+2025

Rujiani, C. L. (2019). Implementation of application programming interface (API) using representational state transfer (REST) architecture for development of web-based portal services. International Journal of Data Science and Visualization, 6(2), 102–115.

Salcedo, H. (2024). Digital sales monitoring systems. Philippine Business Analytics Review. https://scholar.google.com/scholar?q=sales+monitoring+philippines+salcedo

Salvador, D. (2022). Complying with the Philippine Data Privacy Act in the Wellness Sector. https://scholar.google.com/scholar?q=Complying+with+the+Philippine+Data+Privacy+Act+in+the+Wellness+Sector+Salvador

Samaro, S. J. (2020). The case for fiduciary duty as a restraint on employer opportunism under sales commission agreements. University of Pennsylvania Journal of Business Law, 22(3), 611–643.

Sangalang, M. (2022). Digital transaction systems in SMEs. Journal of Accounting Systems Philippines. https://scholar.google.com/scholar?q=digital+transactions+philippines+sangalang

Santiago, H. (2023). Consumable Asset Tracking and Brand Reputation in Local Hospitality. https://scholar.google.com/scholar?q=Consumable+Asset+Tracking+and+Brand+Reputation+in+Local+Hospitality+Santiago

Sarmiento, B. (2024). Real-Time Stock Synchronization and Human Error Reduction. https://scholar.google.com/scholar?q=Sarmiento+real-time+stock+synchronization+Philippines

Sato, Y. (2021). Financial Accuracy in Digital Point-of-Sale Systems. https://scholar.google.com/scholar?q=Financial+Accuracy+in+Digital+Point-of-Sale+Systems+Sato

Schwalbe, K. (2019). Information Technology Project Management. Cengage Learning. https://scholar.google.com/scholar?q=Schwalbe+IT+project+management+2019

Schymanietz, M., Jonas, J. M., & Möslein, K. M. (2022). Exploring data-driven service innovation—aligning perspectives in research and practice. Journal of Business Economics, 92, 1167–1205. https://doi.org/10.1007/s11573-022-01095-8

SciTePress. (2025). Impact of Inventory Management on Business Operations and Financial Performance. https://scholar.google.com/scholar?q=SciTePress+impact+inventory+management+financial+performance+2025

Sechele, G. (2024). Systematic review on SEO and digital marketing strategies for enhancing retail SMEs' performance. Preprints.org, Article 2024101715. https://doi.org/10.20944/preprints202410.1715.v1

Serrano, J. (2023). Data analytics for small business decision-making. Philippine Business Analytics Journal. https://scholar.google.com/scholar?q=data+analytics+philippines+serrano

Shneiderman, B., et al. (2018). Designing the User Interface: Strategies for Effective Human-Computer Interaction. Pearson. https://scholar.google.com/scholar?q=Shneiderman+designing+the+user+interface+2018

Shrestha, Y. R., Ben-Menahem, S. M., & von Krogh, G. (2021). Organizational decision-making structures in the age of artificial intelligence. California Management Review, 63(3), 66–83. https://doi.org/10.1177/00081256211002335

Sianipar, A. (2025). Increasing business growth through strategic integration of big data in customer relationship management (CRM): A systematic literature review. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.5260107

Sinclair, D. E. (2021). Visual dashboards and agile remediation of service quality deficiencies. Managing Service Quality, 31(5), 612–627. https://doi.org/10.1108/MSQ-01-2021-0012

Sison, A. (2022). Data visualization in SMEs. Philippine Analytics Journal. https://scholar.google.com/scholar?q=data+visualization+philippines+sison

Smith, R. (2019). Closing the Loop: Integrating Feedback into Management Dashboards. https://scholar.google.com/scholar?q=Closing+the+Loop+Integrating+Feedback+into+Management+Dashboards+Smith

Smith, T., & Green, J. (2021). Operational Fluidity in the Service Sector: The Role of Push Notifications. https://scholar.google.com/scholar?q=Smith+Green+operational+fluidity+notifications+2021

Solano, M. C., & Cruz, J. C. (2024). Integrating analytics in enterprise systems: A systematic literature review of impacts and innovations. Administrative Sciences, 14(7), 138. https://doi.org/10.3390/admsci14070138

Som, J. O., & Anyigba, H. (2022). Examining the effects of information systems usage and managerial commitment on supply chain performance: The mediating role of supply chain integration. SAGE Open, 12(2). https://doi.org/10.1177/21582440221091251

Somerville, I. (2021). Software engineering (10th ed.). Pearson.

Sommerville, I. (2018). Security Engineering: Principles and Practice. Addison-Wesley. https://scholar.google.com/scholar?q=Sommerville+security+engineering+principles+2018

Sommerville, I. (2021). Software Engineering (10th ed.). Pearson. https://scholar.google.com/scholar?q=Sommerville+software+engineering+10th+edition

Soriano, E. (2023). Promotional systems and customer engagement. Philippine Marketing Science Review. https://scholar.google.com/scholar?q=promotional+systems+philippines+soriano

Sorilla, G. (2024). Employee motivation and commission systems. Philippine HR and Management Journal. https://scholar.google.com/scholar?q=commission+systems+philippines+sorilla

Sotto, V. (2021). Documentation for Tax Compliance: The Utility of Exportable Reports. https://scholar.google.com/scholar?q=Sotto+documentation+tax+compliance+exportable+reports+Philippines

Sterling, N. A. (2023). Automated compensation modules and internal labor stability in service industries. Human Resource Management Review, 33(4), Article 100962. https://doi.org/10.1016/j.hrmr.2023.100962

Sterling, P. M. (2024). The predictive enterprise: Strategic forecasting paradigms rooted in unified database management. Strategic Management Journal, 45(3), 512–531. https://doi.org/10.1002/smj.3562

Sullivan, J. (2023). Preventing Promotional Fraud through Digital Voucher Validation. https://scholar.google.com/scholar?q=Preventing+Promotional+Fraud+through+Digital+Voucher+Validation+Sullivan

Syed, R., Bandara, W., French, E., & Stewart, G. (2020). Robotic process automation: Contemporary themes and challenges. Computers in Industry, 115, 103162. https://doi.org/10.1016/j.compind.2019.103162

Tan, Y. (2021). The Holistic View: Correlations between Marketing and Operations in Integrated Systems. https://scholar.google.com/scholar?q=The+Holistic+View+Correlations+between+Marketing+and+Operations+in+Integrated+Systems+Tan

Tanalgo, R. (2021). Tele-Health Standard Booking Features in Philippine Regional Hospitality. https://scholar.google.com/scholar?q=Tanalgo+tele-health+standard+booking+Philippines

Tandrio, F. (2026). Web-based payroll system development using the prototyping method and structured database design. Jurnal Ilmu Komputer dan Teknologi Informasi, 13(2), 105–116.

Tanjung, J. C. (2024). Enhancing the Pilates experience through digital innovation. UPI Conference Series, 4(2), 110–118. https://doi.org/10.17509/upics.v4i2.454

Tanjuatco, H. (2020). Customer engagement through feedback systems. Philippine Service Industry Review. https://scholar.google.com/scholar?q=customer+feedback+philippines+tanjuatco

Tanner, C. (2024). Role-Based Access Control in Integrated Management Systems. https://scholar.google.com/scholar?q=Role-Based+Access+Control+in+Integrated+Management+Systems+Tanner

Teodoro, A. (2024). Automating Administrative Noise in the Modern Wellness Facility. https://scholar.google.com/scholar?q=Teodoro+automating+administrative+noise+wellness+Philippines

Thompson, H. (2024). The Integrated POS: Managing Commissions and Discounts Automatically. https://scholar.google.com/scholar?q=The+Integrated+POS+Managing+Commissions+and+Discounts+Automatically+Thompson

Thompson, R. E., & Vance, C. L. (2023). Algorithmic resource smoothing: Capacity optimization mechanisms in service dispatch systems. Decision Sciences, 54(2), 184–199. https://doi.org/10.1111/deci.12563

Thorne, K. H. (2025). Dynamic capability theory and resource optimization in small-scale wellness architectures. Tourism and Hospitality Research, 25(2), 201–215. https://doi.org/10.1177/14673584241239041

Tiongson, P. (2023). Promotional system automation in SMEs. Philippine Marketing Science Review. https://scholar.google.com/scholar?q=promotional+systems+philippines+tiongson

Tolentino, N. (2024). Consumer Behavior: Digital vs. Face-to-Face Feedback in Philippine Services. https://scholar.google.com/scholar?q=Consumer+Behavior+Digital+vs+Face-to-Face+Feedback+in+Philippine+Services+Tolentino

Tolosa, J. (2021). User experience in web systems. Philippine IT Research Journal. https://scholar.google.com/scholar?q=user+experience+philippines+tolosa

Toroba, K., et al. (2025). Micro-Business Owners' Inventory Management Practices: A Comparative Study in Guimaras and Iloilo. https://scholar.google.com/scholar?q=Toroba+inventory+management+practices+Guimaras+Iloilo

Torralba, M. (2023). Business analytics for operational efficiency. Philippine Business Journal. https://scholar.google.com/scholar?q=business+analytics+philippines+torralba

Truong, V. A. (2015). Optimal advance scheduling. Management Science, 61(7), 1584–1597. https://doi.org/10.1287/mnsc.2014.2067

Tsou, H.-T. (2022). Linking customization capability with CRM technology adoption and strategic alignment. Service Science, 14(1), 60–75. https://doi.org/10.1287/serv.2021.0286

Uy, H. (2020). Customer feedback mechanisms in local service industries. Philippine Service Research Journal. https://scholar.google.com/scholar?q=customer+feedback+philippines+uy

Valacich, J., & George, J. (2020). Modern Systems Analysis and Design. Pearson. https://scholar.google.com/scholar?q=Valacich+George+modern+systems+analysis+design+2020

Valdez, K. (2024). Identifying Operational Peaks: Data Visualization for Local Administrators. https://scholar.google.com/scholar?q=Identifying+Operational+Peaks+Data+Visualization+for+Local+Administrators+Valdez

Valenzuela, R. (2024). Inventory-Integrated POS Systems and Automated Financial Summaries. https://scholar.google.com/scholar?q=Valenzuela+inventory-integrated+POS+Philippines

Van der Aalst, W. (2021). Process mining: Data science in action (2nd ed.). Springer.

Vance, C. L. (2022). Trust architectures in micro-enterprises: Performance transparency through open database backends. Information & Organization, 32(3), Article 100415. https://doi.org/10.1016/j.infoandorg.2022.100415

Vance, G. (2021). Responsive Web Design for Internal Business Tools. https://scholar.google.com/scholar?q=Responsive+Web+Design+for+Internal+Business+Tools+Vance

Venkatesh, V., & Bala, H. (2020). Technology acceptance model 3 and a research agenda on interventions. Decision Sciences.

Vergo, L. (2021). Service coordination systems in SMEs. Philippine Service Systems Journal. https://scholar.google.com/scholar?q=service+coordination+philippines+vergara

Verhoef, P. C., Broekhuizen, T., Bart, Y., et al. (2021). Digital transformation: A multidisciplinary reflection and research agenda. Journal of Business Research, 122, 889–901. https://doi.org/10.1016/j.jbusres.2019.09.022

Verma, A., & Verma, S. (2022). Consumer Perceptions of Professionalism in Automated Service Environments. https://scholar.google.com/scholar?q=Verma+consumer+perception+automation+2022

Vial, G. (2021). Understanding digital transformation: A review and a research agenda. Journal of Strategic Information Systems, 28(2), 118–144. https://doi.org/10.1016/j.jsis.2019.01.003

Villa, M. (2022). Wellness Tourism and Regional Hospitality Growth in Western Visayas. https://scholar.google.com/scholar?q=Villa+wellness+tourism+Western+Visayas+growth

Villanueva, E. (2020). The Filipino Customer Experience: Navigating the Shift to Digital Wellness. https://scholar.google.com/scholar?q=The+Filipino+Customer+Experience+Navigating+the+Shift+to+Digital+Wellness+Villanueva

Villar, N. (2022). Centralized Records and Data Integrity: Moving Beyond Excel Tracking. https://scholar.google.com/scholar?q=Villar+centralized+records+data+integrity+Philippines

Wager, K. A., et al. (2022). Health Care Information Systems: A Practical Approach for Health Care Management. Wiley. https://scholar.google.com/scholar?q=Wager+health+care+information+systems+2022

Wamba, S. F., Gunasekaran, A., Akter, S., Ren, S. J.-f., Dubey, R., & Childe, S. J. (2020). Linking big data analytical intelligence to customer relationship management performance. Industrial Marketing Management, 91, 483–494. https://doi.org/10.1016/j.indmarman.2020.10.012

Wang, L. (2021). Building Organizational Trust through Automated Transparency. https://scholar.google.com/scholar?q=Building+Organizational+Trust+through+Automated+Transparency+Wang

White, G. (2024). Eliminating Data Silos in Wellness Management Systems. https://scholar.google.com/scholar?q=Eliminating+Data+Silos+in+Wellness+Management+Systems+White

Williams, B. (2023). Customer Relationship Management (CRM) in the Spa Industry. https://scholar.google.com/scholar?q=Customer+Relationship+Management+in+the+Spa+Industry+Williams

World Economic Forum. (2025). Global Cybersecurity Outlook 2025: Navigating the AI Paradox. https://scholar.google.com/scholar?q=World+Economic+Forum+Global+Cybersecurity+Outlook+2025

WVSU / Blanza & Catilo. (2025). Leveraging Technology for Sustainable Growth: MSMEs in Iloilo Post-Pandemic. https://scholar.google.com/scholar?q=Blanza+Catilo+Iloilo+MSME+technology+2025

Wysocki, R. K. (2019). Effective Project Management: Traditional, Agile, Extreme, Hybrid. Wiley. https://scholar.google.com/scholar?q=Wysocki+effective+project+management+2019

Yoshi, K. (2022). The Importance of Offline Data Portability in Small Businesses. https://scholar.google.com/scholar?q=The+Importance+of+Offline+Data+Portability+in+Small+Businesses+Yoshi

Youn, S., Geismar, H. N., & Pinedo, M. (2022). Planning and scheduling in healthcare for better care coordination: Current understanding, trending topics, and future opportunities. Production and Operations Management, 31(12), 4439–4463. https://doi.org/10.1111/poms.13867

Zamora, K. (2023). Inventory efficiency in service industries. Philippine Operations Research Journal. https://scholar.google.com/scholar?q=inventory+efficiency+philippines+zamora

Zhang, X. (2023). Just-In-Time Inventory for Perishable Service Goods. https://scholar.google.com/scholar?q=Just-In-Time+Inventory+for+Perishable+Service+Goods+Zhang

Zhao, Y. (2022). Empirical metrics of organizational trust rooted in open accounting data layers. Information & Management, 59(6), 103–118. https://doi.org/10.1016/j.im.2022.103672

Zheng, L. (2023). Incremental Development Models for Financial Software Components. https://scholar.google.com/scholar?q=Incremental+Development+Models+for+Financial+Software+Components+Zheng

Zhong, Y., Birge, J. R., & Ward, A. R. (2024). Learning to schedule in multiclass many-server queues with abandonment. Operations Research, 73(6), 3085–3103. https://doi.org/10.1287/opre.2022.0197

Zhu, Y., Liu, Z., & Qi, X. (2024). Nonsequential appointment scheduling with a random number of requests. Production and Operations Management, 33(1), 184–204. https://doi.org/10.1177/10591478231224926

Zygiaris, S., Hameed, Z., Ayidh Alsubaie, M., & Ur Rehman, S. (2022). Service quality and customer satisfaction in the post pandemic world: A study of Saudi auto care industry. Frontiers in Psychology, 13, Article 842141. https://doi.org/10.3389/fpsyg.2022.842141

Żyminkowska, K. (2025). The role of artificial intelligence in customer engagement and social media marketing—Implications from a systematic review for the tourism and hospitality sectors. Journal of Theoretical and Applied Electronic Commerce Research, 20(3), 184–201. https://doi.org/10.3390/jtaer20030012